Part B

History of Changes

DoA_v1_01.09.2014

An explanation of the subcontracting costs of SPECTRAL ENGINES was included in Section 4.2.

An explanation of the third party of the Universitat de Barcelona (UB) was included in Section 4.2.

Recommendations from the Ethics Screening were completed in Section 5.

4 additional risks were included in the Critical Risks Table in the Part A of the DoA, relating to IPR issues, Commercial, Organisational and Managerial Risks.

DoA_v2_04.09.2014

General editing through the Part B document to correct typos and language related mistakes

In relation to the text on Background per beneficiary in 2.2.2.1 the follows was added on request of the legal team at MBN "Each beneficiary has the right to exclude or limit specific Background from each other's access by informing the other partners about this exclusion."

With regard to Open Access publications mentioned in 2.2.2.1, we added that we will publish in either Green or Gold. Before it just said Gold.

CV of Pau Puigdollers was added in the profile of IRIS.

An additional risk in relation to a partner leaving the consortium were added to the Critical Risks table in the Part A of the DoA. Removal of the Consortium Agreement Deliverable (D11.2) from the deliverables list in the Part A Deleted D11.2 from the Gantt on page 41 of the Part B.

DoA_v3_20.09.2014

Part B put into the Formal DoA template in keeping with the "Template with Instructions" Document.

DoA_v4_15.10.2014

Track changes accepted and the History of Changes section added.

DoA_v5_12.03.2019

List of changes in Annex 1 Part B

Location	Type of change	From	To
Section 4.1	IRIS name	Innovacio i Recerca Industrial i Sostenible	IRIS Technology Solutions SA
Table 3.4b	IRIS costs	166.000.00	226 120.00
Table 3.4b	UNIVLEEDS costs	187.787.00	219 067.00
Table 3.4b	TUD costs	24.000.00	27 000.00
Table 3.4b	DECHEMA costs	40.000.00	37 000.00
Table 3.4b	MBN costs	46.000.00	51 000.00
Table 3.4b	Grecian costs	55.000.00	60 000.00
Table 3.4b	GSK costs	76.000.00	50 000.00

List of changes in Annex 1 Part A

Page	Type of change	From	To
5	PM change in WP7	127.00	128.50
5	PM change in total	687.50	689.00
9	Beneficiary of D7.4	IRIS	UNIVLEEDS
10	Beneficiary of D8.1	DECHEMA	TUD
36	PM change for IRIS	14	30
36	PM change UNIVLEEDS	5	21
37	PM change for GSK	33.50	1
37	PM change for GM	5	6
37	PM change for MBN	4.50	5.50
37	Beneficiary of D7.4	IRIS	UNIVLEEDS
39	PM change for TUD	5.50	7.50
39	PM change for ZHAW	6.60	8.60
40	PM change for DECHEMA	4	0 (DECHEMA removed)
40	Beneficiary of D8.1	DECHEMA	TUD
59	PM change for IRIS	WP7:14, Total:102	WP7:30, Total:118
59	PM change UNIVLEEDS	WP7:5, Total:102.70	WP7:21, Total:118.70
59	PM change for GSK	WP7:33.50, Total:54	WP7:1, Total:21.50
59	PM change for GM	WP7:5, Total:13.50	WP7:6, Total:14.50
59	PM change for MBN	WP7:4.50, Total:14	WP7:5.50, Total:15
59	PM change for TUD	WP8:5.50, Total:29	WP8:7.50, Total:31
59	PM change for DECHEMA	WP8:4, Total:17.70	WP8:0, Total:13.70
59	PM change for ZHAW	WP8:6.60, Total:37.60	WP8:8.60, Total:39.60

List of changes in Annex 2

Type of change	From	To
IRIS Direct personnel costs	520.000.00	574 200.00
IRIS Other costs	166.000.00	226 120.00
LEEDS Direct personnel costs	469.543.00	521 543.00
LEEDS Other costs	187.787.00	219 067.00
GSK Direct personnel costs	324.000.00	129 000.00
GSK Other costs	76.000.00	50 000.00
MBN Direct personnel costs	145.200.00	152 800.00
MBN Other costs	46.000.00	51 000.00
GM Direct personnel costs	120.832.00	129 832.00
GM Other costs	55.000.00	60 000.00
DECHEMA Direct personnel costs	127.890.00	99 890.00
DECHEMA Other costs	40.000.00	37 000.00
TUD Direct personnel costs	152.250.00	163 250.00
TUD Other costs	24.000.00	27 000.00
ZHAW Direct personnel costs	zero	14 000.00

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1. Excellence

1.1. Objectives

The European process industry is uniquely positioned and represents the economic roots of the European economy (by transforming raw materials into intermediate and end-user products but also using energy, water, and possibly producing byproducts, emissions and wastes)[1]. It thus sits at the core of most industrial value chains via discrete manufacturing into a whole host of sectors[1]. The process industry is characterised by having a high dependence on resources (energy, raw materials and water) in production. Optimising the effectiveness of process control systems and strategies is **vital to improve quality, reliability and efficiency[2], as well as the sustainability of operations**.

Following the SPIRE philosophy, the ProPAT project aims to development an **integrated process control approach** (figure 1) embedding a novel monitoring system based on low cost analysers combined with smart sensors for monitoring in real time the critical control parameters of key processes in the minerals, ceramics, non-ferrous metals, and chemical process industries. The data from the sensors and analysers will be seamlessly managed via a versatile Global Control System Platform that will acquire, process and mine the data for closed loop control of the process or for decision support. The ProPAT platform is highly scalable for the integration of numerous analysers in the future and its flexible database enables data to be combined with information coming from business systems to guarantee structural consistency with the whole process or compatibility with existing supervision systems.

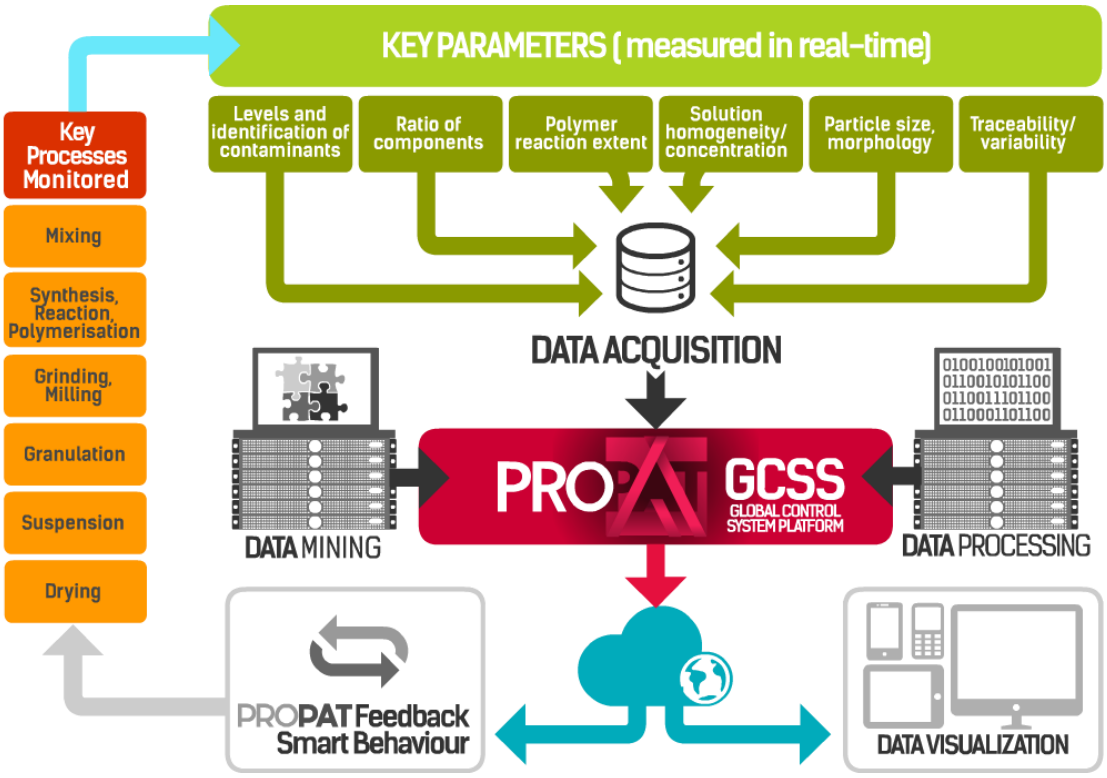


Figure 1- ProPAT integrated process control approach

The ProPAT development will deliver the following key innovations:

a) Affordable high-end probes (low-cost NIRs and particle size analyser) that will enable their use in several critical points along the same process[2].

b) MEMS-based standard sensors (temperature, flow rate, pressure) which may, on the long term, be powered by energy harvesting approaches (battery-less), wirelessly connected to the data logging

[1] Pablo Tello & Ron Weerdmeester. SPIRE Roadmap. http://www.spire2030.eu/uploads/Modules/Publications/spire-roadmap_december_2013_pbp.pdf [2] http://www.spire2030.eu/mediaroom/36/21/Process-Control-Industrial-Automation-Summit-2014

[2] Note that although the range of the used IR sensor will be extended towards MIR (mid IR) & TIR (thermal infrared spectroscopy) as required during the project, it is referred as NIR throughout the document; and the particle analyser is called Imaging Scatterometer (ISI).

subsystem. This will open up the possibility for some of them to form part of the process ("flow tracker sensors") in order to report bulk properties that are very difficult

to measure by static sensors.

c) Process-oriented development and validation of the ProPAT platform at several subsequent scales for assisting in the production of **minerals, ceramic, non-ferrous metals, chemical (plastic and pharmaceuticals)**.

d) An integrated **PAT platform** with tailored hard- and soft- solutions according to the process and applications of interest. Analytical sensors will be combined with data handling tools, including classical MSPC (multivariate statistical process control), **process modelling and data fusion for multisensor interpretation when needed**.

The **specific objectives that** will be fulfilled over the course of the ProPAT project are discussed below (effort has been made to ensure that these objectives are S.M.A.R.T.-*Specific, Measurable, Achievable, Realistic and Timely*- and clearly linked to the work plan and milestones).

1. **To perform an analysis of the quality control measures currently available to or implemented by the target mineral, ceramics, non-ferrous metals and chemical industries *(WP1)*.**

   This analysis will identify the critical control parameters governing variability along the process chain of the target industries (including major processes such as polymerisation, milling, mixing, granulation, drying). The currently used control strategies will be identified. Other available approaches will be researched from the state of the art. Initial recommendations for improvements in process efficiency will be made, while at the same time aiming to improve final product quality, as well as resource and energy efficiency. Variability will be qualified in terms of: raw material variability, in-process variability, final product variability, cycle time variability, product wastage and energy consumption variability. An assessment of the proposed ProPAT sensors and analysers will be made and discussed with the end-users in terms of integration requirements: sampling requirements, speed of analysis, cleaning-in-place, sensor fusion, etc. Similarly, an analysis of requirements in terms of data analytics and integration with current data management systems will also be carried out. Emphasis will be placed on the potential adoption of 'Green PAT' strategies to reduce energy and waste during processing due to advanced process control PATs. The required and desired features of a PAT strategy for the target applications will be produced and used to draw up the development and adoption strategies of ProPAT. A concept design and an integration strategy will be established for the different versions of the ProPAT platform to be rolled out in the selected target production lines.

   Performance indicator: By M3, end user specifications will be defined to enable PAT development work to commence and by M6, the project will have a roadmap in place to effectively guide the efforts in relation to potential PAT strategies for various control parameters along the process chains of the target industries.

2. **To set up and calibrate the novel scatterometer analyser for real-time and on-line measurements during granulation (*WP2*).**

   This novel sensor will be capable of measuring particle sizes above a micron offering a very low cost and rapid response. At the project start, the scattometric sensor will be integrated in a suitable test rig to enable the off line calibration of the sensor vs. standard measurements in the lab using relevant sample sets selected by the end users in the pharma, ceramics, non-ferrous metals, and minerals industries. In parallel, a replica sensor will be optimised for its use in an industrial environment.

   Performance indicator: By M12, the granulometric sensor will be calibrated vs. conventional sensors offline and by M16, an optimised robust version will be delivered for its integration with rest of the hardware in WP4.

3. **To set up and improve the novel low cost MEMS-NIR spectroscopic analyser (WP3).**

   A novel **MEMS-NIR spectroscopic analyser** will be integrated to measure compositional data at low cost vs. commercial NIR instruments and offering a rapid response. At the project start, the NIR sensor will be integrated in a suitable test rig to enable the off line calibration of the sensor vs. standard measurements in the lab using relevant sample sets selected by the end users in the pharma and chemical industries. In the test rig, three basic requirements of the NIR sensor will be tested and optimised: 1) high accuracy to obtain reproducible measurements, useful to develop

calibration models or to be fed in other control-oriented data analysis tools, 2) design and housing in order to work in demanding environmental conditions, and 3) inputs for the sampling method that will be developed in WP4 and needs an exact adapted opticalmechanical system. Various techniques will be used depending on the compositional parameters of interest as defined in the state of the art review and feedback from the survey of participating industries in WP1. In parallel, IR sensors with different wavelength ranges will be optimised for its use in an industrial environment so it can be delivered for its integration with rest of the hardware in WP4.

Performance indicator: By M19, the different versions of the robust MEMS-NIR sensor will be delivered with adjusted wavelength ranges for each product of interest for their integration with rest of the hardware in WP4.

4. **To develop chemometric models for individual sensors and methods for analysing multisensor inputs** (related to WP3 and 4).

   Development of software for individual sensors involves those providing a multivariate response (specially, the NIR sensor and, to a lesser extent, the scatterometer). For the NIR sensor, two main axes can be envisioned: process monitoring and control and process modelling. For the first one, Multivariate Statistical Process Control (MSPC) based on exploratory analysis of spectra can be implemented as well as control of product properties, through suitable multivariate calibration tools based on Partial Least Squares (PLS) or other algorithms when needed. Special attention will be focused on data preprocessing, wavelength selection methods, adequate sampling, and correct upgrading of models from well-controlled laboratory conditions to pilot-plant scale, where an adaptation process (known as model transfer) and new validation will be required. Complementing the process control aspect, NIR sensors could also be equipped with process modelling tools that could provide (when applicable) information on the development of a reaction (in synthesis processes) through the use of hard-modelling tools based on physicochemical models or to the description of the general transformation of a system, e.g. in blending processes, through soft-modelling methodologies, such as Multivariate Curve Resolution (MCR). Hybrid hard- soft-methodologies can also be envisioned to deal with non-ideal response behaviours or with systems in which the response variation cannot be fully described by a first principles model. Some multivariate raw data from scatterometer, e.g., particle size images, particle size vs. measurement angle, etc. will be explored with multivariate data analysis tools to test the possibility to obtain additional information on the variation of particle size along the development of the process. Data fusion approaches can be envisioned in the framework of multisensor platforms. To do so, strategies to weight the relevance of the different sensor responses in the global response will be considered. Data fusion at different levels and multiset or multiblock analysis methods to deal with data from different origin will be tested to find the best approach for the different applications. URV and UB will jointly coordinate for these tasks. Through URV-UB, software for the suitable sensor platforms will be developed. URV will focus mainly on developing chemometric models for the MEMS-NIR sensor for all the products of interest for laboratory data and models will be developed for pilot-plant scale and an optimised version will be delivered for its integration with the rest of the hardware in WP4. UB will mainly focus on process modelling tools and multivariate analysis tools to interpret the scatterometer output will also be provided. Tailored data fusion strategies for multisensor platforms will be designed.

   Performance indicator: By M19, chemometric models will be developed for the MEMS-NIR sensor for all the products of interest for laboratory data. Suitable software for the sensor platforms will be developed. By M22, models will be developed for the multisensors platform and ready to be integrated with the rest of the ProPAT system in WP5 and tested in pilot-plant scale in WP6.

5. **To develop and integrate a suite of smart sensors for use within ProPAT hardware delivering cheap and robust, yet high performance technology with a wide range of applications** (***WP4***).

   A promising suite of smart sensors and components will be selected and integrated, such as:

   a) MicroElectroMechanical systems (MEMS)-based components;
   b) Smart sensors that have intelligent readout electronics and the microsensor in a single chip;
   c) Sensors for in batch and in continuous processes
   d) Zero power ‘flow tracker’ sensors.

These new miniaturised sensors will match the existing solutions in performance, while enabling size reductions, low power consumption, and low unit costs. These sensor platform technologies will be integrated into ProPAT prototypes (a tailored prototype will be designed and delivered for each industrial application) together with the granulometric and NIR-MEMS sensors from previous WPs. The sampling units and interfaces for the installing the system in the different target industries will also be built.

Performance indicator: By M22, a novel platform of low cost sensor solutions ready to be integrated in the selected process industry operations will be developed. System performance such as accuracy and stability will be quantified against current standards.

6. **To develop data management strategies to analyse the vast data arising from PAT adoption and the Global System Control in order to successfully integrate and implement the ProPAT systems in the target process industries** (***WP5***).

   One important reason why many companies shy away from developing NIR-based and other spectroscopic sensors is the **lack of effective data management systems**. With the inputs of the chemometrics developed in WP3 for NIR and the scatterometer sensor, as well as the data fusion strategies to handle the signals of all additional sensors from WP4, a flexible system will be developed which facilitates data collection from an array of sources and interprets the data for appropriate decisions to be taken. Robust data processing protocols for sensor methodologies, which identify and explain critical sources of process and product variability will be developed. An effective framework for interpretation of multisensor data fusion will be created to give more accurate on-line process performance monitoring. A model-based supervisory control system suitable for the target process industry applications will be produced.

   To successfully implement a PAT system, a **Global Control System Software** (GCSS) - provided with built-in multivariate analysis or data mining tools for real-time prediction and assisted decision routines - is required. This will holistically provide the connectivity management, interface with users and other business systems, and complete data management at a higher level. Managing PAT data is highly complex because of the huge information flow and because it includes mixed data formats from different types of sensors and instruments. This data needs to be combined with the information coming from the business systems to guarantee structural consistency with the whole process or compatibility with existing supervision systems - a **flexible database**. The database architecture should also be suitable for integration with the open architecture software of the Global Control System. The PAT paradigm makes sense only if sensors and software resources are harmonically integrated within a specific production line. The conditioning of sensors to make them compatible with production line specificities, and performing minor adaptations in the production line are critical steps. A communication network needs to be implemented for data collection and exchange. Finally, the GCSS should be housed in suitable hardware, all the elements need to be integrated in a common and coherent setup, and pre-industry validation tests must be performed. The control system will be versatile to serve the purposes of the different target industries and processes.

   Performance indicator: By M24, a Global Control System Software will be developed linking the sensor software and global data management, and by M25 the ProPAT Global Control System will be validated at lab conditions and ready to be installed in a pilot plant.

7. **To validate the ProPAT system for online process monitoring at pilot scale (*WP6*)**

   First of all, the target processes, downscaling those of participating industries, will be set up and tested using standard quality procedures. The ProPAT system will be integrated for online process monitoring in high temperature chemical pilot plants, fine chemistry mixing/granulating as well as milling pilot facilities. The synthesis of pharma, resin and solid products from among ceramics, metals and minerals will be monitored first of all in manual mode whereby the indicators of quality such as the advance of the polymerisation will be provided to the user to analyse their validity and safety for the process to proceed, with the process conditions being changed manually. Chemometric models that were developed in WP3 based on offline tests of samples taken at different points of the process will be updated (either by model transfer when possible, or by new modelling from newly taken samples). The predictive models included in the PAT approach in WP5 will be validated. Following this pre-validation

work, the system will be used in closed loop. Any operational issues identified for improvements in terms of hardware or software will respectively feedback into WP4 and 5 for further optimisation.

Performance Indicator: By M24, at least 3 model processes (high temperature resin synthesis, mixing and granulation of fine chemicals, dry and or wet milling) will be in place at pilot scale. By M36, the ProPAT system will be validated at pilot scale and guidelines ready for its testing in industrial scale and related reports will be ready at M39.

8. **To perform in-process verification and testing of the integrated PATs before PAT and QbD implementation case studies** (***WP7***).

   The participating end users from the target mineral, ceramics, non-ferrous metals and chemical industries will test the PATs and control strategies developed and optimised by ProPAT under normal production conditions to assess their efficacy and performance within the process environment. Design issues arising from the industrial tests will be fed back to the consortium members to facilitate further optimisation, and again the chemometrics models will be fine-tuned as needed. Such tests will reproduce in the industrial facilities the polymer synthesis previously performed at pilot scale, as well as easier (non-reactive) processes such grinding, suspension or dispersion processes. Finally, structural parts will be engineered converting the previous polymers and will be controlled via the ProPAT sensors e.g. in terms of any accessible compositional parameter.

   Performance indicator: By M36, the ProPAT platforms will be successfully installed in the 4 test sites of the process industries. By M45, a series of trials testing the ProPATs under process conditions will validate the performance of the ProPAT system for obtaining in-process measurements and control.

9. **To perform PAT and QbD implementation case studies in order to evaluate the impacts on cost reduction, resource and energy efficiency and demonstrate the advantages of using the system (WP8)**

   Based on the results of WP6 and WP7, once a statistically significant number of batches have been produced the cycle time and quality of the resulting product using ProPAT to control the process will be compared will the existing methods. This comparison vs. existing quality control will fuel into recommendations for future standardisation. The evaluation of the results and their impact on the waste reduction, cycle time, energy consumption, etc. will also fuel into the commercial feasibility study for the systems in different industrial environment scenarios. Finally, a life cycle analysis of productions using the ProPAT system vs. previous set ups will allow the environmental impact of the new practice to be demonstrated, along with the contribution to the sustainable chemistry platform pillars.

   Performance Indicator: By M48, the system will be demonstrated in terms of environmental, economic and standardisation potential.

10. **To design and implement a comprehensive knowledge transfer and training programme aimed at the European process industry in order to overcome major barriers of expertise in relation to the implementation of PAT and QbD in industry (*WP8*).**

    Performance Indicator: By M48, at least 4 training sessions will have been delivered reaching over 200 members of the target sectors of the SPIRE community.

11. **To maximise the innovation impacts of the project by effectively managing the knowledge generated during the project to allow for the wide dissemination and use of the results to benefit the ProPAT consortium and the wider European process industry *(WP9).***

    In order to ensure a swift route to market for the results of the developed technologies, the protection of the knowledge generated will be of paramount importance. With a view to gearing up for commercial adoption of ProPAT in the industry, standardisation and regulatory issues will be addressed to ensure compliance with process conditions. The industry partners will draw up an exploitation plan for the use of the project results as well as dissemination activities to raise awareness about the research among the industry and the general public, in order to stimulate future market demand. The issue of branding and communication will be given careful attention.

    Performance Indicators: By M48, the protection procedures for the projects results will be in place, an agreement on the ownership, sharing and use of the results will be signed among all the partners, a business plan outlining the business model for commercial exploitation of the results will have been drawn up by the partners, and a comprehensive series of dissemination activities will have been carried

out that will have successfully reached out to the process industry, plant and equipment providers, policy makers, educational bodies, the research community and the general public.

1.2. Concept and approach

The overall concept underpinning the project

The SPIRE Public Private Partnership, which brings together cement, ceramics, chemicals, engineering, minerals and ores, non-ferrous metals, steel and water sectors (4 of which are represented in the project which together provide a range of materials to all SPIRE industries and beyond), has a strategic roadmap to rejuvenate the European process industry, reducing resource and energy inefficiency and the environmental impact of industrial activities, leading to growth and increasing the competitiveness of Europe in a global market, of which **Process monitoring, control and optimisation is one of the Key Actions** (KA 2.3)[1].

Today, the industry faces major challenges including globalisation, environmental regulation, and shortening product life cycle. Meeting these challenges has required the development of innovative technologies and alternative approaches geared towards **reducing costs and improving the environmental and economic profile of processes**. Breakthroughs in process operations and process modelling have been necessary for achieving energy and material efficiency gains. These breakthroughs are made possible by the use of process analytical technologies optimally designed for **real time in situ process monitoring, characterisation and accurate control**[3] that offer benefits such as minimising waste, improving reaction output, increasing energy efficiency, decreasing the formation of by-products, and minimising the potential for accidents[4].

The Process Analytical Technologies (PAT) approach, which according to the Food and Drug Administration (FDA), is a system for designing, analysing, and controlling manufacturing through timely measurements of critical quality and performance attributes of raw and in-process materials and processes with the goal of ensuring final product quality. The hypothesis behind PAT is that the quality of the products can and should be incorporated by process design and not by postproduction quality testing. In addition to pharmaceutical manufacture, the potential impact for PATs in other processing sectors, such as chemical/polymers, pharmaceutical, mineral and ceramics, as well as the other SPIRE sectors, for product quality and safety, process and resource efficiency, yields, and consequently competitiveness is tremendous.

PAT adoption has to date been confined to large multinational companies, however there is a major opportunity for the development of new PAT tools that are affordable, easily integrated into existing processes, simple to use with clearly demonstrated advantages for implementation. However, barriers exist for wide deployment within the European processing industry, including; sensor costs, dedicated sensors targeted at the unique operations encountered in the processing industry, dedicated data management systems, strategy awareness, gaps in knowledge and internal expertise, and validation of the PAT philosophy for the target applications.

The ProPAT platform (Figure 1) will overcome these barriers to adoption by developing a suite of affordable, high performance ProPAT sensors and analysers - low cost MEMS-NIR spectroscopic analyser, novel scatterometer probe, smart sensors that have intelligent readout electronics and the microsensor in a single chip, sensors for in batch and in continuous processes, and zero power 'flow tracker' sensors- integrated into a Global Control System Platform containing the chemometric tools and the predictive software to deliver an integrated process control platform. This will allow for the implementation of "near real time" closed-loop process control concepts making it possible to operate industrial processes at their optimum both economically and ecologically while ensuring a high level of safety, but it will also enable the uptake of the QbD approach for continuous process improvement. The platform provides self-learning and predictive capabilities aimed for dramatically reducing overcosts derived from even slight deviations from the optimum process.

Through its use for monitoring **milling, granulation, mixing, drying, synthesis** processes, which are broadly used in all SPIRE industries and beyond, the versatile ProPAT system will be validated in the

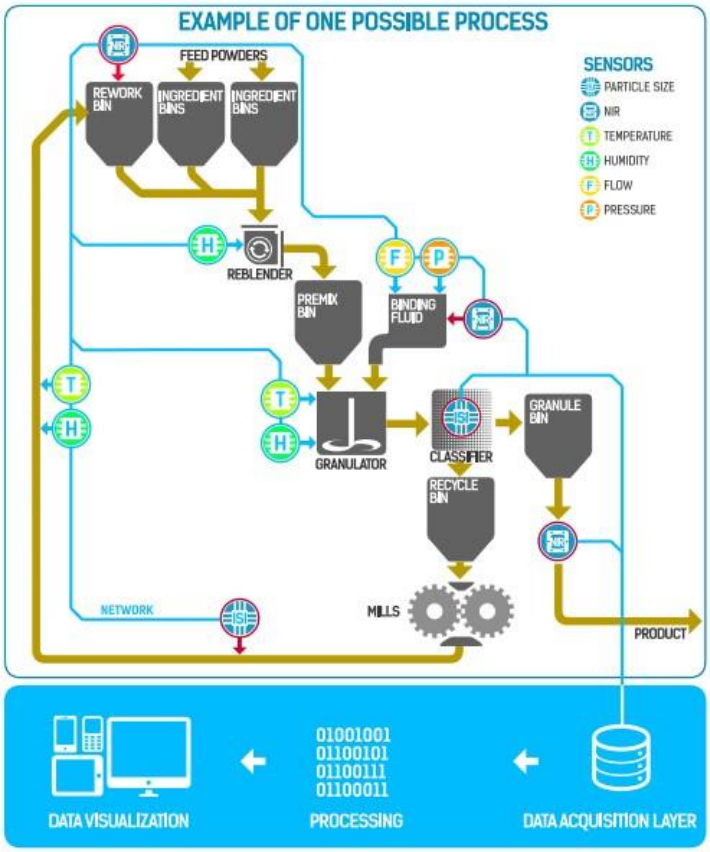
[3] http://us.mt.com/us/en/home/events/webinar/ondemand/odw-going-green-pat.html

minerals, ceramic, non-ferrous metals, chemistry (plastic and pharma) production, and will represent a major step forward towards more reliable and sustainable industrial operations in the European process industry, whereby there is **high potential for the cross-sectorial transfer of the ProPAT technologies**. Indeed, to facilitate this, the project will include a comprehensive **knowledge transfer and training component to bridge the expertise gaps** that are currently blocking the uptake and deployment of PAT and QbD approaches in the European Process Industry.

Figure 2 shows the implementation of ProPAT in a typical pharma process. A number of our low cost NIR and granulometric sensors will be required along the different production steps, in order to first of all ensure that the mixture of the excipients and active principles is correct, to optimise the granulation process and obtain the optimum particle size in solid or liquid state, as well as to optimise the time to obtain it, before spray drying during which both the particle size and the residual water amounts will be monitored to again adjust the time. All in all, having access to real time data as opposed to offline analysis that can take up to several hours, as well as obtaining feedback on the process, will allow quality to be maximised, minimising energy and production time. It will also speed up the time for designing new products, and maximise the valorisation of the capital investments into production capacity.

From its inception the PAT philosophy has been one of collaboration, between sensor manufacturers, process & control engineers, quality control scientists and regulatory agencies. It was recognised that such an approach would benefit each party, be that through equipment provision, improved processing or reduced risks. “Analytical” has developed to include chemical, physical, microbiological, mathematical, and risk analysis conducted in an integrated manner. Consequently PAT research initiatives have almost universally brought a diverse grouping of specialities together.

Spread within the EC are emerging areas of expertise, which, when combined, meet the diverse objectives of developing a framework for innovative manufacturing and quality assurance. ProPAT brings such international expertise together to complete the considerable aims of the project. This expertise will not only improve the processes of end-users throughout Europe but also lead to the development of innovative solutions both hardware and software. European PAT solution providers will be in a position to bring such innovation to the global market.



The ProPAT mobilises multidisciplinary knowledge combining expertise in sensor development and integration, optical engineering and integration, numerical and statistical modelling, software and web development, telecommunications, mechanical and electronic engineering, as well as process expertise in relation to the critical process of the target industries. This expertise is mobilised from the academic and scientific communities (ICFO, VTT, UNIVLEEDS, URV, ZHAW, TUD, UB, IMAR), from technology providers (IRIS, SPECTRAL, MEMSFAB), as well as the process industry end-users from the SPIRE sectors (GM, GSK, MEGARA, MBN) and completed with the unrivalled PAT expertise of DECHEMA that groups over 5500 members from the European process industry.

Figure 2- Schematic of the ProPAT Technology illustrated in a typical pharmaceutical process

Positioning of the project

As was discussed, one of the fundamental differentiating factors of the project is to integrate novel sensors developed previously by ProPAT project partners with potential to be more affordable than commercial

equivalents. Both of them are currently at TRL3 and each has been successfully tested at lab scale. During the project these sensors will be scaled up and optimised for their integration in a robust version that can be validated by the industry. To this end, the work performed in the project will advance the readiness of these sensors increasing their robustness for the demanding industry environment in which they will be tested and validated by firstly performing exhaustive tests in a close loop approach at pilot scale in UNIVLEEDS before being demonstrated for providing useful and reliable data in industrial facilities (TRL 5 to 6). The Global Control System Platform will also be developed to TRL5-6, whereby it will be validated and demonstrated in an operational environment. This integrated PAT platform incorporating a multitude of smart sensors and hard- (or model driven) and soft- (or data driven) solutions tailored to the process and applications of interest will support the manufacturers to optimise their processes in a close loop fashion on the long term.

National or international research and innovation activities which will be linked with the project

Fundamentally for this project, continuous links will be ensured with the SPIRE PPP, in that it aims at integrating, demonstrating and validating systems and technologies capable (across all SPIRE sectors) and ProPAT will leverage SPIRE to reach the European Process Industry. Links will also be built with other European initiatives such as the Resource and Energy Efficiency Partnership "REP", which is a group comprising a broad range of major European Industry Associations (such as the European Engineering Industry Association, the Industrial Minerals Association, Glass for Europe, the European Coil Coating Association, the European Copper Institute, the European Aluminium Association, Eurométaux, the Confederation of European Paper Industries, the European Ceramic Association, the European Cement Association and the Nickel Institute, among others) and European Technology Platforms (such as the European Steel Technology Platform, the European Platform for Sustainable Chemistry, among others).

In order to **maximise the European Knowledge Area**, the ProPAT project will consider the results of relevant past and ongoing projects under past Framework Programmes, as well as national or regional projects, in which the partners are/were involved, such as: F^3 Factory, Flexible, fast and future production processes (FP7-NMP), one of the lighthouse projects from which SPIRE is a follow-up[4], PARTICLEPRO- Development of a hybrid process analytical technology to characterise the physical and chemical properties of particles (FP7); OPTICLEAN- Effective and reliable optical system for cleaning validation in pharmaceuticals manufacture (FP7); OPTIJECT- (A novel NIR-based instrument for in-line monitoring during injection moulding (FP7); MUSE-TECH (Multi sensor technology for management of food processes), IQFUEL (development of MEMS processing technology for NIR FPI for fuel quality sensing); CoNanoMet (European Project on Nanometrology development and standardization); Spanish grant: CTQ2012-38616 (Development and application of chemometric tools for process and hyperspectral image analysis); NANODEFINE (Development of an integrated approach based on validated and standardized methods to support the implementation of the EC recommendation for a definition of nanomaterial); in addition to a number industrial innovation activities, e.g. implementation of Continuous processes in AstraZeneca globally; Chemometric maintenance of on-line NIR analyser technology at REPSOL refinery, Spain; among others.

Moreover, in the context of the European Structural and Investment Funds through the smart specialisation strategies, ProPAT will engage with policymakers to communicate the benefits of the project and its results, as well as its replicability across Europe, in order to mobilise **investment of resources which would stimulate knowledge-driven growth**.

Overall approach and methodology

In relation to the overall approach and methodology that will be followed in order to successfully succeed in the project's objectives, five distinct project activity phases have been envisaged: 1) Defining the ProPAT specifications; 2) Developing the ProPAT Technologies; 3) Validation and Demonstration; 4) Training and Education; 5) Deployment in the European SPIRE industries (a post-project phase, but which drives a cross-cutting or horizontal series of activities throughout the project, namely 'Gear up for Deployment').

In relation to the first project phase, **Defining the ProPAT specifications**, the requirements of the targeted end-user minerals, ceramic, non-ferrous metals, chemistry (plastic and pharmaceuticals) industries will be

[4] Søren Bøwadt, DG Research and Innovation, Directorate Industrial Technologies; Presentation at Suschem Brokerage event, 31.01.14, Brussels

studied in order to define the constraints of their current processes that will need to be taken into account by the ProPAT integrated process control system both in terms of potential improvements of process consistency and product quality (reduced batch-to-batch variation), and in terms of the constraints that might influence on the proposed sensoring systems (e.g. temperature and pressure to handle, number of sampling points, etc.). This phase will finish with the drawing up of the concept design of the different versions of PAT platform for each end user as well as their integration strategy.

In relation to the second project phase, **Developing the ProPAT Technologies**, based on the specifications that will fed in from Phase 1, the ProPAT suite of low cost, robust and reliable ProPAT sensors and analysers (including the novel tailored NIR and particle size sensors) will be further developed and optimised for their ability to provide fast, in-line process measurements. The Global Control System Platform will be designed and built to enable smooth data acquisition for subsequent processing and mining for real-time prediction and assisted decision routines for closed-loop. The Control Platform will also manage connectivity, interface with users and other business systems, as well as complete data management at a higher level. The flexible database will enable data to be combined with the information coming from the business systems to guarantee structural consistency with the whole process or compatibility with existing supervision and control systems.

Once the ProPAT hard- and software technologies have been outputted from the previous phase, **the third project phase can commence, Validation and Demonstration.** This will involve a preliminary set up of model processes that will mimic those of the participating end users in terms of polymer synthesis, pharma mixing, granulation, drying and minerals or ceramics milling before testing exhaustively the developed sensors and PAT system. This is a fundamental phase of the project, in that the qualification of equipment and validation of analytical methods are critical components for the implementation of process analytical technology (PAT)[5]. Furthermore, the sensors will be demonstrated at each of the end users where the value of the data they bring both in terms of product and process will be verified, as well as the ability of ProPAT to act as a decision support system leading to significant quality improvements.

Training will also be instrumental to a successful ProPAT approach. Indeed at the 2014 edition of the Annual PAT & Quality by Design Conference[6] for the pharmaceutical sector, 80% of companies surveyed said they plan to invest in Quality by Design and PAT solutions, yet 72 % said they do not have a fully established process in place, with 59% of respondents citing the reason for this gap being due to internal lack of expertise, which highlights the critical need for training. To this end, in addition to the comprehensive training activities that will be organised in the ProPAT work plan delivered to the SPIRE community, engagement with “teaching” experts is also planned to ensure that the knowledge generated during the project which is identified as an innovation outcome relevant for teaching activities can be channelled effectively to the education community. This will be done through the organisation of dedicated workshops and events for the sharing of knowledge and best practices at appropriate stages of the ProPAT project. This will have a major impact in terms of raising skills in Europe in the area of PAT and QbD which will greatly benefit implementation in industry. In relation to securing suitable skillsets in the future workforce, through DECHEMA and the University partners, the project will outreach to education stakeholders in Europe in order to build new skills in Europe through enriching education programmes.

Finally, the **Deployment in the European SPIRE industries** of the ProPAT technologies in industry will be the true measure of the success of the ProPAT project. Indeed, with sectors such as minerals, ceramic, non-ferrous metals, chemistry (plastic and pharmaceuticals) that show a large number of important production operations throughout Europe, the **SPIRE innovation and retrofitting potential** is enormous[1]. As such the current manufacturing facilities establish an enormous leverage base for SPIRE innovation, in some cases, such as new process control, offering double digit replication potential compared to the initial single pilot project in which a novel SPIRE technology will be proven and demonstrated[1]. As the chain of development goes through many steps with corresponding increased amounts of investment, support measures such as the PPP are needed to cross the valley of death from laboratory level technology to mini-plant or pilot and further to the build-up of demonstration capability. Once there is successful technology demonstration in SPIRE, the most expensive step of deployment (typically with factors of 5 to 10+ higher than for a demonstration capability) will be made by industry[1]. It is as such fundamental that there are well planned activities during the project to gear up for the post-project

[5] Process Analytical Technologies (PAT) and Quality by Design (QbD) Overview 2014. Pharma IQ.

deployment phase (such as managing knowledge and IPRs, business and post-project industrialisation planning, gear up for regulatory approvals, etc.)

Sex and/or gender analysis

In relation to **gender analysis**, the ProPAT consortium understands that inventions and developments that incorporate the smartest aspects of gender can open new markets and enable innovation in products, processes, services, or infrastructures. Indeed, products that meet the needs of complex and diverse user groups enhance global competitiveness and sustainability[6]. Inspired by how taking gender into account has yielded new research results and sparked creativity, opening new avenues for future research for those who give it serious attention[7], during the work plan, **gender analysis will be addressed in relation to the User Interface of the ProPAT platform**, whereby graphic designs of the interface will be consulted with both men and women during the design phase in WP5, and posterior alpha and beta versions will be trialled by both men and women.

Moreover in relation to **business behaviour** gender analysis will also be important, especially in view of growing trends and commitments to see more women occupy high level and decision-making positions in industry. For example, in Europe, Norway a decade ago mandated a 40% quota for female director's at large companies, and according to reports has now achieved or even exceeded this target. Spain is aiming for a similar target by 2015[8]. In June 2011, Belgium's parliament voted to force stateowned businesses and stock market-listed firms to award 30% of managing board seats to women. Large companies have five years to comply, SMEs eight years. In 2012 started, the parliament of the Netherlands was poised to debate similar rules. By 2017, at least 40% of managing and supervisory board members at companies listed on France's CAC-40 stock exchange and all other firms with sales of more than €50m and at least 500 employees must be female. Germany and the UK established voluntary targets for female representation in 2011[9]. All of this points towards a strong movement for change.

Indeed, with more young women than ever now leaving universities with chemistry degrees[9], the ProPAT consortium, which is headed by a senior female researcher with a chemistry background, Dr. Elodie Bugnicourt, will benefit greatly from effectively addressing gender analysis issues in that there is a distinct European trend in process industries, and in particular the chemical industry, in **boosting role of women decision makers**. Some examples from the chemical industry include, among others: Rotterdam-based petrochemicals giant Lyondell Basell, appointed Karyn Ovelmen as chief financial officer in 2011; in October 2010, Sandra Peterson was named CEO of Bayer Crop Science and BASF has pledged to increase the proportion of women in what it terms 'solid-line' German management to 15% by the end of 2020, from just below 10% in 2010; up to 2015, the Bayer group aims to 'gradually' increase the share of women in senior management from about 21% at present 'toward a target of 30%' and its goal for the supervisory board is 20%; German chemicals firm Merck made two key appointments in December 2011 with Friederike Rosch, being named head of corporate accounting and Isabel De Paoli, head of corporate strategy.

In relation to the **gender dimension of the ProPAT consortium**, the team of researchers, technologists, industrialists and policy influencers has a significant cohort of both sexes (ca. 40% female; 60% male). Moreover, a recruitment policy is in place designed to ensure a reasonable level of balance between the sexes. This is important in ensuring the education and training of the next generation of scientists, technologists and others needed in the EU and elsewhere. It is noteworthy that a number of the key management and decision-making positions within the project organisational structure have been assigned to highly qualified women, not least the role of the Project Coordinator (Dr. Ing. Elodie Bugnicourt), the IPR Manager (Dr. Silvia Carrasco), the data manager (Ms Carla Kapikian) and the Dissemination Manager (Dr. Renata Körfer). Interestingly, additional females are among the key drivers of this ProPAT project (Oonagh Mc Nerney, Dr. Cristina Serrano, Dr. Anna de Juan, Dr. Luisa Freitas Dos Santos, Dr Anna Rissanen, Dr. Poppy Krassa, Dr. Athina Meidani, Dr. Agnes Zoller, etc.).

[6] http://ec.europa.eu/research/science-society/gendered-innovations/index_en.cfm
[7] Londa Schiebinger (2008). Gendered Innovations in Science and Engineering. ISBN: 978080475814
[8] http://www.soci.org/Chemistry-and-Industry/CnI-Data/2012/2/Closing-the-gender-gap

1.3. Ambition

Advance beyond the state-of-the-art

The process industry transforms material resources (raw materials, feedstock) during a batch or (semi)continuous conversion into a new material that has significantly different physical and chemical performance than the starting substance. This material is then usually shaped by discrete manufacturing into an end user product or intermediate component; often this requires combining several different and discrete manufacturing operations to result in the final consumer product[1].

Process manufacturers require a high degree of automation, monitoring, and advanced simulation and control for the more challenging operations. Emphasis is on continuous or batch mixing, reaction and separation of materials to produce other materials of higher value. Advancing globalisation and stronger competition, demand the process industry to have **seamless process control, greater flexibility and cost efficiency** and many process manufacturers are becoming aware that they can benefit from investing in **technologies to automate their often complex manufacturing processes**[9].

The **proposed project is highly ambitious** in that it aims to open up a world of process understanding and continuous on-line control for improved product quality and safety, process efficiency, yields, resource and energy efficiency, and ultimately profitability. Indeed, research-based development and implementation of PAT in the European process industry (as a consequence of projects such as ProPAT) is **focused on improved control, rapid final product quality evaluation, increased resource and energy efficiency and productivity**, will have a knock-on effect in terms of industrial innovation and enhancing competitiveness. The key innovations brought by ProPAT are, on the one hand, the use of novel low cost NIR (MEMS based) and granulometric (imaging scatterometer) sensors in combination with smart sensors capable of measuring traditional parameters in a time/space resolved fashion depending on the individual processes, and on the other hand the calibration and validation of the integrated PAT platforms in a number of process applications including minerals, ceramic, nonferrous metals, chemistry (plastic and pharmaceuticals, etc.).

The ground breaking integrated process control approach that will be developed during ProPAT embeds a **highly novel monitoring system**. It will deliver: a) **affordable high-end probes** for use in several critical points along the same process; b) MEMS-based standard sensors (temperature, flow rate, pressure, etc.) powered by energy harvesting approaches (batteryless), wirelessly connected to the data logging subsystem to allow their use in the actual process ("flow tracker sensors") to report bulk properties; c) process-oriented development and validation of the ProPAT platform at several subsequent scales for assisting in the production of chemical/polymers, pharmaceutical and mineral products.

How the ProPAT novel sensors advance the state-of-the-art

The fundamental differentiating factor of the project is to integrate novel sensors (particle size analyser ISI and MEMS-NIR spectrophotometer) with the potential to be more affordable than their commercial equivalents. For instance, the MEMS-NIR probe -upgraded to be used as a PAT tool- would be priced around €7K, compared to an equivalent NIR system costing in excess of €25K. Likewise, an off-the-shelf on-line particle size analyser could cost over €30K, whereas the prospective price for the ISI system will be in the region of €4K. As such, **ProPAT is disruptive** in its ability to overcome the high cost of analysers, which is currently one of the barriers to **limiting the widespread implementation of the PAT paradigm in industry**.

Comparison of the novel Imaging Scatterometer (ISI) sensor with prior-art solutions for particle size analysis

Particle sizing methods can be separated into three basic classes: ensemble methods, counting methods, and separation methods. Separation methods are disregarded from this comparison as they do not satisfy basic requirements because they imply destructive test methods. Ensemble methods collect mixed data from all of the different size particles in a sample at the same time, and then digest the data to extract a distribution of particle sizes for the entire population. Common ensemble techniques are Low Angle Laser

[9] http://industrialautomation.wikia.com/wiki/Process_vs._Discrete

Light Scattering, Photon Correlation Spectroscopy, Back-scattering Spectroscopy, and others (compared in table 1).

Table 1: Comparison of conventional particle size analysers based on ensemble methods

Ensemble methods	Physical parameter measured	Time/ Read	Dynamic range	Previous knowledge of sample required?	Mixtures	Size distribution data	Measurement
LALLS	Forward scattering light power in multiple angles(low angles)	2 min	VERY HIGH 0.1 µm- 1 mm	YES	NO	YES	At-line/off-line
PCS	Forward & Back scattering light intensity variation	1 min	LOW 0.3 µm – 3 µm	NO	YES	NO (only mean size and standard deviation)	At-line/off-line
BBS	Backscattering light power in multiple angles(high angles)	1 min	HIGH	YES	YES	YES	In-line (noninvasive)
DLS	Forward & Back scattering light intensity variation	10min	VERY HIGH 1.5nm–10µm	YES	YES	YES	Off-line

The counting methods all characterise the sample distribution one particle at a time, basically by accumulating counts of particles with similar sizes. Some common counting methods are *the electrozone counter, the light counter, the time* of *flight counter, and the microscope (optical or electron).* In each of these methods, particles are classified and placed in "size bins", one particle at a time. These methods (compared in table 2) must each ensure that multiple particles are not counted together and cause errors in the reported size distribution ("co-incident counting"). The accuracy and resolution of these methods depend on how accurately the size of each particle can be characterized during the (usually) very brief time that it is counted.

Table 2: Comparison of conventional particle size analysers based on counting methods

Counting methods	Physical parameter measured	Time/ Read	Dynamic range	Previous knowledge of sample required?	Mixtures	Size distribution data	Measurement
EZC	Variation in electrical resistivity in an orifice	2 min	HIGH 0.5 µm-300 µm	YES (previous preparation of the sample)	YES	YES	At-line/off-line
LC	Light power attenuation by Absorption & Reflection of the sample	3 min	VERY HIGH 0.5 µm - 2 mm	YES (previous preparation of the sample)	YES	YES	At-line/off-line
TFC	Time function of Mass and Volume of sample	1 min	HIGH 0.2 µm-700 µm	YES (previous preparation of the sample)	YES	YES	Off-line
MC	Imaging by Light Absorption & Reflection of the sample	5 min	HIGH 0.5 µm-500 µm	YES (previous preparation of the sample)	YES	YES	At-line/off-line

The applied standard of particle size analysers in the market is mostly the ASTM D4464 - 10 Standard Test Method for Particle Size Distribution of Catalytic Material by Laser Light Scattering.

In relation to a review of the precision of currently available commercial analysers[10], median precision scores revealed that there is a marked difference between techniques that use laser diffraction[11] and those that rely upon on other physical principles. Laser-based techniques all showed precision scores lower than 0029, the Fritsch Analysette 22C achieving the best overall score. In contrast, the next most precise technique is over an order of magnitude less precise. Of the other instruments, settling methods were the next most precise[13], followed by electro-resistance[12], dynamic image analysis[13], time-of-transition[14] and static image analysis[15].

ICFO has developed different generations of particle size analysers: Imaging Brigthfield Cytometer (CYT-B), Imaging Glossmeter (HZM-G) and Imaging Hazemeter (HZM-H). The CYT-B is a compact, low-cost cytometer based on a lens-free microscope combined with a propietary antibubble fluidic system (microfluidic, microscope slide or chamber slide), capable of monitoring most common particles and cells, minimizing false readings, with 1-2 µm as lower particle size detection limit and 10 particles per 1ml as lower concentration limit. It can analyse more than 100.000 particles/cells in one image, thanks to a huge field of view of 20mm^2. The HZM-G is a compact, low-cost glossmeter based on a lens-free microscope combined with a propietary optical system that allows measuring power backscattering diagram at low angles (<10 degree) in reflection configuration, gloss values with 0.1% precision, analyse and counting microscopic particles and detecting imperfections with 1-2 µm resolution. The HZM-H is a compact, low-cost hazemeter similar to the HZM-G but that operates in transmission configuration, that measures haze values with 0.1% precision, analyse and counting microscopic particles and detecting cells, particles, or imperfections with 1-2 µm resolution.

The **Imaging Scatterometer (ISI)** developed at ICFO, encompasses the technologies of the above mentioned prototypes, has a new design and method to measure the scattering of solid, gel and liquid samples, to measure its associated gloss, haze or turbidity properties, using low-cost and large-consumer components. The design consists basically in a light source, a spatial filter or spatial filter array absorbing in the spectrum of the used light source, an image sensor combined with image processing based on lens-free holographic algorithms. The combination of both elements offers a visualization system in which the images can be interpreted by means of image processing (counting and analysing) obtaining particle size statistics and the scattering map of the analysed area. As a consequence, the apparatus offers microscopic images of the sample at the same time that it offers the scattering power value at different predefined angles with an associated haze value if the measurement is in transmission from a gel or solid sample, an associated gloss value if the measurement is in reflection from a gel or solid sample, or an associated turbidity value if the measurement is in transmission from a liquid sample. This novel patent pending[16],[17],[18] instrument is capable of measuring power scattering parameters such as haze, gloss, turbidity, scattering diagram at low angles and, at the same time, analysing and counting, by lensfree imaging post-processing, microscopic particles in aerosol or suspension in a non-opaque medium. The concept is not limited to the visible part of the spectrum; it can be implemented along the whole optical spectrum –depending on the sensing technology and the relative optical materials that are used in each case- allowing different particle size resolutions. In the current configuration, which has been successfully demonstrated at lab scale, the resolution is 1-2 µm; by using shorter wavelengths and denser sensors arrays, the resolution could be improved towards the sub-micrometer range. Likewise, the current field-of-view (FOV) is around 20-mm diameter, which is suitable for the most common applications. Larger or smaller FOVs are compatible with the concept. The samples to be monitored can be isolated from the entrance aperture by a standard optical window. Its typical dimensions –around 15 cm sides (it will be reduced in this project miniaturizing the

[10] Goossens (2008)- "Techniques to measure grain-size distributions of loamy sediments : a comparative study of ten instruments for wet analysis",
Sedimentology (2008), Volume: 55, Pages: 65-96, ISSN: 0037-0746, ISBN: 1365-3091, DOI: 10.1111/j.1365-3091.2007.00893.x
[11] Coulter LS 200 (Beckman Coulter Inc.), Fritsch Analysette 22C, Horiba Partica LA-950, Malvern Mastersizer S (Malvern Instruments Ltd, UK) [13] Atterberg cylinder and Sedigraph 5100 (Micromeritics Instrument Corporation, Norcross, GA, USA)
[12] Coulter Multisizer 3: Beckman Coulter Inc., Fullerton, CA, USA
[13] Eyetech: Ankersmid B.V., Oosterhout, The Netherlands
[14] CIS-100: Galai Production Ltd, now maintained by Ankersmid B.V.
[15] Histolab, Microvision Instruments
[16] Apparatus for measuring light scattering, V. Pruneri et al., patent application EP14382043 (2014)
[17] Optically transparent support for bubble-free specimens, V. Pruneri et al., patent application P201331446 (2013)
[18] Compact system for optical measurements using a discretized plate, V. Pruneri et al., patent application ES201330270 (2013)

electronics and light source)- as well as its robustness (also to be optimised in ProPAT for industrial implementation) are compatible with non-lab applications, like in-line monitoring. Finally, the estimated price is significantly lower than the price of the most off-the-shelf particle size analysers.

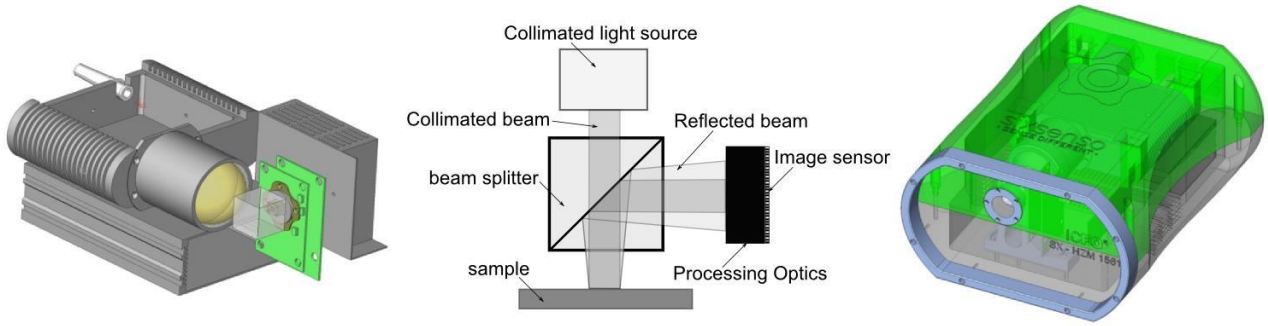


Figure 3. General scheme and prototype elements of ISI sensor in reflection configuration

Haze, glare and gloss are monitoring parameters mainly used in roll-to-roll plastic sheet, glass or film production. These parameters are measured with different apparatus, with high cost and large dimensions, in order to detect imperfections and inhomogeneities in the samples. After imperfection is detected, an image inspection is needed in order to determine the nature of it. Three different systems are being used for one unique purpose, i.e. to detect and analyse imperfections in material production. The Imaging Scatterometer is a portable and low-cost apparatus that can perform all the measurements needed to detect and analyse any artefact, imperfection or detail in a solid sample.

With respect to commercially available and prior-art solutions, the present product allows:

- The image sensor and the lensfree image post-processing offer amplitude and phase image with 2 micron resolution with a ten times higher FOV than most off-the-shelf microscopes. It does not have lenses offering a cost-effective solution for online applications.
- The light backscattering from the sample is processed in the processing optics allowing instantaneous scattering measurements. It increases the performance of particle size analysers offering scattering diagram (power scattering per angle), particle size distribution data (independent of particle shape or geometry), microscopic imaging with high dynamic range (0.5um-2mm) and at the same time still providing optical parameters like gloss, glare, haze or turbidity values of the sample.
- High field of view (FOV) in microscopic images, providing high surface scanning rate (1cm^2 per second) of the sample in roll-to-roll or continuous processes.
- Measurements in transmission, reflection or both configurations.
- In-line, remote and non-invasive measurements, without previous sample preparation needs. A significantly lower cost than the price of the most off-the-shelf particle size analysers

Table 3- Performance of the ICFO Imaging Scatterometer

Physical parameter measured	Time/ Read	Dynamic range	Previous knowledge of sample required?	Mixtures	Size distribution data	Measurement
Forward&Back scattering light intensity variation combined with Imaging of light absorption & reflection of the sample	1 seg scattering read(early alarm of imperfection); 1 min imaging & statistics	VERY HIGH 1 µm-5mm	NO	YES	YES	Inline (noninvasive)

Comparison of the novel MEMS Spectrophotometer with prior-art solutions among NIR spectroscopic analysers

Industry needs new low cost sensor solutions to increase the efficiency of manufacturing processes and quality of products. In many fields smaller and low cost spectral devices have a high potential to enable these targets. FTIR systems present state-ofthe-art measurement systems in MIR (Mid Infrared Spectroscopy) spectroscopy. FTIR systems are broadly used in laboratories, but they are still rarely utilised in process measurements. Big size, complexity, long measurement times and robustness have been the main reasons why Fourier Transform Infrared (FT-IR) systems have not been used successfully in real-time process monitoring. The attraction to use Fabry-Perot Interferometers- Infrared (FPI–IR) technology instead of FT-IR technology comes from being able to utilize the power of spectroscopy in a **small, low-**

cost package with high environmental tolerance. The price of an FPI spectral sensor is less than €5000 whereas FTIR spectrometer to process environment is approx. €50,000.

Frost&Sullivan has forecasted in their Technology Insight Advances in Process Spectroscopy (2012) the impact and certainty of new technological trends in process spectroscopy industry. The most significant trend is **'Cost-effective microspectrometers',** which is represented here by SPECTRAL's devices, incorporating the novel MEMS technology from VTT Technical Research Centre of Finland. The trend of miniaturisation is being tackled by several companies, such as JDSU, Pyreos, Axsun, Infratec, Insion, Hamamatsu and Horiba Jobin-Yvon. Most of these companies approach the miniaturization challenge by taking a conventional grating spectrometer arrangement and trying to find ways to make things smaller. This method works to some extent, but it is limited by the obligatory usage of infrared linear detector arrays, whose noise performance and cost pose severe challenges in meeting cost and performance requirements. An InGaAs detector array for instance costs approx. €500 and can only cover the wavelength range of up to 1.7 µm. To go further in the wavelengths one needs to have a cooled ex-InGaAs array, which adds to the cost and complexity and still does not go further than the Near Infrared Region. It is therefore a great benefit to be able to use **a single point detector instead of an array**.

Miniaturisation of spectral sensors utilizing single-point detectors is commonly done using MEMS technology.[19] The main types are Fourier Transform Spectrometers, tunable gratings and Fabry-Perot Interferometers. Generally speaking, an FTS provides wide wavelength range and good resolution. However, resolution is linked to the movement of a reference arm mirror such that in practice 100's of micrometres of movement has to be organised by MEMS. This is not an easy task and to date has been tried by two means: out-of-plane oscillating mirrors and in-plane comb electrode actuators. The former results in cumbersome monitoring systems and vacuum packaging and therefore **has not been commercialised**. The latter is currently being commercialised by the Japanese company Hamamatsu. However, in industrial sensor applications there are many challenges remaining, such as how to control wavelength stability with a long mirror stroke, how to couple light efficiently in and out from a planar structure, and how to make the system signal-to-noise ratio adequate due to small optical surfaces. Much of these problems are also present in tunable grating/mirror systems. The wavelength stability requirements for many NIR applications is down to 0.01 nm/˚C, so anything with large stroke/angular movement **poses great challenges for wavelength stability monitoring and often results in costly additional measurement modules**, such as stabilised reference lasers. For example, the MEMS-based products of ARCOptix and ThermoFisher Scientific cost approx. € 20,000, making the cost of MEMS itself a very small portion of the total cost.

Table 4. Current key players in research/business for miniaturised spectrometers utilising a single-point detector.

Type	Current players in spectral sensing utilising SINGLE-POINT detectors	Features
Fabry-Perot Interferometer	Spectral Engines	Industrial-grade spectral IR sensors based on VTT's MEMS technology
	VTT Technical Research Centre	Robust, fast and batch-producible MEMS-filter technology
	Vaisala	Industrial-grade CO2 sensor based on VTT's MEMS technology
	Axsun Technologies	Commercial tunable narrow-band light sources using multiple, high-order MEMS-FPIs
	Infratec	Commercial variable colour detectors incorporating a MEMS tunable filter.
Fourier Transform Spectrometer	ARCOptix	Commercialized FTIR instrument incorporating in-plane MEMS FTIR engine (MEMS product obsolete)
	Hamamatsu	Commercializing miniaturized FTIR spectrometer for NIR using Si-Ware's technology.
	Si-Ware	Miniaturized, in-plane FTIR engine technology
	Fraunhofer Institute	Out-of-plane resonance mirror technology
Tunable MEMS gratings/mirrors	HiperScan	Commercial tunable MEMS mirror technology
	CSEM	Rotating MEMS grating

[19] Antila, J., Tuohiniemi, M., Rissanen, A., Kantojärvi, U., Lahti, M., Viherkanto, K., Kaarre, M. and Malinen, J. 2014. MEMS- and MOEMS-Based NearInfrared Spectrometers. Encyclopedia of Analytical Chemistry. 1–36.)

	ThermoFisher Scientific	Commercial portable NIR analyser utilizing a MEMS programmable grating

To fully utilize the cost and miniaturization potential of MEMS technology, a solution needs to be found where only a few components are needed in addition to the chip itself. Fabry-Perot Interferometers (FPI) offer this possibility, as the out-of-plane movement is only on the order of 1 µm or even less and the device can be stacked directly on a detector, making the system very compact and easy to align.

VTT has developed structures and manufacturing methods for FPI enabling **compact, low cost, electronically adjustable spectrometers** working from the visible (VIS) wavelengths up to thermal infrared (IR). Surface micromachined FPIs for the IR spectrum measurements have been developed at VTT since the 1990's[20]. The first commercial application has been in carbon dioxide gas concentration measurements by dual-wavelength absorption measurement which is commercialized in Vaisala's environmental monitoring solutions. Multiple new core technologies have been developed in recent years[21,22,23,24,25] for new sensing needs and these are now being commercialized by Spectral Engines Oy. The hydrocarbon gas sensor prototype developed by VTT and Spectral Engines received a Frost&Sullivan 2013 New Product Innovation Leadership Award for the "Gas Sensor".

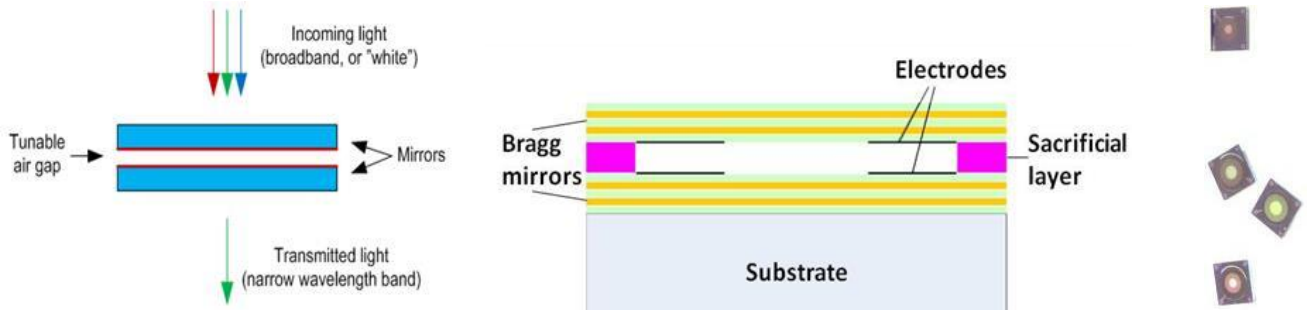


Figure 4. working principle of a Fabry-Perot Interferometer (left) -General structural schematics of VTT's MEMS FPI platform (middle)- MEMS FPI chips by VTT, size of 3x3 mm² with various apertures around 1-2 mm in diameter (right)

In short, the **Spectral Engine's MEMS Spectrophotometer** (fig. 4) for the SWIR to MWIR range, is compact, robust, sensitive, small and affordable. The instrument is not larger than a few centimetres including readout electronics, data transfer and housing- and costs significantly less than the cheapest off-the-shelf instrument currently commercially available for such a spectral region. The wavelength selector is based on an electronically tuneable Fabry-Perot filter, i.e. it is not a dispersive analyser; consequently, its signal/noise ratio and dark current are better than the typical ones for equivalent diffraction-gratingbased spectrophotometers. In addition, due to the robust core technology, the device is highly immune to temperature changes and shocks and vibration, making it ideal for industrial solutions.

[20] Blomberg, Martti; Torkkeli, Altti; Lehto, Ari; Helenelund, Christer; Viitasalo, Mauri; Electrically tuneable micromachined fabry-perot interferometer in gas analysis; Physica Scripta. Vol. T69 (1997), 119 - 121

[21] Tuohiniemi, Mikko; Näsilä, Antti; Mäkynen, Jussi ; Characterization of the tuning performance of a micro-machined Fabry-Pérot interferometer for thermal infrared; Journal of Micromechanics and Microengineering. IOP Publishing. Vol. 23 (2013) No: 7, Article no. 075011

[22] Tuohiniemi, Mikko; Blomberg, Martti; Akujärvi, Altti; Antila, Jarkko; Saari, Heikki; Optical transmission performance of a surface-micromachined Fabry-
Pérot interferometer for thermal infrared ; Journal of Micromechanics and Microengineering. Vol. 22 (2012) No: 11, 115004

[23] Rissanen, Anna; Kantojärvi, Uula; Blomberg, Martti; Antila, Jarkko; Eränen, Simo; Monolithically integrated microspectrometer-on-chip based on tunable visible light MEMS FPI; Sensors and Actuators, A: Physical . Elsevier. Vol. 182 (2012) No: August, 130 – 135

[24] Rissanen, Anna; Akujärvi, Altti; Antila, Jarkko; Blomberg, Martti; Saari, Heikki; MOEMS miniature spectrometers using tuneable Fabry-Perot interferometers; Journal of Micro/Nanolithography, MEMS, and MOEMS. SPIE. Vol. 11 (2012) No: 2, 023003-1-6

[25] Blomberg, Martti; Kattelus, Hannu; Miranto, A.; Electrically tunable surface micromachined Fabry-Perot interferometer for visible light; Sensors and
Actuators A: Physical. Vol. 162 (2010) No: 2, 184 – 188

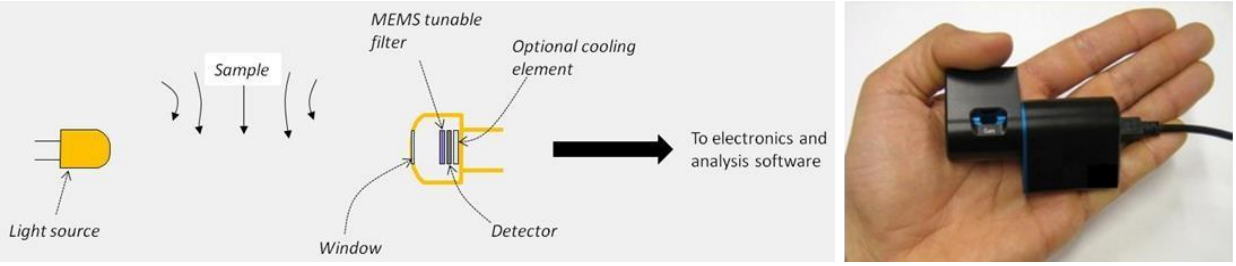


Figure 5. General principle of Spectral Engines' sensor concept. The "sample" can be gas, liquid or solid and the light from the light source can be coupled onto the sample and the detector either open space, or via a fibre probe.

A plurality of spectral ranges are available (and additional ones, tailored to the target applications, will be developed in the project and especially towards MIR), which implies the possibility of using many similar devices in different control points along a production line for determining different analytes. This feature is highly valuable in terms of using these devices as "spectral probes" for PAT applications. Current standard, and more expensive, spectrophotometers have wider spectral ranges although the significant analytical information is often limited to a specific spectral zone, as will be verified for several products in ProPAT. In order to serve customers having needs to measure multiple parameters, such as ethanol, glucose, glutamine, ammonium, hydrocarbons, CO_2, moisture, protein, fats and active pharmaceutical ingredients (API), multiple spectrometer modules need to be developed and combined into a single multiparameter sensor. New wavelength ranges and combinational sensors will be developed in ProPAT to match the user application requirements.

Novelty of the ProPAT smart sensors

By definition, **smart sensors** integrate the sensing elements, the readout electronics, the microprocessor provided and the digital communication channels in one package. They are simpler than cyber-physical devices because they usually are focused in only one parameter.

There are several major advantages of smart sensors compared to traditional ones, namely: compactness -which means lower sensitivity to drifts and noises and more mechanical **robustness**-, **smaller** size -which implies easier integration-, built-in calibration and signal processing -which provides **reliable** and ready to use numerical data and dramatically simplifies further data processing- and, finally, smart sensors are serially manufactured by means of mature technologies derived from the semiconductors industry -which is the reason of their **reduced cost**.

Although there are a significant variety of commercially available smart sensors, most of them are not adapted to specific or even generic applications; they are bare chips which should be powered, housed and matched to the measurement points. For this reason they are not yet widely used in industrial environments. ProPAT aims at demonstrating the advantages of using smart sensors instead of traditional ones in high-end applications like PATs, whereby data redundancy and consistency can be increased by adding more sensing points without increasing the overall cost. Furthermore, the ProPAT data management strategy is based, among other concepts, on distributed data processing as the optimum way to reduce both data failures and handling a big amount of real time data through data buses. Among the considered commercially available smart sensors for the different target applications in the project are: pressure, temperature, flow rate, etc.

The ProPAT online approach vs. conventional measurement techniques

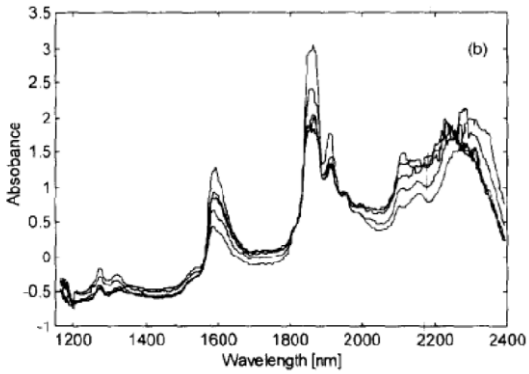
Nowadays, the different products selected for the project (polymers, pharma, ceramics, nonferrous metals, minerals) are mainly analysed offline along the production. One of the main challenges for ProPAT is to provide similar data online or at least a set of data that can still allow the respective manufacturers to take robust decisions on the process to optimise it without needing anymore offline controls.

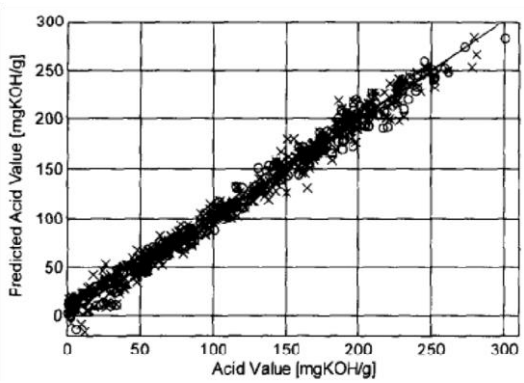
In polymer analysis, molecular weight (Mw) is one of their key features which can be determined by two complementary analytical techniques, including Gel Permeation Chromatography (GPC) and Matrix Assisted Laser Desorption Time-of-Flight Mass-Spectrometry (MALDI-ToF-MS). GPC, also referred to as Size Exclusion Chromatography (SEC), is a mode of liquid chromatography in which the components of a mixture are separated on the basis of size. GPC is mainly used for samples with a molecular weight above

2000 g.mol^{-1} although it is also in use for oligomer separations. There is in principle no upper limit in the molecular weight; even polymer analyses with molecular weights of several millions are possible. Currently, GPC is one of the most powerful and versatile analytical techniques available for analysing polymer. More recently, MALDI-ToF-MS has emerged as a powerful alternative / complementary technology for polymer analysis and has become increasingly popular. Besides it offers simple sample preparation procedures, fast analysis and high-throughput. It also achieves a very high mass resolution and high upper mass limit. Both techniques together can provide a powerful analytical technique for polymer analysis to support product characterisation. For lower mass polymers, HPLC based approaches, may also be of interest, as they achieve higher resolution and hence superior chemical information. Furthermore, this technique may be extended to microHPLC and possibly implemented in-line but the survey of commercial technology indicates there is no equivalent for GPC, i.e. suitable for greater Mw.

In terms of following **thermosetting resins** (e.g. the unsaturated polyester resins (UPR) used for powder coatings and Alkenyl Succinic Anhydride, ASA, used for sizing paper, improvements to which would be very significant for MEGARA's competitiveness)**,** which are often obtained by polycondensation (not the case of ASA), and which polymerisation are generally followed by titration of the end groups, ProPAT can provide the required information in terms of free end groups through NIR (or even MIR once the range of the MENS spectroscopic sensors will be extended), as well as humidity content indications. The platform would also couple different Smart Sensors for temperature, colour (or MEMS-VIS), and absolute pressure. In order to have complementary information, off the shelf inline viscosimeters[26] could also be used in the sampling system.

In terms of prior art using NIR for **monitoring resin reactions**, searches have been made regarding the two aforementioned. UPRs are the most commonly used thermoset resins around the world. These versatile resins are produced by reaction between dicarboxylic acids and diols leading an ester group. Determination of acid (AV) and hydroxyl value (OHV) is particularly used in quality control of these kinds of resins during the industrial process. These values are usually determined by titration method[27] in laboratory but this consumes time, reagents and depends on the collection of samples from the reactor, which implies a delay between the measurement and what is happening in the reactor. Spectroscopic methods are being increasingly used as analytical techniques and tend to replace wet methods such as chemical titrations used at MEGARA, offering shorter analysis times, non-destructiveness and greater accuracy and reliability when associated with chemometrics. NIR allows chemical groups on polymers to be identified and quantified and is sensitive to the O-H bonds and C=O abortions, therefore hydroxyl and acidic value determination is a common application. At academic level polyesterification reactions have been successfully monitored by NIR spectroscopy[30]. In addition, the ability to reliably study the value of acid an hydroxyl groups by NIR has been demonstrated as is shown in Figure 6[28],[29]. NIR methods at lab scale has demonstrated being an alternative to conventional techniques to determine the classical quality parameters, in addition is a robust tool for monitoring in-line which makes it suitable for process control purposes.





[26] http://www.hydramotion.com/XL7_ht.html
[27] Ubranski J, C. W. (1977). Handbook of analysis of synthetic polymers and plastics. London 30 P-Dailin. (1997). Process Control and Quality, 167-171.
[28] M. Blanco, J. C. (2008). Control production of polyester resins by NIR spectroscopy. Microchemical Journal, 118-123.
[29] R. A. Heikkaa, K. T. (1997). Determination of acid value, hydroxyl value and water content in reactions between dicarboxylic acids and diols using nearinfrared spectroscopy and non-linear partial least squares regression. Analytica Chimica Acta, 287-296.

Figure 6. a) evolution of the NIR spectra during between dicarboxilic acid PA and and diol PG during polyestererification reaction;
b) experimental and predicted acidic values for calibration with no linear PLS[32]

In terms of monitoring the **production of ASA** which react covalently with cellulose and are used an internal sizing agent for paper. ASA are produced through a reaction between maleic anhydride and C12-C16 olefins. Regardless of the reaction pathway all products bear succinic anhydride functional groups. Currently, the content of maleic anhydride is monitored at lab scale as quality parameter. Titration leads to quantification of the anhydride groups or their hydrolyzed forms but is time consuming, therefore alternatives need to be evaluated. At academic level some studies have been developed to quantify maleic anhydride grafted onto polypropylene by FTIR[30] and have been correlated with values obtained by titration. Similarly, the presence of the carbonyl stretching bands at 1786-1863 cm^{-1} from succinic anhydride which is not present for olefin itself can be useful to monitor the reaction[31].

In addition, for **pharma products**, which will be considered more from a mixing process stand point in the project rather than from the initial synthesis of the active molecules, many other analytical methods are used: for chromatography (gas, ion- thinlayer and liquid-chromatography with various detectors), optical spectroscopy, NMR, (IR, VIS-UV; NIR, Fluorescence, Raman, Flow-Injection, etc.), elemental analysis (XRF, ICP), electrochemistry (calorimetry, potentiometry) and on-line volatile analysis (Proton-Transfer-Reaction Time-of-Flight Mass-Spectrometry), to name the most important ones. The planned ProPAT monitoring strategy in terms of combining novel NIR (providing data on composition during mixing and moisture during drying) and ISI sensors (providing dada on particle size during granulation) was previously described with figure 6. Similarly to resins, smart sensors could also be added as required. Previously indicated micro-HPLC based techniques could be of great value in pharma applications, as the molecular masses are generally lower than for polymers, and would perfectly fit the ProPAT smart sensor concept. However, this is not expected to be required during the project since no synthesis will be monitored among the target pharma products and chemometrics assisted NIR will be sufficient to determine composition of mixtures.

Besides, going significantly beyond the previously described literature, during the ProPAT project, the data obtained will be optimally analysed by a combination of soft-modelling methods that only take into account the NIR spectra changes, such as Multivariate Curve Resolution (MCR)[32], and hard-modeling methods[33] that take into account the real physicochemical (first principle) model underlying the resin production. Hybrid models could eventually be used when non-ideal behaviour in production plants has to be envisioned[34].

Finally, for the compositional analysis of **minerals, as well as other solid particles**, elemental analysis and Ion
Chromatography, WD-XRF, -ED-XRF, ICP-OES, continuous source AAS, conventional AAS with hydride system (for Hg, As, Sb), polarography and ion chromatography are among the suitable analytical techniques that will be used as references for the project. Although for those at least from partner GM's operation consisting in dry milling of products resulting from direct mining, size distribution is the most important data. In terms of ceramics and nonferrous metal products, compositional information may be required (IR may apply to certain mixtures only) for the more elaborated and variable mixtures from partner MBN, including for example **ceramic-metal materials** used for wear-resistant coatings and based on Tungsten Carbide with Cobalt as binder. MBN produces particles that are nanostructured through high energy ball milling, a top-down technique yielding a wide particle size distribution. Subsequent spray drying then leads to greater size aggregates presented as spherical particles. Depending on the applications, different particle size outputs of the milling machines are needed either in the nano, in the micro or coarser

[30] Roovers, D. (1995). Journal of Polymer Science. Part A, 829.
[31] M. Sclavons, P. F. (2000). Quantification of the maleic anhydride grafted onto polypropylene by chemical and viscosimetric titrations, and FTIR spectroscopy. Polymer, 1989-199.
[32] A. de Juan, S.C. Rutan, M. Maeder, R. Tauler. Chapters in Multivariate Curve resolution in 'Comprehensive Chemometrics' (eds. S. Brown, R. Tauler, B.
Walczak).pp 249-260,325-344,473-506 (2009) Elsevier
[33] M. Maeder, Y.M. Neuhold. 'Practical Data Analysis in Chemistry'. Elsevier (2007)
[34] A. de Juan, M. Maeder, M. Martínez, R. Tauler. Combining hard- and soft-modeling to solve kinetic problems. Chemometrics and Intelligent Laboratory
Systems, 54 (2000) 123-141

ranges. The required fractions are then separated by sieving, centrifugation or by means of cyclones. The use of a system such as ProPAT will allow MBN to have greater control over their coarser products. As previously mentioned the extension (postproject) of the ISI sensors to X- rays, could allow the thinner products from MBN to be captured.

Innovation potential

The innovation potential of our ProPAT project is high, especially in context that there are **many drivers for the uptake of the ProPAT technologies**.

The **process analytical technology (PAT) initiative** is improving product **quality by process design** through knowledge of the fundamental scientific principles behind it, and continuous online control of a process. PAT enables quality cost reduction by moving quality control closer to the production-line (in-line, on-line, at-line) and away from the post-manufacturing quality labs[35]. Indeed PAT is now the buzzword in the pharmaceutical industry and it is gaining interest from other sectors as a new framework for better understanding processes, whereby central to PAT is improving final product quality by process design through knowledge of the fundamental scientific principles behind it, and continuous online control of a process. PAT is now an established platform for good manufacturing practice within the pharmaceutical industry. It is a powerful tool that when linked with processing machinery, allows companies **to actually monitor the process itself** and to come up with real time release of the process to cut down on the amount of testing downstream and also really link the PAT with the process to ensure the quality of the material is optimised at the end[36]. Quality by Design (QbD) is designed to give the flexibility to change downstream. According to Brett Cooper at Mercke Sharpe and Dome: "So if you can demonstrate that you have a wide working margin then you can find ways to optimise and improve your process downstream while remaining within the QbD space that you've designed"[40]. In particular, a more efficient process results in **greater production capacity, easier and more predictable processing, less waste generated, and cost savings**.

In addition, industry is beginning to equate PAT with reducing the environmental impact of processes[37]. As PAT is a system for designing, analysing and controlling manufacturing processes through measurements of critical quality and performance attributes of raw and processed materials to ensure final product quality, the idea of which is to become more efficient while **reducing over-processing, enhancing efficiency and minimising waste**. Indeed, "GreenPAT" is an emerging term within the pharma and petrochemical sectors[38] leading to **higher production sustainability**.

There are also the benefits of cycle time reduction and increased process efficiency (e.g. increased yield, throughput and equipment utilisation) that are achieved by eliminating the delays associated with off-line assays. These competitive benefits are tangible and readily measureable: for example, in the pharmaceutical industry, **Eli Lilly achieved a ten-fold increase in throughput enabled by moving HPLC assays on-line**. According to Darryl Ertl, Manager of GlaxoSmithKline's PAT group in relation to their gravitation towards PAT to reduce cycle times, save energy, and cut costs: "*We used to have to push these initiatives, but now the pilot plants are saying, 'We want to do these things. We can save a significant amount of money.' It's a pull vs. push mentality*."[42]

However, despite its potential benefits for quality control, resource, energy and cost efficiencies, its wide uptake in industry has been hampered by the absence of PAT tools that are affordable, easily integrated into existing processes, simple to use with clearly demonstrated advantages for implementation. As managing PAT data can be complex due to the huge information flow and because it includes mixed data formats from different types of analysers, the PAT paradigm makes sense only if analyser and software resources are harmonically integrated within a specific production line. Nevertheless, the traditional approach is even more complicated since the different off line data are not even centralised and may lead the operator to paradox information jeopardising the good decision making process.

[35] Process Analytical Technology: An industry perspective. European Pharmaceutical Review. Issue 4, 2010.
[36] Brett Cooper Research Fellow at Mercke Sharpe and Dome Development Laboratories at the 2014 PAT and QbD Conference
[37] Paul Thomas (2010) Green Processes: PAT and QbD Take Root. Pharmaceutical Manufacturing. MAY 11, 2010
[38] Thomas P (2009) Is It Time for PAT to Go Green? Pharma Manufacturing. http://www.pharmamanufacturing.com/articles/2009/148.html.

ProPAT through its approach will address existing technology and market barriers that have been preventing the wide update of PATs. All in all, the capabilities and versatility of the ProPAT platform offer enormous advances for implementing integrated process control in in the SPIRE community and beyond (e.g. food, biotech industry). During the project, the system will have been demonstrated for the following product parameters and processes (Table 5):

Table 5. Key product parameters and processes for which the ProPAT technologies will be validated

Product Parameters	Key processes
Polymer reaction extent	Mixing
Levels and identification of contaminants	Synthesis, Reaction, polymerisation
Ratio of components	Grinding, milling
Solution homogeneity/ concentration	Granulation
Particle size, morphology	Suspension
Traceability/ variability	Drying

While the ProPAT technologies will be validated for adopting the PAT paradigm in a number of critical processes in minerals, ceramic, non-ferrous metals, chemistry (plastic and pharmaceuticals), there is an **enormous leverage base** as the SPIRE sectors are the potential client-base for the outputs of the ProPAT project, who are eight world-leading European industrial sectors (cement, ceramics, chemicals, engineering, minerals, non-ferrous metals, steel and water) that through the SPIRE Public Private Partnership are committed to **integrating, demonstrating and validating systems and technologies across sectors**. This unique co-operation will enable massive leveraging of costs and faster implementation of technologies through **closer connection and integration of innovations in industry and their value chains**.[39]

The key differential feature of ProPAT in reference to similar platforms is basically the affordability. In fact, the ProPAT data sources comprise a new generation of soft sensors, particularly, MEMS-based IR spectrophotometers and a scattering-based particle size analyser which are being developed as low cost solutions. The existing PAT platforms -like the ones supplied by ABB or SIPAT from Siemens- are oriented to integrate expensive analysers commercialized by well established companies. Consequently, these PAT solutions can hardly be afforded by SMEs. ProPAT could represent the significant first step towards the democratisation of high-end tools for achieving widespread sustainable production processes.

A comprehensive patent search has been performed which has confirmed the novelty of the ProPAT technologies and the freedom to operate. Various patented technologies were identified for the in-line monitoring processes, although none are specifically focused on the systems that are aimed at being studied during the project.

In terms of monitoring **polymerisation processes**, in general the patent search has demonstrated the interest of plastic companies, especially large multinationals, in on-line control process, most of them based on spectroscopy methods, such as: An improved interfacial polycarbonate manufacturing process for online measurement of monomer concentration, specifically bisphenol A based on vibrational spectroscopy[40]; A method for online measurement of crystallinity of semicrystalline polymers by Raman spectroscopy enabling on-line adjustment of the crystallinity of the polymer[41]; A process plant of halo-butyl rubber with online control of process parameters based on a FTNIR spectrometer, fibre optic cables, a visosimeter and a resistance temperature device. An on-line real time analyser predicts the properties of the polymer and provides process control[42]; An online continuous detection method of oil polymer resin

[39] http://www.spire2030.eu/mediaroom/34/21/European-Commission-signs-first-ever-7-year-innovation-partnership-with-Europe-s-process-industry

[40] Method for monitoring monomer concentration in interfacial polycarbonate manufacturing process. DOW GLOBAL TECHNOLOGIES, Application number CN2010813218 20100224

[41] Online measurement of crystallinity in semicrystaline polymers by Raman Spectroscopy. DOW CHEMICAL CO. WO1998US25170 19981124

[42] On-line control of chemical process plant. EXXON CHEMICAL. Application number US19960034614P 19961231

material viscosity based on various instruments in a pipeline for realtime continuous detection of mass flow, pressure difference, temperature of the material in the pipeline and environmental temperature. Real-time field data acquisition is carried out and the viscosity of the material is calculated using a model present[43]; A method and systems for online analysis (melt index, density, viscosity, molecular weight, molecular weight distribution, weight ratios of different polymers) of a polymeric material within a mixing device by Raman spectroscopy analysis to enable process control for adjusting operating parameters of the mixing device and/or an associated polymerisation process[44]; A system for rapid determination of molecular weight and/or size, and/or other properties of importance to the polymer quality in a polymer process based on fast sample extraction, preparation and delivery techniques to convert complex polymer sample solutions into the diluted polymer analyte solution which passes through the flow-through detectors[45]. Similarly, numerous inventions seek to determine the **end point of chemical reactions** through infrared spectroscopy[46, 47]. A method was patented for online monitoring carbamate synthesis process which includes real-timely monitoring of the amine, dimethyl carbonate and carbamate concentrations in the reaction system by FTIR, and determines the degree of the reaction[48]. A process for controlling a polyphenylene sulphide resin by online IR has also been patented[49]. In general, the control of polymerization by IR requires the use of chemometrics. A few chemometrics models can be found in the patent literature but no conflict was identified with the project and in particular none of the patents covered the specific resin selected for the project.

In terms of using smart process control in the **ceramic** sector, an online detector for ceramic ball mill has been patented. The information is collected by an optical fibre and processed with the purpose of real-time detection of concentration and granularity of milled clay raw material inside the ceramic ball mill drum.[50] Other inventions refer to a method for monitoring and controlling milling and mixing processes in the limestone, cement and mineral material industry or other matters that are beyond the scope of this project.[51 52].

With respect to the **pharmaceutical industry**, a patent describing a method for monitoring on line a mixing process of pharmaceutical production locating an optical sensor on a mixing tank which provides IR absorbance spectrums of mixed materials was found. The invention can process a plurality of characteristic sequences during the mixing process of pharma products and automatically reach a conclusion on degree homogeneity, which guarantee quality of the mixing process and makes the method suitable for automated processing of pharmaceutical production[53]. However the patent is in Chinese, and only granted there, and it is difficult to assess its novelty and extent. In any case, our project goes further and not only seeks to monitor the mixing process by IR but also combines particle size sensors, as well as any required smart process sensor and makes use of most advanced chemometrics approaches. Other patents regarding e.g. granulation process monitoring[54],[55] where returned but were not relevant to the technologies used in ProPAT.

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[43] Hydrocarbon resin polymer material viscosity on-line continuous detecting method. SINOPEC LANZHOU PETROLEUM CHEM. Application number CN2008118135 20080425

[44] On-line properties analysis of a molten polymer by Raman Spectroscopy for control of a mixing device. EXXONMOBIL CHEM PATENTS INC. Application number WO2006US23102 20060614

[45] Online monitor for polymer processes. GENERAL ELECTRIC. Application number US19980183698 19981030

[46] Online detection method for pyrolysis reaction end point in Octogen production process. Z.Gao. Patent application CN20131367463 20130821

[47] Online detection method of crystal-transformation reaction endpoint in production process of cyclotetramethylene tetranitramine. Z. Ghao et al. Patent application CN20131367461 20130821

[48] Method for online monitoring carbamate synthetic process. X. Zhenzhen et al. Patent application. CN20121540231 20121213

[49] Synthesis process for polyphenylene sulfide resin. J. Liu el al. Application number CN20111229715 20110811

[50] Milling process online real-time detector for ceramic ball mill. T Li et al. Application number CN20092290572U 20091225

[51] A method for the control of milling and mixing processes in the limestone, cement and mineral material industry. R. Thomas. Application number EP19960660015 19960524

[52] Advanced vertical mill control system based on model identification and predictive control

[53] Method and system for on-line monitoring medicaments production mixing process. D. Ruxu.el al. Application number CN2007177549 20071130

[54] Sintering granulating moisture control parameter tuning method for effectively inhibiting raw material flow fluctuation. W. Min et al. Application number CN20111106430 20110427

[55] Online automatic powdery and granular material grain size analyser. TA. Hideo et al. Application number JP19860018953 19860130 [59] http://www.pharmainfo.net/reviews/process-analytical-technology-pat-boon-pharmaceutical-industry

All in all, the generic use of process monitoring and feedback to optimise the process is not novel and many specific systems are reported but in general they lack such a global approach as in ProPAT. In addition, ProPAT's novelty lies in the use of a combination of very specific sensors and their embedding in the global process control platform. Therefore, ProPAT is expected to lead to a number of patent opportunities, particularly surrounding the proposed technological advancements in sensor components and data handling software.

2. Impact

2.1. Expected impact

Contribution to the expected impacts set out in the work programme, under the relevant topic

The proposed ProPAT integrated process control platform that will be developed, validated and demonstrated in the process industry during this project will contribute abundantly to the expected impacts of the SPIRE-1-2014 call.

Improved capabilities for valid, reliable and real-time measurement of the properties and quality of process streams and final products for existing and for more flexible process operation concepts.

The PAT initiative focuses on building quality into the product and processes, as well as continuous process improvement. Essentially, integrating on-line measurement and/or modelling of critical quality attributes with automated feedback control of the process parameters impacting these attributes can ensure more **efficient control of processes**, reducing product variability, which will subsequently **reduce the risk of releasing off-spec product into downstream discreet manufacturing**, and increase customers' satisfaction, thereby preventing products being rejected (and discarded) further down the supply chain.

ProPAT aims to open up this world of process understanding and continuous on-line control for improved product quality and safety, process efficiency, yields, resource and energy efficiency, and ultimately profitability through and affordable, robust, versatile platform, that will be validated in key process industries for critical unit operations. Indeed, research-based development and implementation of PAT in the European process industry (as a consequence of projects such as ProPAT) is focused on improved control, rapid final product quality evaluation, increased resource and energy efficiency and productivity, and will have a knock-on effect in terms of **industrial innovation and enhancing competitiveness**.

Moreover, process understanding results in an appropriate management of variability and improved operational efficiency (e.g.,
"Lean Manufacturing," "Right First Time" strategies)[59] leading to **more flexible process operation concepts**. A defined design space for production processes offers flexibility for raw materials used and even process controls[29]. Because the influence of raw material variations is well-known, the process control strategy allows adaptation to the variability of the raw material quality attributes[29]. This also leads to **safer processes** because the control strategies are optimised both from a manufacturing and an operator point of view.

ProPAT will develop and validate versatile and affordable sensors to provide the industry with quantitative, real-time in-line control of critical quality and performance attributes for raw and in-process materials during processing. The key innovations brought by ProPAT are, on the one hand, the use of novel low cost NIR (MEMS based) and granulometric (imagining scatterometer ISI) sensors in combination with smart sensors capable of measuring traditional parameters in a time /space resolved fashion depending on the individual processes, and on the other hand the calibration and validation of the integrated PAT platforms in a number of process applications including minerals, ceramic, non-ferrous metals, and chemicals (plastic and pharmaceuticals).

ProPAT will develop improved **measurement tools,** which will be validated in the targeted process industries (minerals, ceramics, non-ferrous metals, polymers and pharmaceuticals), to move analysis closer to the process from off-line measurements via at-line to on-line and in-line analysis. This change allows for the introduction of the principles of PAT where the process industries can deliver a product that will not require post-process testing because it is based on **active process control to achieve an in-spec**

product. In order to validate and demonstrate the reliable, real-time measurements of the properties and qualities of process streams and final products, an analysis of the current control protocols, both off-line and online, currently employed by the targeted end-user process industries in the unit processes that will be validated by ProPAT will be performed (Figure 7). As such in WP2, the novel sensors will be calibrated offline against established quality control and/or standardized methods used by the industry to characterise products. Subsequently, they will be tested for inline operations in WP6 and 7.

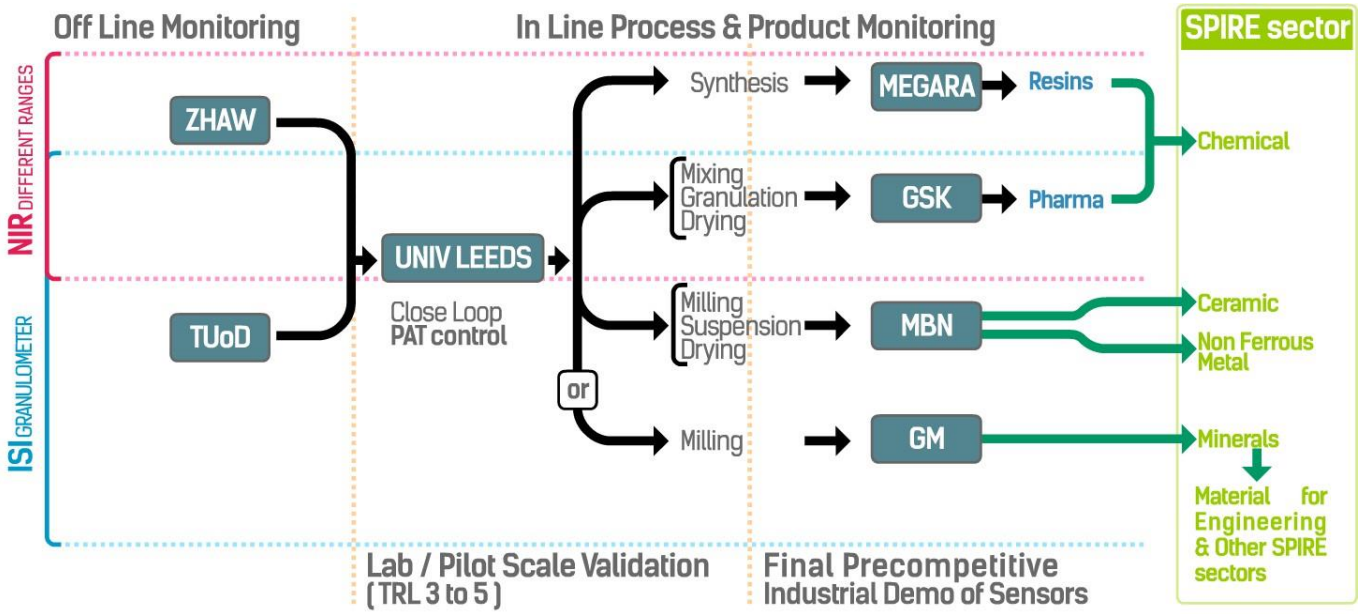


Figure 7. Process and Product specific validation of the ProPAT system

The impact and novelty of ProPAT solutions arise not only from the development of the innovative analysers and sensors, but also the overarching data management solutions, Global Control System Software and proposed integration strategies.

More sustainable plant operations due to the extensive usage of all information available from validated PAT and PPMT measurements for model based control.

Ultimately, large amounts of PAT data have to be collected, analysed, displayed, transmitted, processed, and stored in a manner that provides timely information about manufacturing process variables. ProPAT will develop an appropriate data management system to accommodate all these factors. One important reason why many companies shy away from NIR-based and other spectroscopic sensors is the lack of effective data management systems. The project will develop a versatile data management system to act as the single front end application for the developed sensors and analysers. The system will facilitate collection of data from a range of sources (e.g. spectral analysers, process data, LIMS-data, other databases). The ProPAT interface that will be developed will contextualise the collected data from different sources and time, align the data to bring together all relevant information from different sources.

Key to the approach will be the performance of calculations in real time (e.g. transformation of multivariate spectral data into discrete CQA-values (Critical to Quality Attribute-values). Combined with integration of commonly used chemometrical software packages (e.g. Unscrambler, Matlab) the necessary tools to allow for APC (Advanced Process Control) will be developed. The ProPAT software will facilitate data mining for offline analysis, to provide information that will lead to a better understanding of the process.

A flexible system will be developed which facilitates data collection from an array of sources and interprets the data for appropriate decisions to be taken. Robust data processing protocols for sensor methodologies which identify and explain critical sources of process and product variability will be developed. An effective framework for multi-sensor data fusion will be created to give more accurate on-line process performance monitoring. A model-based supervisory control system suitable for the target process industries will be produced.

An important driver for PAT is the pressing need within the industry to **reduce production costs** and speeds[59]. In short, ProPAT is an enabler for the extensive usage of all information available will contribute to **more sustainable plant operations** through measurements of critical quality and performance

attributes of raw and processed materials to ensure final product quality, the idea of which is to become more efficient while reducing over-processing, enhancing efficiency and minimising waste. The conversion rate vs. cycle time of the polymeric resins will be improved by the access to real time information on the progress of its synthesis reaction as opposed to, sometimes, long offline titrations. The off spec minerals or ceramic particles sizes that would be separated by sieving or possibly reworked or sent to lower value applications will be reduced. The time required for drying powder resulting from wet milling or granulation will be optimised. These are just a number of examples among the very broad capabilities of the ProPAT system that will be demonstrated in the project.

Improved monitoring and control of continuous plants

As process manufacturing is distinguished by a production approach that has minimal interruptions in actual processing in any one production run, or between production runs of similar products. Process understanding is the basis for process control and assured end product quality. PAT is an approach to monitoring manufacturing (and other) processes on a continuous rather than discrete basis[29]. According to Uwe Kirschner, General Manager at Sentronic GmbH, Dresden and responsible for the Process Analyser business unit: "PAT is a door opener for continuous production processes, where in-line control is essential to achieve constant high quality."[56]

One of the prominent features of ProPAT is online monitoring, which means it is not only recording information, but it's also closing the loop and making adjustments to the process as the product is being manufactured. In the context of the applications that will be targeted in ProPAT in terms of close loop process control that will be tested in UNIVLEEDS facilities for 3 models processes that will mimic those later implemented during industrial demonstration: 1) Solid/slurry processing: granulation, wet milling; 2) pharma/fine chemistry processing (mixtures); 3) high temperature processing (resin synthesis). This will allow the demonstration of: 1) the impact of the sensor on the process (e.g. benefits on quality, cycle time, yield, waste reduction, etc.), in a statistically reliable fashion; 2) the robustness of the sensors to scaling up/working at process conditions (extend of recalibration required for each scale); 3) Robustness of consistency of measurement in spite of batch to batch variations or in prolonged time.

Improved support of the operators leading to safer, more reliable and sustainable process operation improving process efficiency.

Management of processes by operators not fully understanding the process can result in unsafe, unreliable and unsustainable process operations. PAT technologies can lead to **improving operator safety** and **reducing human error**[61]. Through the adoption of ProPAT in process industries, operators will be empowered, through access to real-time process data, to monitor progress and halt and correct a problem, or even once the ProPAT system has been fully validated to work in close loop, rather than passing it on to a final quality inspector.

To this end, processes could become **safer and more flexible through the use of ProPAT**, whereby the operator would be supported with new technology to correct or prevent a problem in the production process based on critical process parameters (CPPs) and an analysis of the critical operator data (COD) necessary and/or for integrated system control or required by an operator to effectively manage the process. Indeed, in addition to process parameters already available, new real-time parameters will also be measured and managed with the process data from smart sensors in integrated fashion via the ProPAT system. The identification and confirmation of CPPs will be accomplished by using neural net, mathematical modelling and statistical software to find correlations between key quality attributes and measured real-time process parameters. By using real-time methods, the process endpoint no longer needs to be a fixed time, but rather can be the time required to reach a specific state or condition. Some examples of expected CPPs are the polymerisation extent of the resins vs. its temperature of synthesis and reaction time, particle size vs. mechanical inputs and time, drying vs. temperature and time, whereby in each case the aim is to find best compromises of process in terms of quality, efficiency, sustainability, etc.

In terms of **contribution to safety**, on the one hand ProPAT will allow wastes resulting from off-specs materials to be reduced, some of them having to be treated as hazardous wastes. Moreover, for the workers, the need to carry out sampling along the process is sometimes hazardous (e.g. when handling

[56] Featured in: http://www.industry.siemens.com/topics/global/en/magazines/process-news/pharmaceutical/pat-industry-perspective/pages/default.aspx [61] Blevins, T. & Beall, J. (2007) Monitoring and Control Tools for Implementing PAT. Pharmacuetical Technology, March 2007.

hot resins, unreacted monomers, powdery particles, etc.) so the shift to online monitoring will be a significant advance. Likewise, in terms of sample management or substitution of quality control tests (e.g. when using solvents for titrations, etc.). Finally, the particle size sensor will allow the process to be optimised and when required for safety reasons, reducing the nano-sized fraction of the particles contributing to health and safety of workers and users of the products.

Besides in terms of **contribution to flexibility of processes**, faster process set up and better control thanks to ProPAT will ensure the same facility can be used for a greater number of productions once the ProPAT system is calibrated for each of them. The present inflexibility of the process industry is often reported as one of its weaknesses, which is related to one of the goals of the F^3 project.

Better process operations with respect to resource and energy efficiency.

In line with the SPIRE Roadmap, this project will deliver a **short to medium term impact** in relation to the improvement in measuring systems, process control and process optimisation that can be integrated to significantly increase energy and resource efficiency in the "installed base"[1]. "GreenPAT", an emerging term within the pharma and petrochemical sectors[57], will be a core objective of ProPAT leading to **higher production sustainability**. SPIRE aims to implement its research and innovation roadmap through six Key Components that are **at the core of a resource and energy efficient process industry**, of which one is 'Process', which is focussing on solutions for more efficient processing and energy systems for the process industry. Within the Process Key Component, is the Key Action **KA 2.3 Process monitoring, control and optimization.**[63]

As novel and radically improved production processes are **key to increasing the energy, resource and CO_2 efficiency in industrial value chains**[1], the adoption of the ProPAT technologies by process industries will **improve production efficiency in terms of energy consumption and resources wastage** through advanced process control that will lead to improved yields, reduced over-processing, reduced cycle times, energy savings, and essentially reduced costs. Moreover, integrating on-line measurement and/or modelling of critical quality attributes with, on the long term once fully demonstrated, automated feedback control of the process parameters impacting these attributes, such as is proposed by ProPAT, will ensure more efficient control of processes, reducing product variability, which will subsequently reduce the risk of releasing off-spec product into the discrete manufacturing, thereby preventing products being rejected (and discarded) further downstream. This will help to reduce or eliminates production scarp, contributing to reducing waste.

This is very much in line with the EU's EUROPE 2020 growth strategy for the coming decade that will lead the way to transform the EU into a smart, sustainable and inclusive economy. Indeed, without the contribution of a Sustainable Process Industry, it becomes difficult to imagine that the objectives of Europe 2020 will be achieved[1].

In situ real-time monitoring of reactive processing of the **chemical industry** in general, and polymer in particular, is of crucial importance since it allows measuring continuously changes in physico-chemical properties during the process while in the current situation, resorting to off-line QC methods, there is a lag between the result and what is occurring in the reactor. In addition, off-line methods are time-consuming. For example, MEGARA has reported that process control of Alkenyl Succinic Anhydride (ASA) production requires the quantification of maleic content during the 20 hours of production. This measurement takes 40 minutes. Overall, they expect that online monitoring will save a total of 4 hours, i.e. **20% reduction of cycle time** (cumulating the savings in each stage of the synthesis) and that such immediate actions which result in a **20% increase in productivity and energy savings**. The prompt ending of the reaction is also expected to help in achieving a better **colour** for the final product, a very desirable attribute for the final application as paper sizing. In addition, it is expected to **avoid the decomposition of maleic anhydride (MA)**, eliminating the need for filtering the final product, so the properties of the product will also be improved and further time and energy will be saved. This improvement will not be exclusive of this process. ProPAT intends to also guide the manufacturers to optimise the production which will result in the **minimization of the off-spec batches** which currently account for 5% of the total annual production on polyester resins at MEGARA. This will be possible by the monitoring of the reaction process leading to

[57] Thomas P (2009) Is It Time for PAT to Go Green? Pharma Manufacturing. http://www.pharmamanufacturing.com/articles/2009/148.html. [63] http://www.spire2030.eu/

timely intervention in the early stages of the reaction with appropriate corrections (addition of glycols/acids) when needed (inaccurate feeding, high glycol losses etc.).

In terms of fine chemistry and more especially secondary pharma production, as planned to be used as a case study in GSK, benefits are expected from the installation of the ProPAT system in the different steps of the process, and an even greater potential is envisaged while expanding the deployment of ProPAT also in primary pharma production and synthesis than only in the selected model processes that will be tested during the project. An important asset leading to significant cost savings is expected from the **minimisation of drying times** by detecting the drying end point on each batch through access to the residual moisture in the powder in the spray dryer in real time. In addition, similarly as for resins, the inline access to **compositional information of mixtures** will allow corrective actions to be taken immediately by the operator or in closed loop on the long term. Furthermore, the **real time detection of particle size** is important for optimizing the process, reducing energy consumption, as well as for product quality but also for operative questions, e.g. to avoid clogging filters in case of big agglomerates in the spray drying system. The listed benefits in monitoring those 3 main processes will all contribute in an increased productivity due to better control of the industrial processes as well as reducing the off-spec product quantities and all in all ensure high standards of final product quality. The main assets of ProPAT vs. existing systems available on the market or already present at GSK are the use of lower cost sensors with good accuracy combining chemical and physical information. Finally a system like ProPAT would also allow reducing the time to market by assisting in new product development and allowing the production facilities to be more flexible.

Meanwhile the **minerals as well as the ceramics /metals industries** represented in the consortium can obtain benefits using ProPAT though the motorization of particle size during classical processes as milling.

In terms of **minerals**, GM reported the possibility to use ProPAT to follow on-line a vertical pendular milling process, which is currently controlled though off line analysis taking around 1h. It is expected that the on-line monitoring system will increase the efficiency of milling by optimising the control parameters. The standardised control will make even small batches efficient both cost- and production-wise. In addition, **quicker milling parameters adjustment** will also decrease the amount of out-of-spec products (reducing current scrap levels by around 5%), increase yield (around 10%) and save energy (around 10%).

In terms of metal and ceramics, similarly MBN have expressed its interest in monitorisation of milling, using high energy ball mills. Currently, the process is monitored off line through size particle analysis which can take up to 2 hours. Considering also the time needed for gradual passivation of particles (20-60 min), the time for discharging the milling chamber, for the classification and preparation for the second batch, on-line monitoring of the process is expected to **reduce the cycle time by 30-40%** leading to considerable energy savings too. In addition, high energy ball milling is a process that can induce different effects on the material like welding, work hardening, diffusion, heating, fracturing, flattening, alloying, mixing, etc. which affect the particle size distribution. The on-line monitoring of the process can facilitate the development of products (which is very strategic for MBN who has a lot of specialty grades) and the prompt detection of a malfunctioning of the machine before the process output is affected. In such applications, it is estimated that the overall **process yield could be increased by 10%** by arriving at the optimum conditions on line.

Significant decrease in greenhouse gas emissions.

According to the IEA the energy sector (41%), transportation (22%) and industry (20%) are the main contributors to global CO_2 emissions, followed by the building sector (7%). In the industrial area steel (6%), cement (5%) and chemicals (3%) are the largest source of CO_2 emissions[58]. To this end, the SPIRE process industries are indeed in a position to make a significant contribution to energy and resource efficiency within their own industries (representing 20% of the global CO_2 emissions). The aforementioned case scenarios contributes to SPIRE goals in terms of reduction in fossil energy use of up to 30% (via optimum process conditions and cycle time), reduction in non-renewable, primary raw material use of up to 20% (by reduction of scrap), all together leading to a reduction in CO_2 equivalent footprint up to 40%. In addition, overall, although during the project the system will be demonstrated for a finite number of model process scenarios, a very broad range of benefits already appears both in terms of product and process consistency, process set up time, certain unique product features, operational advantages and

[58] Energy technology transitions for industry, IEa 2009 - http://www.iea.org/co2highlights/co2highlights.pdf

risk prevention, etc. The potential of ProPAT is even much greater, considering the post-project calibration of the system (generating business in system exploitation and technical assistance) to virtually any process industry in the SPIRE sectors. A life cycle analysis will consider effects along the value chain and will demonstrate the sustainability advantage.

Strengthening of the competiveness of the European industry both in the domain of PAT technologies and control solutions and with respect to economically sustainable industrial processes.

ProPAT will strengthen the competiveness of the EU industry in the domain of PAT technologies and control solutions in that it will overcome many of the barriers to the adoption of PAT in industry, mainly the high costs of analysers, the lack of a flexible platform for the integration of different analysers into existing plants, and the large volumes of data to be managed. These advantages will contribute to **technology push for the ProPAT technologies**. In addition to overcoming the technological barriers, ProPAT will also prepare the European process industry for PAT adoption through comprehensive knowledge transfer and training activities, as well as by demonstrating the benefits through the validation work that will be performed by UNIVLEEDS as well as through the participating process industry representatives in this consortium.

PAT represents a **silent revolution in industrial quality control**[59], whereby it enables a shift away from the historical quality assurance approach (i.e. testing quality after the process is complete) to a true quality control approach where the process is continually adjusted to maintain product quality and consistency, as well as to optimise process efficiency. This can be a huge enabler for the competitiveness of the European process industry in that such industries do not lean towards a defect-oriented manufacturing. The approach in the process industry is **direct, real-time control**, whereas in discrete manufacturing its afterthe-fact statistical analysis to get continuous improvement. Moreover, most process industries (e.g. resins, pharma, powdery products obtained in suspensions) deal with liquid (or solid liquids) processing, which is **more complex and variable** than discrete manufacturing.

Indeed, research-based development and implementation of PAT in the European process industry as a consequence of ProPAT is focused on improved control, rapid final product quality evaluation, increased resource and energy efficiency and productivity, which will have a knock-on effect in terms of industrial innovation and enhancing competitiveness. In the pharmaceutical industry, PAT has enabled processes to be actively managed to achieve a high degree of repeatability and efficiency, quality assurance to become continuous, and real-time activity, which has brought such tangible benefits as, improving asset uptime and availability for pharmaceutical unit operations by up to 40 %, costs are reduced by up to 30 % while fundamental product quality is maintained[60]. In addition, there are the benefits of cycle time reduction and increased process efficiency (e.g. increased yield, throughput and equipment utilisation) that are achieved by eliminating the delays associated with off-line assays. These competitive benefits are tangible and readily measureable: for example, in the pharmaceutical industry, Eli Lilly achieved a ten-fold increase in throughput enabled by moving HPLC assays on-line. Obtaining similar tangible benefits in the European process industry will lead to savings, profitability, growth and jobs. It will also contribute to **market pull for the ProPAT technologies.**

In short, the competitive benefits that will be derived for the European process industry, together with the advances that the
ProPAT project will make in terms of bringing disruptive analysers in terms of cost, and a highly flexible and powerful Global Process Control Platform will greatly **stimulate demand for the ProPAT technologies** in the EU process industry and beyond. Given that one of the fundamental goals of SPIRE is boosting the deployment of innovative resource and energy efficient solutions and practices among the partners facilitating their adaptation, transfer and take-up in a cross-sectorial way[1], ProPAT will leverage the enormous potential of the SPIRE community as a platform for post-project deployment of the ProPAT results in the industry. Given that the sectors united in SPIRE represent a major part of the manufacturing base in Europe (EU27), including more than 450,000 individual enterprises, the market opportunity for

[59] van den Berg, F., et al., Process Analytical Technology in the food industry, Trends in Food Science & Technology (2012), doi:10.1016/j.tifs.2012.04.007
[60] Buijs, T. Process Analytical Technology: Integrated PAT solutions for the Life Sciences industry. Special Report Automation Systems ABB Review [67] Lingman et. al. (2013). European Roadmap for Industrial Process Automation. ProcessIT.eu

deployment of the results of the ProPAT technology in industry are enormous, which will lead to growth and jobs for companies dedicated to the supply of PAT technologies and control solutions (discussed below).

Retention and creation of jobs for the European measurement and automation and process industries.

Growth will lead to jobs being retained in the EU in industrial process automation, which is an important sector for Europe, and one in which many European companies are world leaders in terms of systems and applications development, supply and usage[67]. If we consider that the world market for the automation of process industries is €26.3 bn, with the EU market representing € 10 bn[67]. Should the ProPAT technologies secure even a 0.1% of the EU market in the first years of market activity, this would represent € 10 M worth of annual turnover. Using this as our starting point, and assuming a time-to-market, of 18-24 months after project completion (i.e. market entry by 2020), by 2025 € 50 M in turnover could be generated through the supply, integration and maintenance of the ProPAT technologies and some 84 new jobs could be created in the European measurement and automation industries over that same period. However, if we consider expansion into the global market, these figures could well be doubled. Importantly these new jobs would be in technological SMEs (IRIS, SPECTRAL, MEMSFAB), which would strengthen the positioning of SMEs in a sector that is dominated by large players. For example, ABB is a world leader in this area with a share of 22%, followed by Siemens and Schneider Electric.

If we consider the retention and creation of jobs in the European process industries, the sectors united in SPIRE employ over 6.8 million employees, generating more than 1,600 billion € turnover. As such they represent 20% of the total European industry, both in terms of employment and turnover[1]. Should ProPAT be uptaken in 3 % of the EU process industries by 2025
(these industries would represent € 48 bn of turnover in the EU process industry), and envisaging savings and profitability impacts through the competitive benefits of implementing PAT and integrated process control strategies in their operations of 1 % of turnover, this would equate to €48 M per annum, which would lead to the creation of 400 new jobs per annum in the sector (for system manufacturing and implementation including creating specific chemometrics models). On the long term, by 2030, ProPAT could have generated 2000 new jobs in the process industry in Europe, and 10,000 new jobs by 2050. Beyond these direct jobs, ones needs to consider the indirect employment impacts from the downstream effects in the European discrete manufacturing industries.

Contribution to improving the innovation capacity and the integration of new knowledge

Key to improving the innovation capacity of companies is to strengthen their competitiveness and growth by developing innovations that **meet the needs of European and global markets**. As was discussed, when the ProPAT technologies are commercialised the market opportunities will be promising, yielding competitiveness for Europe, jobs and growth.

ProPAT will play a positive role in the integration of new knowledge in the process industry. The development and adoption of PAT tools in the EU industry could play a major role in assisting companies, including SMEs, in increased process efficiency and production sustainability, reduced operating costs, reduced wastes, increased process validation and ultimately improved final product quality. To date, PAT adoption has been relatively slow due to the constraints of technology costs, complexity, etc. However given the nature of the process industry in the EU, it is critical that new PAT tools are developed which are affordable, easily integrated into existing processes, simple to use with clearly demonstrated advantages for implementation. For achieving the resource and operational efficiencies, as well as the sustainability target of SPIRE, it is imperative that the EU industry has the know how to implement an effective PAT strategy to assure product quality, increase process efficiency and ultimately maintain its competitiveness in the global market.

It is clear from the SPIRE Roadmap, that the SPIRE process industries, which have a large number of production operations throughout the EU, have important requirements for innovation and retrofitting in line with further specialisation of the sectors (e.g. specialty chemicals) and to comply with stricter (future) environmental standards[1]. These companies offer an **enormous leverage base for SPIRE innovation;** in some cases (e.g. new process control) **offering double digit replication potential compared to the initial single pilot project in which a novel SPIRE technology will be proven and demonstrated**[1]. Following successful technology demonstration at market scale and market acceptance the step of

commercialising the technology needs to be taken. This is the most expensive step, typically with factors of 5 to 10+ higher than for a demonstration capability, even for a single commercial plant. **This is the investment that the industry needs to make and will be very willing to make for the successful projects that emerge from SPIRE as they will want to see a return on the significant R&I investment (200M euro/year) made in the PPP**[1].

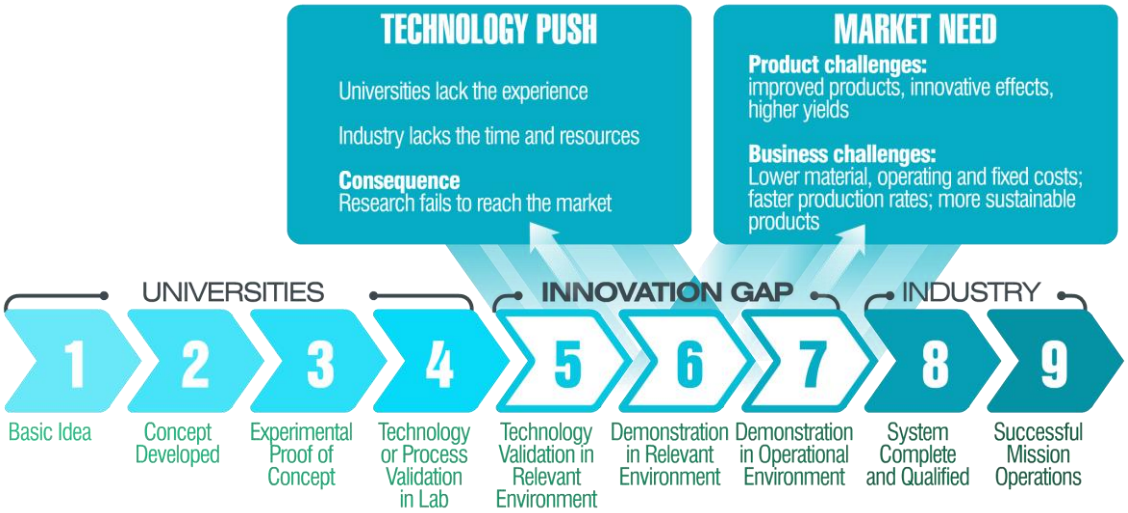


Fig. 8- The Innovation Gap

In the case of our project, the strong industrial presence in the ProPAT consortium is testimony to the high commercial relevance of project and its outcomes. Importantly our industry mix consists of innovation drivers (IRIS, MEMSFAB and SPECTRAL) and innovation adopters (GM, GSK, MEGARA, MBN). On the one hand, the technological SMEs have a strategic business interest and capability to exploiting the project results and performing further development to towards industrialisation and commercialisation. They will stimulate technology push. On the other hand, the participating early adopters of the ProPAT system will adopt PAT in their plants, thereby creating market pull. This relevant presence of industry will contribute to bridging the “innovation gap” (Fig. 8) that frequently prevents research outputs from reaching the market. Moreover, ProPAT has also generated wider interest from SPIRE members such as BASF, DSM, Solvay, Total, DOW, as well as from the Suschem platform, with whom we will engage as part of our wider communication and industry outreach activities, as unfortunately not all industry end-users could be included in the bid due to various constraints such as previous commitments, budget constraints and eventually the impending deadline.

To maximise the impact of ProPAT, the technologies developed will be tested and validated over as many suitable unit operations and products as possible (figure 7). Also it is envisaged that many of the ProPAT technologies developed within the project **will have application within the wider process industries** and indeed the food, dairy, infant formula and other comparable process industries. The presence of a market actor, DECHEMA, will also ensure that any legislative or regulatory barriers to market entry are managed and overcome to pave the way forward for commercial use of the project outcomes.

Barriers/obstacles, and any framework conditions

In order to ensure that the ProPAT results do not encounter any regulatory barriers that would prevent industry adoption, the work plan includes a number of activities to plan the work and its outcomes in line with standardisation to will ensure that the project outputs are geared up for future regulatory compliance and to meet with all relevant industry standards. Right at the project outset, in *Task 1.4*, DECHEMA who sit on the DIN 055, CEN 332 and ISO 48 technical committees and have been the one founders of the PAT initiative, will provide the consortium with the standards of all the related methods used in the target applications of ProPAT, which will ensure that the ProPAT sensors can be calibrated against well-established methods. This is a very important step that will ease ProPAT’s post-project standardisation. TUD will also contribute knowledge and expertise at this level as they extensive experience in setting up ISO and DIN standards regarding particle size analysis. This will give the project every chance of success in relation to the fulfilment of regulations and standards to pave the way forward for uptake of the ProPAT integrated process control system in the European process industry. In addition, during *Subtask 10.3.2*

Regulatory Compliance and Standardisation, under the guidance of DECHEMA a CEN Workshop Agreement (CWA) as best practice document that industry can refer to, which will be announced and distributed via CEN and all national standardisation bodies. Moreover, quality working manuals and quality working procedures, datasheets, guidelines etc. based on quality standards such as ISO9000 and ISO14000 will be drawn up for the ProPAT technologies.

2.2. Measures to maximise impact

2.2.1. Dissemination and exploitation of results

To realise the expected project impacts of contributing to the "strengthening of the competiveness of the European industry both in the domain of PAT technologies and control solutions and with respect to economically sustainable industrial processes" and the "retention and creation of jobs for the European measurement and automation and process industries", at this planning stage of the ProPAT project, we have devised our initial strategy for the dissemination and the exploitation of the project results, with a view to increasing the innovation and business impacts of the project. In relation to a dissemination plan, this initial plan is embodied in the Communication/Dissemination Plan discussed later in this section, whereby the target audiences are defined and along with how they will be reached and what will be communicated to each. While the Communication plan discusses communication actions that will be carried out during the project, this will be the starting point for post-project dissemination activities that will be designed to boost the commercial exploitation of the project results.

Importantly, a first exploitation/business plan has been drawn up considering the contribution of the results to patents, to new or existing products, services, infrastructures and knowledge bases, to ensure the project innovations are delivered to the market. Exploitation of the project results is the most direct way of gaining socio-economic benefits such as improved processes and efficiencies, jobs and competitiveness. The **industrial use of the project knowledge** and outputs for increased competitiveness and growth are where the innovation impacts of this project lies. The expected impacts from this project can only be realised through a strong commitment among the ProPAT partners to plan a robust post-project exploitation strategy.

In terms of the proposed exploitation routes, it is envisaged that the results will be used for: a) **Commercial exploitation**; b) **Education and training**; and c) **Further research**. Each partner has communicated clear exploitation expectations (table 6), which will be planned in greater detail during *Task 10.3- Preparation for the practical implementation of the project results.*

Table 6- Summary of exploitation intentions of the ProPAT partners

No.	Short Name	Intention for use of project results
1	**IRIS**	Use of the knowledge generated in relation to the development of global process control systems for PATs for their commercial PATBox^{TM} solution (a process control platform for integrating PAT technologies in the food, pharmaceutical and process industries).
2	**ICFO**	Use of the knowledge generated in relation to the ISI analyser in the post-project industrialisation stage towards a commercial device
3	**VTT**	Use of increased knowledge of PAT in the industry in order to start follow-up projects in this field, and to transfer this knowledge to other industries, such as pulp and paper and biorefineries
4	**UNIVLEEDS**	Use the knowledge generated in relation to PAT technologies to improve the analytical capabilities of IPSE and iPRD in
		industrial process control. Specifically, developed advanced understanding of integrated inline sensor and control systems will be exploited in various R & D projects being conducted at the University in multiple sectors, such as current collaborations with chemical, pharmaceutical, personal care, water and nuclear industries.
5	**URV**	Use of the knowledge generated in relation to modelling of novel data types, novel preprocessing techniques and multivariate quality control for publication in scientific journals, presentation in international conferences and organization of courses for industries. Development of a doctoral thesis.
6	**ZHAW**	Use the knowledge generated in relation to PAT technologies to improve the analytical technologies currently used in research projects
7	**TUD**	Use the knowledge generated in relation to PAT technologies to improve the analytical technologies currently used in research projects

8	**UB**	Use the knowledge of data fusion from a range of sensors and PATs generated in this project will extend the groups expertise, resulting in further development of postgraduate (and undergraduate) courses provided by them, including close collaboration with project partners and other potential partners on future student and research projects. Development of a doctoral thesis.
9	**IMAR**	Use of the knowledge generated in relation to smart sensors to start follow-up projects in this field, and to transfer this knowledge to other industries such as dairy processing and infant formula
10	**SPECTRAL**	Use of the knowledge generated in relation to the MEMs-NIR for the post-project industrialisation stage towards a commercial device
11	**MEMSFAB**	Use of the knowledge generated in relation to the MEMs-NIR filter to be able to commercially manufacture chips for integrated SPECTRAL devices developed in this project
12	**DECHEMA**	Use of increased knowledge for advisory, training and education programmes to the wider European process industry
13	**GM**	The generated knowledge will be exploited for own use in milling installation for monitoring and quality control of final products
14	**GSK**	Preferential use of the knowledge generated for adoption of the ProPAT system in their pharma processing facilities after the post-project industrialisation phase (and collaboration in the industrialisation phase as required)
15	**MEGARA**	Preferential use of the knowledge generated for adoption of the ProPAT system in their chemical processing facilities after the post-project industrialisation phase (and collaboration in the industrialisation phase as required)
16	**MBN**	Preferential use of the knowledge generated for adoption of the ProPAT system in their ceramics processing facilities after the post-project industrialisation phase (and collaboration in the industrialisation phase as required)

In terms of the measures that will be in place in order to effectively plan for the exploitation of the results of the project, first of all an **Innovation Committee** is established, consisting of qualified representatives from each of the beneficiaries (as was discussed in *Section 2.3.2*). This Committee will be responsible for effectively planning for the post-project use of the results through both the commercial and new research routes. In order to assist in this effort and to ensure that both exploitation routes are effectively addressed and prioritised over the course of the project, an **Innovation Manager,** Alejandro Rosales from the SME IRIS, will lead the planning of the industrial exploitation route in the process industry, as well as in others to which the technology could be transferred (food, drug pharmaceuticals, cosmetics, greenhouses, refrigeration, air-conditioning, as discussed). The **Research Manager**, Prof. Frans Muller from UNIVLEEDS, will lead the consortium in planning for future research directions that could emerge in order to build on the knowledge generated during the project. These managers will be responsible for leading the Exploitation Committee in implementing the measures for planning for the post-project exploitation of the results. They will both work closely with the **IPR Manager,** Dr. Silvia Carrasco from ICFO, to ensure that the results are properly protected in a timely manner and that access rights are defined and agreed. They will also work closely with the Dissemination Manager, Dr. Renata Körfer from DECHEMA, to ensure that the dissemination activities serve as powerful tools for stimulating future use of the ProPAT results (for example: disseminating in relevant international trade shows in the case of facilitating future commercial exploitation; disseminating in SPIRE events, and Horizon 2020 events, tradeshows, as well as conferences for attracting and identifying further research opportunities to build on the results; etc.). In addition the **Data Manager**, Ms Carla Kapikian (ZWAH), will ensure that all the research results data generated/collected during the project are well managed, including standards that will be used, how the data will be made accessible for verification and re-use, curated, preserved.

How the proposed measures will help to achieve the expected impact of the project

The realisation of the impact of this research and innovation action requires the eventual commercial exploitation of these results. There is a **strong innovation dimension** in the ProPAT approach to the work plan with a view to **increasing the likelihood of market uptake of project results** for European industrial sustainability, competitiveness, growth, and jobs.

A key measure for contributing to industry adoption of research results lies in a R&D offering that is founded on industry requirements and robust specifications. It is noteworthy that WP1 of the ProPAT work plan is based on defining industry needs in relation to process monitoring, as well as **defining a strategy for PAT implementation in the industry**. This is a fundamental approach in that, based on the experiences of implementing PAT methodologies in the pharmaceutical manufacturing industry, barriers in relation to

implementation of PATs centre on technical issues such as data centralisation, connectivity to the different data sources (analysers, databases, process control systems, etc.) and data management (collection, alignment, aggregation and contextualisation)[61]. In this context, the development of a global process control system (WP5) will also assist in the physical implementation of PATs in the minerals, ceramic, non-ferrous metals, and chemicals (plastic and pharmaceuticals) industries, and indeed in the wider process industries as the results of this project are rolled out and adopted. Essentially PATs need to get simpler and easier to implement and operate68 and ProPAT will contribute to this.

In addition, **overall cost** (including the investment in equipment) is a major consideration of industry in the adoption of PATs. WP2, WP3, and WP4 will centre on the development of affordable and versatile sensor technology, which can be uptaken by the PAT developments in order to arrive at cost-effective solutions. Each PAT platform will be tuned to the end users including only the required sensors. Moreover, simplified and standardised approaches for calibration and model development (WP5) could help to bring the implementation cost down to levels that are tolerable for the industry.

Importantly, the involvement of end-user industries (GM, MBN, GSK, MEGARA), along with SME sensor developers (SPECTRAL, MEMSFAB) and an SME continuous process control and systems integrator (IRIS) will heighten the industrial relevance and mainstreaming of the project results, as well as ensure a route-to-market for the knowledge generated during the project, while will be greatly strengthened by the presence of a pioneer of the PAT concept in Europe (DECHEMA) who can provide the necessary training to ensure industry is capacitated to get the optimal benefits of developments such as ProPAT.

The validation and demonstration of ProPAT in the target industries are **important steps of applied R&D on its way towards commercialisation.** Ensuring that measures are in place to identify and study the impacts of relevant regulations and the requirements of industry standards will contribute to ensuring that are no regulatory obstacles could prevent the future industry adoption of the ProPAT integrated process control system that will be planned for industrialisation post project. In addition the validation and demonstration work in the industry will **quantify the improvements in terms of improved quality control approach, cycle time reduction and increased process efficiency (e.g. increased yield, throughput and equipment utilisation)**, which will provide valuable data for helping to push the technologies to the market, as well as for stimulating market pull.

There can be a fear of the complexity and multidisciplinary character of PAT and there is a high doorstep for smaller companies[68] although the consortium counts 2 SMEs end users. To overcome this, a structured knowledge transfer and training effort towards the industry on the use of the ProPATs, and on the further development work required (WP9), is fundamental for raising technical skills and the knowledge base among the industry actors in the consortium (and to a wider base of industry through public training sessions), to ensure the will be able to **assimilate and exploit the results**. This is in line with the SPIRE Roadmap, which envisages that the progressive execution of the SPIRE roadmap will address also the **socio-economical dimension** as there will be an increased demand for professionals with (new or adapted) skills, for the process industry but also for the related and spin-off industries, facilitating the transition to a **greener and more competitive European industry**. Importantly, and again in line with the SPIRE Roapmap which emphasises workforce training and education programmes across and within sectors should be part of the business uptake strategies for SPIRE PPP investments[1], such as ProPAT, DECHEMA will ensure that the knowledge generated during the project is used to create valuable content and case studies for their future training and advisory services.

Finally, sound management of the knowledge generated during the project, active planning in relation to post-project exploitation activities for industrialisation of the most market ready results, and further research work for escalating along the TRL, or indeed for application of the platform knowledge generated during this project beyond the targeted selected process industries and into food processing sectors for example, such as dairy, as well as a solid communication strategy to disseminate the non-confidential aspects of the result to the SPIRE community and beyond, as well as to market actors and multipliers, and policy makers, will also increase the likelihood of market uptake of the project results.

The exploitation plans that will be drawn up during Subtask 10.2.1, building on the preliminary exploitation plans outlines in this document, will provide the details on **how the project's results will be used in**

[61] Process Analytical Technology: An industry perspective. European Pharmaceutical Review. Issue 4, 2010.

commercial exploitation activities, for training purposes, and for further research activities, including elements such as: purpose, main features and benefits of each technology or product; innovative aspects in comparison with existing and competing technologies and products; summary of need for further R&D activity (and implied risks); standardisation, approval and policy implications involved; collaboration required for exploitation (technology transfer activities); identification of the potential customers and the factors that affect their purchasing decisions; features of the target market (size, growth rate, share that the technology/product could reach; any factors likely to drive a change in the market; legal, technical and commercial barriers; other technologies likely to emerge in the near future; how each beneficiary entitled to the technology exploitation is positioned (or should be positioned) in the market; competing businesses/applications/technologies.

The presence of three technology companies in the consortium (SPECTRAL, MEMSFAB, IRIS) who are committed to routing the ProPAT analysers and sensors to marker, and who have the in-house R&D capacity to perform further advancement of the TRLs to quickly navigate the 'Valley of Death', will paving the way forward for a timely industrialisation phase to see market adoption of the ProPAT within 18-24 months of project completion.

How the participants will manage the research data generated and/or collected during the project

Effective exploitation of data depends upon proper data management and curation, as such a task has been devoted to data management in work plan and a **Data Manager** has been included in the project organisation structure for the project, Ms Carla Kapikian (ZWAH), who has considerable experience in the research data management and its associated challenges. Under his leadership a formal **data plan** will be drawn up and implemented during the project that will include: which data will be generated during the research; metadata, standards and quality assurance measures; plans for sharing data; copyright and intellectual property rights of data; data storage and back-up measures; data management roles and responsibilities; costing or resources needed. Indeed, as can be seen from figure 9 below, data management will be embedded in the wider knowledge management of the project, and it will follow the Guidelines on Open Access to Scientific Publications and Research data in Horizon H2020.

Strategy for knowledge management and protection

To pave the way forward for commercial exploitation and dissemination of the project results, the consortium will implement an integral strategy for the effective **management of this knowledge** *(Fig. 9*).

Background IPR and access rights

Background- information and knowledge (including inventions, databases, etc.) held by the participants prior to their accession to the EC grant agreement, as well as any intellectual property rights which are **needed** for carrying out the project or for using foreground[62] held by the consortium, or indeed by third parties- is defined in the proposal preparation phase (pre-Grant Agreement). The current status of the Background being brought to the project by the consortium is listed in Table 7. Each beneficiary has the right to exclude or limit specific Background from each other's access by informing the other partners about this exclusion. **Access to Background** that is, or will be found to be, necessary for the implementation of the project will be granted **royalty free** and **background needed to use the results of the project** will be provided either **royalty free or under fair and reasonable conditions**.

[62] REGULATION (EU) No 1290/2013 Of The European Parliament And Of The Council of 11 December 2013 laying down the rules for participation and dissemination in "Horizon 2020 - the Framework Programme for Research and Innovation (2014-2020)" and repealing Regulation (EC) No 1906/2006

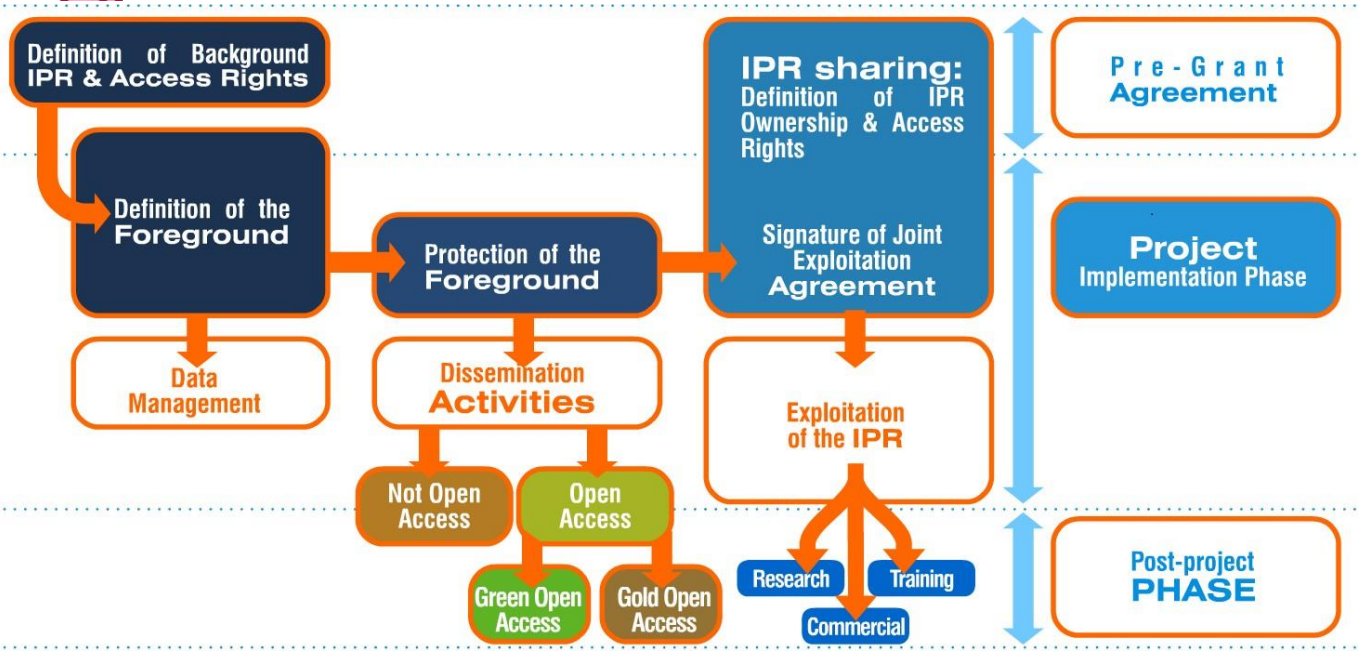


Fig. 9- Integrated Knowledge Management approach

Table 7- Background per beneficiary

No.	Short Name	Background
1	**IRIS**	Knowledge in sensor readings, process logic and control, and modelling, Knowledge predictive and adaptive learning. Knowledge of Wireless Sensor Networks. Database driven web portals and User Interfaces. Knowledge in design and integrated automation hardware technologies; Knowledge in industry-specific solutions for the automation of optical based systems; Knowledge in process interfaces and user interface systems for managing complex operations of control and process monitoring for industrial applications; Knowledge in risk management tools and protocols for industrial systems; Knowledge in product quality analysis.
2	**ICFO**	Patents related to ISI sensor device made at ICFO: Apparatus for measuring light scattering, V. Pruneri et al., patent application EP14382043 (2014); Optically transparent support for bubble-free specimens, V. Pruneri et al., patent application P201331446 (2013); Compact system for optical measurements using a discretized plate, V. Pruneri et al., patent application ES201330270 (2013)
3	**VTT**	FPI technologies represent the core technologies in the VTT's offering. VTT has a vast IP portfolio concerning Fabry-Perot interferometers. The main patents and patent applications are listed below (several other inventions like miniaturized Raman-FPI sensor and method of realizing a large gap FPI mirror structure in filing): PCT/FI2010/050043, EP2394146; PCT/FI2010/050739, FI20095976; PCT/FI2010/050739; FI20095068; EP2394146 ; FI20095068; US Patent US 8,130,380; UK Patent GB 2445956; US2010097613; PCT/FI/2010/050258. Knowledge in the field of optical measurement technologies for industrial applications, especially on-line applications. Knowledge in machine vision and optical spectroscopy, in all wavelength ranges (UV, VIS, NIR, IR, Raman, Fluorescence, etc.). Knowledge in optics, electronics, high precision mechanics, software, embedded software, system engineering, low cost manufacturing techniques, calibration, and various aspects of applied physics and chemistry.
4	**UNIVLEEDS**	Knowledge in Chemical Engineering, multiphase particulate processing and industrial chemistry. World leading expertise in pharmaceutical chemical synthesis and production, milling, granulation and online/in situ sensor development. Knowledge of multiscale analysis and control systems validation. Facilities and expertise to run batch and continuous processes on 20L scale. Facilities and analytical equipment for required complimentary *ex situ* particle characterisation studies; including laser particle sizers, SEMs, morphology and structural analysers.
5	**URV**	Knowledge in multivariate calibration and classification, knowledge in multivariate statistical process control (MSPC), knowledge in experimental design and statistics, knowledge in qualimetrics and quality control, knowledge in Matlab programming.
6	**ZHAW**	Knowledge in chemical analysis techniques. Knowledge in on-line, and time resolved analysis of volatile organic compounds at very high time resolution, mass-resolution and sensitivity. Knowledge in Life Cycle Analysis.

7	**TUD**	Knowledge in process-relevant particle characterisation including selection and optimization of the measuring methods and their validation. Knowledge in international standardisation in CEN (EC-mandate) and ISO.
8	**UB**	Knowledge in the development of chemometric tools for process understanding (modelling and control), notably soft-modelling Multivariate Curve Resolution (MCR) and hybrid hard- soft- modelling. Experience in data fusion strategies for process modeling
9	**IMAR**	Knowledge in smart sensors
10	**SPECTRAL**	Knowledge in the design and realisation of optical sensors based on MEMS Fabry-Perot Interferometers
11	**MEMSFAB**	Knowledge in commercial manufacturing of tuneable MEMS Fabry-Pérot filters
12	**DECHEMA**	Knowledge in PAT adoption in industry.
13	**GM**	Knowledge in excavation, beneficiation (laser/ optical sorters, magnetic separation, sink-float), crushing & sizing, classification and milling of caustic calcined magnesium oxide, dead burned magnesium oxide, magnesium carbonate and dunite
14	**GSK**	Knowledge in the critical control points for pharmaceutical processing….
15	**MEGARA**	Knowledge in the critical parameters to be controlled during the manufacture of powder coating resins, rosin dispersions, alkyd resins, water-based acrylic dispersions, and unsaturated polyester resins.
16	**MBN**	Knowledge on the granulometric requirements for the selected powder systems and relation to material properties needed to be monitored for the production of nanophased materials produced by high energy ball milling.

A comprehensive review of "prior art" and patents (*Section 2.1.3*) has revealed that to date there is **no relevant Background held by third parties** that is needed for implementing the research or using the project results. In order to ensure that the intended ProPAT R&D effort does not trespass into the owned knowledge space of a third party, and to mitigate against expensive infringement claims, during *Subtask 10.1.1: Protection of the IPR,* up-to-date patent reviews will be performed periodically to confirm the novelty of the proposed research and the freedom to exploit the results.

Results definition, protection, ownership and access rights

The expected results from the project are listed in Table 8 below, and will be continually assessed during the project *(*through *Subtask 10.1.1: Protection of the* IPR) for effective knowledge management, whereby **novelty, patentability and protection** will be assessed.

Results shall be owned by the participant generating them.[69] To avoid doubt or confusion over ownership, a protocol of procedures will be defined that each partner will be followed to clearly show the development of the generation of knowledge or results. In cases where the results are generated jointly (i.e. where the separate parts of some result cannot be attributed to different participants), it will be **jointly owned**, with the allocation and the terms of exercising this ownership of the results to be included in a Joint Ownership section of the *ProPAT Exploitation Agreement (or Agreements)* that will be drawn up as part of *Subtask 10.1.2: IPR Ownership and Sharing.* However, an overriding principal will be that the **participating industry partners are always provided with the access that they need to the results to exploit them in keeping with the needs and strategy of their business.** To this end, access rights to foreground for implementing the project will be granted on a royaltyfree basis. Access rights for use purposes will be granted either **royalty-free**, or on **fair and reasonable conditions** to be agreed in the *ProPAT Exploitation Agreement.* Table 8 provides an overview of the expected scenarios of results ownership and access rights.

Table 8- Expected Results ownership and access rights

WP	Description of the Expected Result	Envisaged IPR Tool	Owner(s)	Partners to which access rights will be granted and conditions
2	Improvements to the ISI granulometric sensors	Patent or Know how secured through confidential design	ICFO	Preferential use to IRIS for integration with the GCSS platform under fair and reasonable conditions to be agreed
3	Improved NIR MENS designs	Patent or Know how secured through confidential design	SPECTRAL	Preferential use to IRIS for integration with the GCSS platform under fair and reasonable conditions to be agreed
4	Set of selected smart sensors and their integration in the hardware	Know how secured through confidential design	IMAR	Preferential use to IRIS for integration with the GCSS platform under fair and reasonable conditions to be agreed

4	Chemometric models and multisensor models for the selected sensors	Know how	URV, UB	Preferential use to SPECTRAL and IRIS under fair and reasonable conditions to be agreed
6	Chemometric models and process predictive models for the processes developed at pilot scale	Know how	URV, UB	Free user rights of the models for UNIVLEEDS.
7	Chemometric models and predictive models to optimise process control of resin synthesis	Know how	URV, UB	Free user rights of the model for the end user MEGARA involved in its industrial validation
7	Chemometric models and predictive models to optimise process control of pharma product mixing, granulation and drying	Know how	URV, UB	Free user rights of the respective model for the end user GSK involved in its industrial validation
7	Chemometric models and predictive models to optimise process control of minerals obtained by milling	Know how	URV, UB	Free user rights of the respective model for the end user GM involved in its industrial validation
7	Chemometric models and predictive models to optimise process control of ceramic/metals obtained by milling	Know how	URV, UB	Free user rights of the respective model for the end user MBN involved in its industrial validation
5	Decision Support concepts, tools and integration in the user interface	Know-how through software development secured by code	IRIS, UB	Preferential use to the participating end-users under fair and reasonable conditions to be agreed
5	Global Control System PAT platform	Know how secured through confidential design	IRIS	Preferential use to the participating end-users under fair and reasonable conditions to be agreed Free use of the case scenarios studied by LCA granted to the participating industry partners for marketing purposes. Fair and reasonable conditions for future LCAs.
8	Results of the process environmental impacts when using ProPAT control system	Know how	ZHAW	
6	Model process set ups for milling, granulation, high temperature synthesis for demonstrating the potential ProPAT uses	Know how	UNIVLEEDS	Preferential use for IRIS for further tests of the GCSS platform and the end users under fair and reasonable conditions to be agreed
2	Calibration of the granulometric sensor using conventional test methods	Know how	TUD	Free user rights to ICFO and preferential use to IRIS for integration with the GCSS platform under fair and reasonable conditions to be agreed
3	Calibration of the granulometric sensor using conventional test methods	Know how	ZHAW	Free user rights to SPECTRAL and preferential use to IRIS for integration with the GCSS platform under fair and reasonable conditions to be agreed Preferential use to the participating end-users under fair and reasonable conditions to be agreed
4	Sampling device and opto-mechanical interface and probes for coupling ProPAT to the manufacturing facility	Know how secured through confidential design	IRIS	

In addition to defining and agreeing the terms of exercising ownership and the conditions of access rights to the project Results Foreground (using Table 8 as the starting point), under the leadership of the IPR Manager (Dr. Silvia Carrasco), the *Exploitation Agreement(s)* will also define such issues as whether access will be granted to third parties for exploitation purposes; licensing arrangements with other OEMs if necessary; any exploitation restrictions; etc. A first draft of the Exploitation Agreement will be available by **M24** and will be continually revised and edited throughout the project, and a final version will be signed by all partners at **M48**. Over the course of the project, the consortium will tap into external assistance and support from such sources as the **IPR-Helpdesk** (http://www.ipr-helpdesk.org), **Enterprise Europe Network** (http://www.enterprise-europenetwork.ec.europa.eu/index_en.htm), among other sources in the partner countries. An experienced patent attorney will also be approached.

Open access to peer-reviewed scientific publications

All peer-reviewed scientific publications that will result from the project will be made available free of charge through Gold or Green Open access publishing and budget provisions have been made to fund this. First of all and keeping with the nonconfidential character of background state of the art information, one or several review papers will be prepared during WP1.

This may include a review of existing particle size measurements, a review of existing modelling approaches, a review of polymerization tracking methods, etc. In addition, research papers are planned

when dealing with non-confidential information, e.g. for the calibration of the novel sensors vs. established methods, set up of models processes during scaling up activities, etc.

2.2.2. Communication activities

Following a **stakeholder analysis**, the target audience of our communication activities has been defined as follows: SMEs and enterprises from the European process industry (and their Associations), SPIRE Community, sectors in close relationship to the process industries, i.e. end users (and their Associations), scientists and the research community, policy makers, educational stakeholders, mass media (TV, radio and the written press) and multipliers for addressing the general public, Technology Platforms, other SPIRE and H2020 projects, early stage researchers.

We have planned the communication activities (Table 9) that will be performed for promoting the project and its finding during the period of the grant, based on a process of analysing: **Why** – the purpose of the communication action; **What** will be communicated – the message; **To whom** – the audience; **How** – the method; **When** – the timing. Our project communication plan will link with a broader dissemination strategy for the SPIRE programme, to contribute to efficient coordination and dissemination of SPIRE project outputs across the programme to facilitate the market uptake of innovative solutions, whereby early awareness of key innovation themes emerging from the projects to the entire SPIRE stakeholder community as well as sectors in close relationship to the process industries (i.e. end users) is desirable, as is a communication campaign to all process industries that considers how they can adopt a synergic approach for understanding how energy and resource efficiency might shape profitability across operations and produce new sustainable growth opportunities. To this end, the planned communication activities will be best used to maximise the future dissemination and exploitation of the project results.

Table 9- Communication Activities

How (Method)	Why (Purpose)	To whom (Audience)	What (Message)	When	Evaluation
Website	Awareness, Inform, Engage, Promote	All External Stakeholders	General project info., objs, impacts, consortium, progress, events, SPIRE info, feedback form, etc.	Launched by M3. Ongoing throughout project and post-project	Checking the usage logs and number of project consultations
Flyers	Awareness	All External Stakeholders	Project aims, ProPAT approach, expected impacts, consortium	Available from M3, throughout the project	No. of consultations
Commercial Posters/ Banners	Awareness	EU process industry; SPIRE Community; End users industries	ProPAT system and Unique Selling Points, Competitive benefits and ROI, Validation and Demonstration in EU Process Industry	Tradeshows & Exhibitions: IFPAC (International forum and Exhibition Process Analytical Technology), JEC composites show, K-Fair, Plast, PharmaExpo, SensorsExpo, etc.	No of industry consultations
Newsletters	Awareness	EU process industry; SPIRE Community; End users industries; Policy makers; Research community; SPIRE-1 projects	Project progress and updates; Process Industry testimonials and interviews; Contribution to SPIRE Roadmap objectives, etc.	Every 6 months	No. of consultations
Press Releases	Awareness	General public	Public interest content about the impacts of the project for competitivess, growth, jobs, societal and environmental benefits	At KO, and in line with public interest developments in the project, events, etc.	No. of hits on web
Project Blog	Awareness, Inform, Engage	All External Stakeholders	Posts about non-confidential developments, updates, observations, conclusions	At least one post per month	Readership & No. of consultations
Programme meetings	Engage	SPIRE Community, SPIRE1 projects, Other EC/Nat. Projects,	Thematic content in relation to PAT adoption, integrated process control, etc.	At least 4 over course of project	No. of engagements
Conference Presentations	Engage, Promote	Research community	Scientific and research methodology, results and conclusions in relation to validation work, LCA, etc.	Analytics and Control Technology), PLS (International Symposium on PLS and Related Methods), CAC-2016 (Conference on Chemometrics in Analytical Chemistry), EuPAT (European QbD & PAT Science Conference), ICNIRS (International Conference on Near	No. of engagements and No. of hits on web
Conference Posters	Engage, Promote	Research community	Scientific and research methodology, results and conclusions in relation to validation work, LCA, etc.		No. of engagements and No. of hits on web
Journal Articles	Inform	Research community	Peer reviewed articles, e.g. review of SoA on existing particle size measurements, modelling approaches, polymerization tracking methods, etc. Research papers on e.g. the calibration of the novel sensors vs. established methods, set up of models processes during scaling up activities	Critical Reviews in Analytical Chemistry, Analytica Chimica Acta, Powder Technology, Chemical Engineering Science, Applied Spectroscopy, Talanta, Chemometrics and Intelligent Laboratory Systems	No. of citations
Demonstations, Knowledge Transfer & Training sessions	Engage & Inform	EU process industry; SPIRE Community; End users industries; Policy makers; Research community; SPIRE-1 projects	PAT adoption in process industry, use of predictive and chemometric models, use of the ProPAT system, existing process control technologies, smart sensors	At least 4 over course of the project between M28 and M48 to conincide with key validation and demonstration activities	No. of attendees & No. of engagements
Case studies	Inform	EU process industry; SPIRE Community; End users industries; Policy makers; SPIRE-1 projects	The benefits of implementing the ProPAT system for real-time process monitoring and control	4 case studies will be prepared by each of the end users industries	No. of downloads from web & No. consultations
Reports and other project documents	Inform	Available to all external stakeholders	Public deliverables (from the deliverables list)	Upon completion according to the deliverables list	No. of downloads from web & No. consultations

3. Implementation

3.1. Work plan

Brief presentation of the overall structure of the work plan

The ProPAT project aims to develop robust and affordable process control technologies for improving standards and optimising industrial operations. To this end, over the course of the four years of this project, a series of sensors and analysers that can improve the control and efficiency of the process industry will be tested, optimised and integrated in a Global System Control Platform and validated in processing applications to demonstrate the feasibility of ProPAT as an integrated process control solution for future wide European deployment in the process industry. The R&D effort will centre on engineering design, software/algorithm design, data analytics and data management, process engineering and industrial integration and validation.

The project objectives have been used as the starting point for planning the work that needs to be implemented within the project. All the actions that are required to meet the objectives of this research and innovation action, have been collectively analysed by the interdisciplinary consortium. It was agreed how best to distribute the workload among the various experts in keeping with their relevant fields of knowledge, capacity, infrastructure, resources and in the case of the industry end-users, their process activity, monitoring and control requirements. Importantly, to ensure the project boosts the competitiveness of the wider European Process industry, robust knowledge management activities, innovation related activities and a strong communications campaign for reaching different audiences of stakeholders are in place. This bottom-up planning process has greatly facilitated the task of arriving at a **clear, comprehensive and realistic work plan**, which is clearly linked to the objectives set out in *Section 2.1,* which the consortium is confident will serve as a solid platform to ensure that the project's S&T and innovation goals are fulfilled.

Moreover the project work plan contains **five distinct activity phases** that have been discussed in Section 2.1.2: 1) Defining the ProPAT specifications; 2) Developing the ProPAT Technologies; 3) Validation and Demonstration; 4) Training and Education; 5) Deployment in the European SPIRE industries (a post-project phase, but which drives a cross-cutting or horizontal series of activities throughout the project, namely 'Gear up for Deployment'). While it is not necessary for these to be developed sequentially (as can be seen from the Gantt chart), there will be significant interaction between the WP and task leaders to acquire any necessary knowledge and inputs between these distinct phases.

Finally, in relation to the **independencies of the different tasks**, this has been given careful consideration during the preparation of the workplan and is illustrated in the Pert diagram of the components showing their interdependencies further down in this Section.

Illustrations from the Work Packages

From Task 5.1 Global Control System (GCS) development

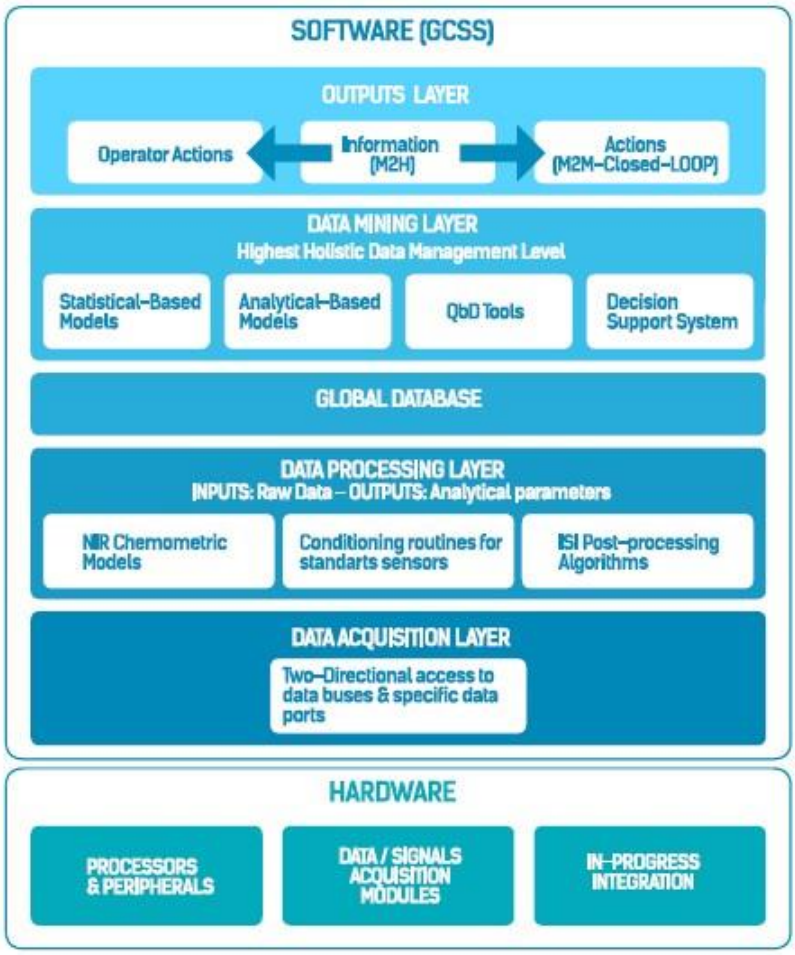
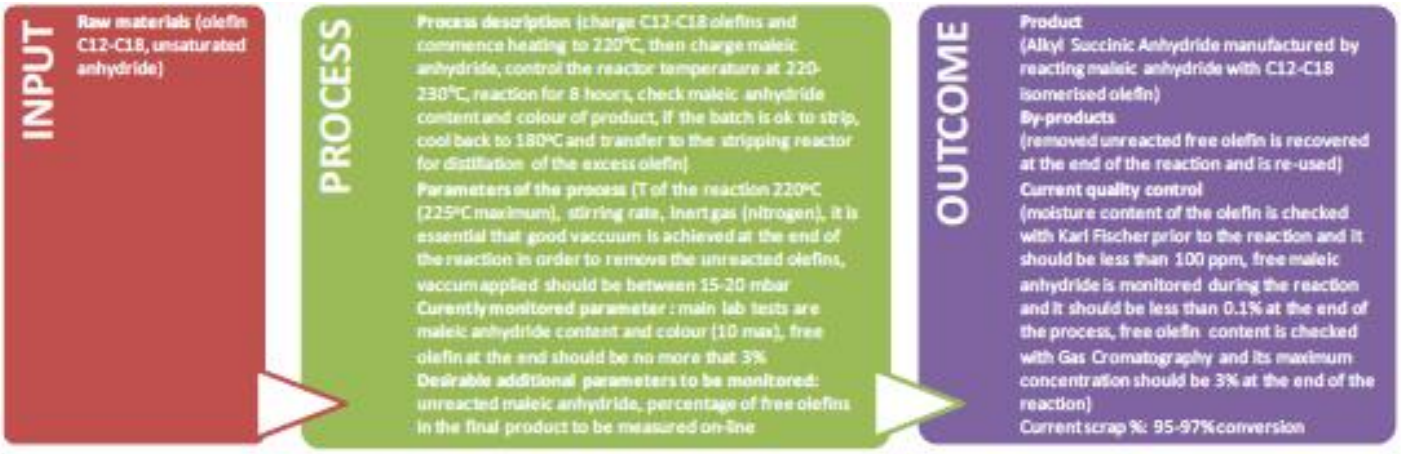


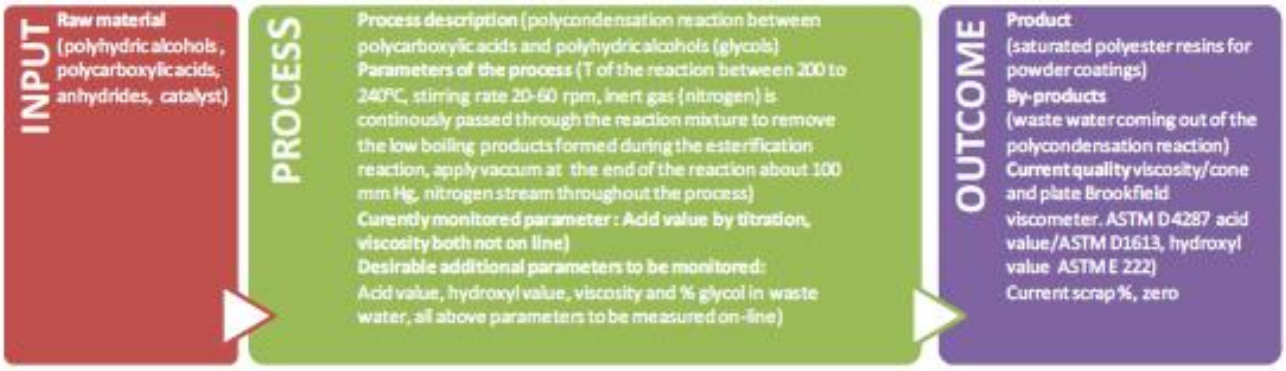
Figure 10-Structure of the GCS

From Task 7.2 Demonstration of the ProPAT system for controlling a polymerization process (Subtask 7.2.1 Monitoring of the process):

Description of 2 possible resin polymerization processes selected by MEGARA (for which ProPAT will be validated in at least one)



Process features for ASA resins production at MEGARA

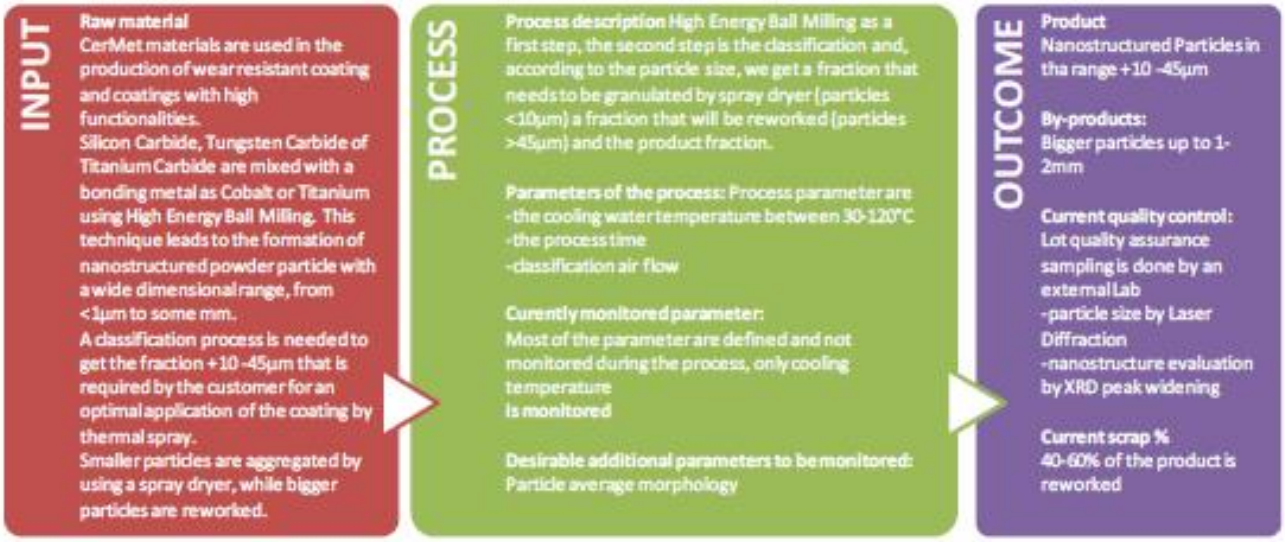


Process features for UPR production at MEGARA

From Task 7.3 Demonstration of the ProPAT system in a pharma mixing and granulation process (Subtask 7.3.1 Monitoring of the process):

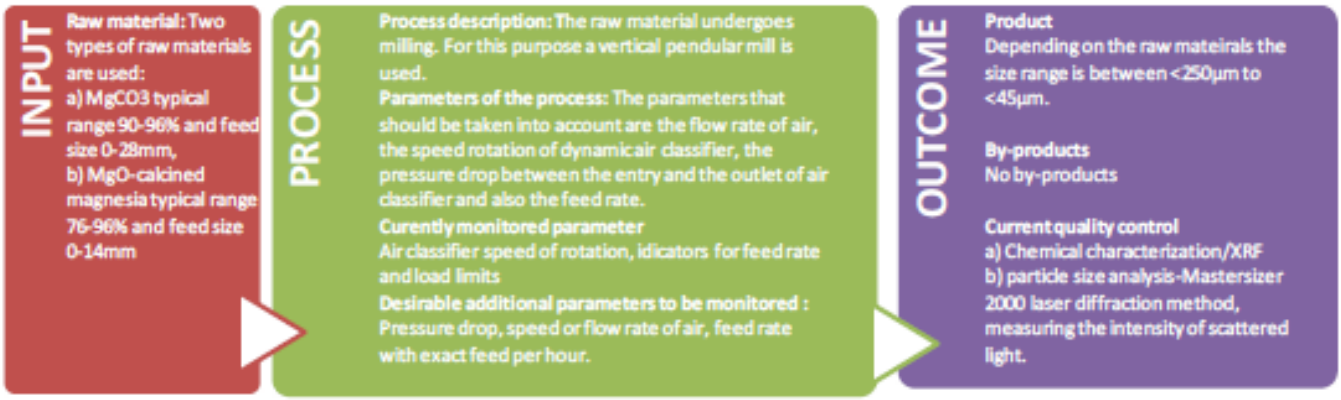
Typical mixing and granulation process in the pharma industry

From Task 7.4 Demonstration of the ProPAT system in a ceramic/metal milling process (Subtask 7.4.1 Monitoring of the process):



Process features for ceramic-metallic blends production at MBN

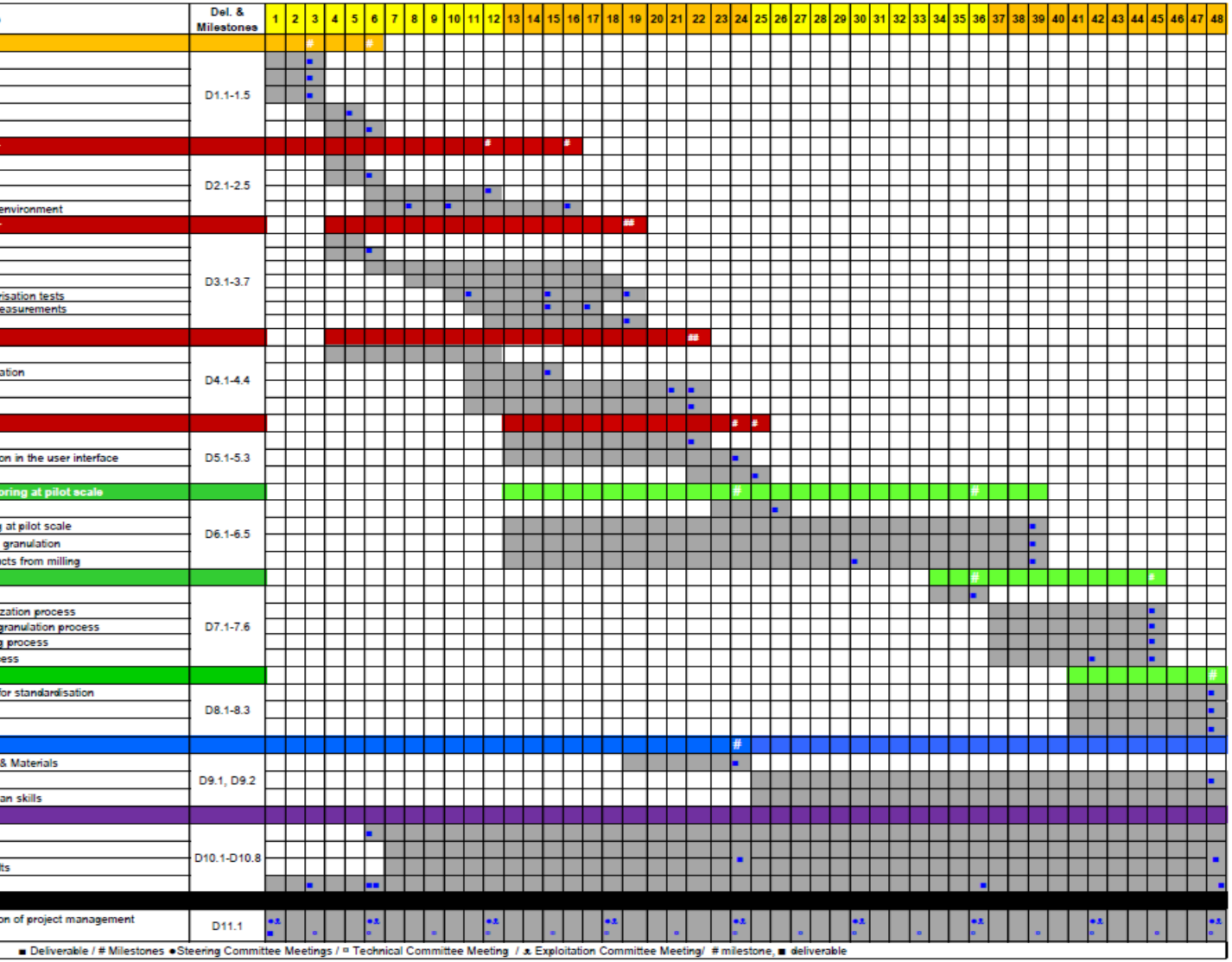
From Task 7.5 Demonstration of the ProPAT system in a minerals milling process (Subtask 7.5.1 Monitoring of the process):



Process features for calcined magnesite production at GM

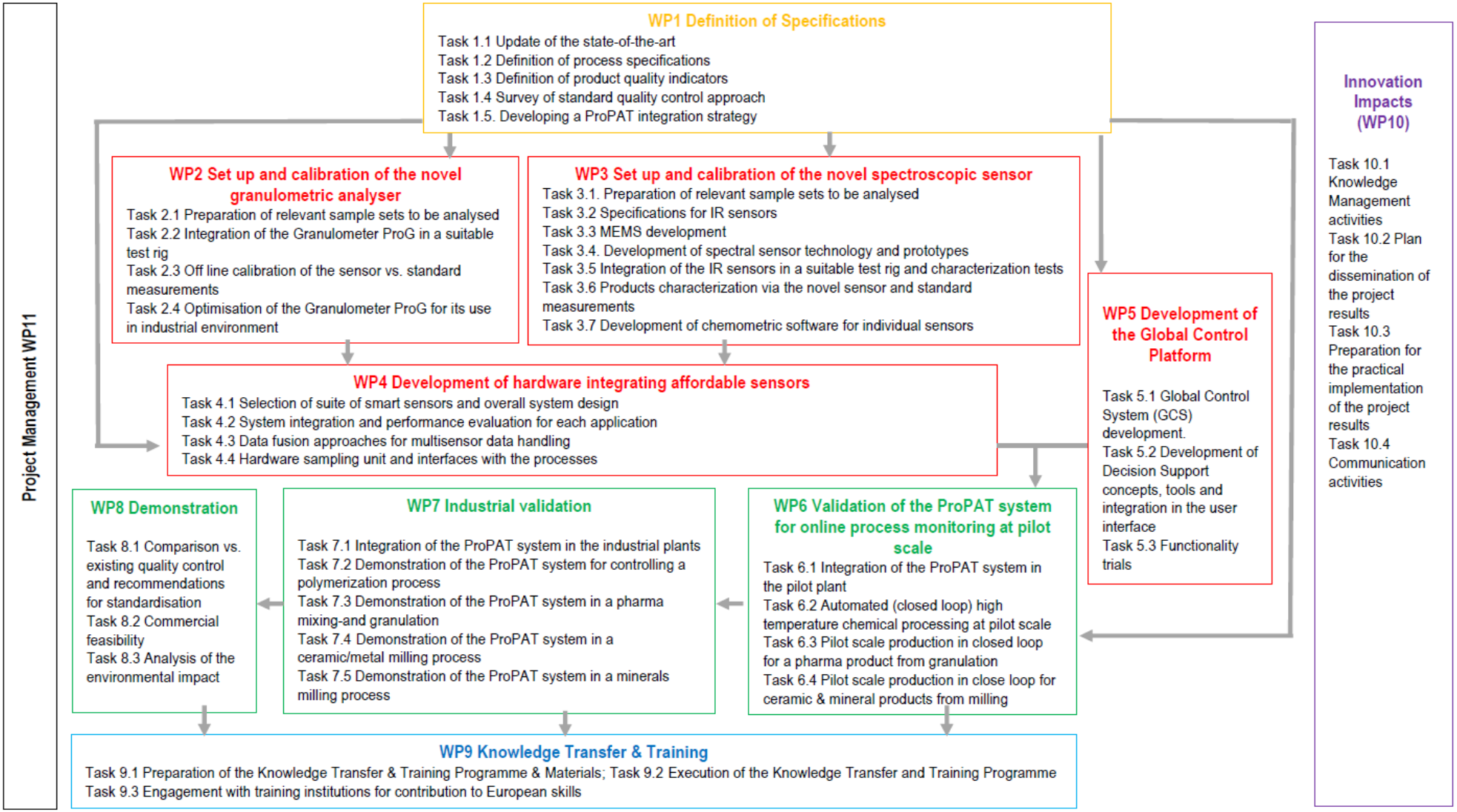
Timing of the different work packages and their components (Gantt chart or similar)

No	Workpackages and Tasks	Del. & Milestones
1	Definition of Specifications	
1.1	Update of the state-of-the-art	D1.1-1.5
1.2	Definition of process specifications	
1.3	Definition of product quality indicators	
1.4	Survey of standard quality control approach	
1.5	Developing a ProPAT integration strategy	
2	Set up and calibration of the novel granulometric analyser	
2.1	Preparation of relevant sample sets to be analysed	D2.1-2.5
2.2	Integration of the Granulometer ProG in a suitable test rig	
2.3	Off line calibration of the sensor vs. standard measurements	
2.4	Optimisation of the Granulometer ProG for its use in industrial environment	
3	Set up and calibration of the novel spectroscopic analyser	
3.1	Preparation of relevant sample sets to be analysed	D3.1-3.7
3.2	Specifications for IR sensors	
3.3	MEMS development	
3.4	Development of spectral sensor technology and prototypes	
3.5	Integration of the IR sensors in a suitable test rig and characterisation tests	
3.6	Products characterization via the novel sensor and standard measurements	
3.7	Development of chemometric software for individual sensors	
4	Development of hardware integrating affordable sensors	
4.1	Selection of suite of smart sensors and overall system design	D4.1-4.4
4.2	System integration and performance evaluation for each application	
4.3	Data fusion approaches for multisensor data handling	
4.4	Hardware sampling unit and interfaces with the processes	
5	Development of the Global Control System Platform	
5.1	Global Control System (GCS) development	D5.1-5.3
5.2	Development of Decision Support concepts, tools and integration in the user interface	
5.3	Functionality trials	
6	Validation of the ProPAT system for online process monitoring at pilot scale	
6.1	Integration of the ProPAT system in the pilot plant	D6.1-6.5
6.2	Automated (closed loop) high temperature chemical processing at pilot scale	
6.3	Pilot scale production in closed loop for a pharma product from granulation	
6.4	Pilot scale production in close loop for ceramic & mineral products from milling	
7	Industrial validation	
7.1	Integration of the ProPAT system in the industrial plants	D7.1-7.6
7.2	Demonstration of the ProPAT system for controlling a polymerization process	
7.3	Demonstration of the ProPAT system in a pharma mixing-and granulation process	
7.4	Demonstration of the ProPAT system in a ceramic/metal milling process	
7.5	Demonstration of the ProPAT system in a minerals milling process	
8	Demonstration	
8.1	Comparison vs. existing quality control and recommendations for standardisation	D8.1-8.3
8.2	Commercial feasibility	
8.3	Analysis of the environmental impact	
9	Knowledge Transfer and Training	
9.1	Preparation of the Knowledge Transfer & Training Programme & Materials	D9.1, D9.2
9.2	Execution of the Knowledge Transfer and Training Programme	
9.3	Engagement with training institutions for contribution to European skills	
10	Innovation Impacts	
10.1	Knowledge Management activities	D10.1-D10.8
10.2	Plan for the dissemination of the project results	
10.3	Preparation for the practical implementation of the project results	
10.4	Communication activities	
11	Consortium Management	
	11.1 Management of consortium resources; 11.2 Implementation of project management procedures; 11.3 Coordination with and Reporting to the EC	D11.1



■ Deliverable / # Milestones ●Steering Committee Meetings / ▫ Technical Committee Meeting / ⚓ Exploitation Committee Meeting/ # milestone, ■ deliverable

Graphical presentation of the components showing how they inter-relate (Pert chart or similar).



3.2. Management structures and procedures

Organisational structure and decision-making

The ProPAT **organisational structure** (Figure 11) consists of the **Project Coordinator**, the interface with the European Commission, who is supported by a **Project Steering Committee**. A **Research Committee** is also envisaged, as is an **Innovation Committee**. The **roles and decision making capacity of each** is discussed below. The project also envisages an **industrial advisory committee** with key flagship companies from the European Process Industry (from the SPIRE Community and SusCHEM platform), who will attend a number of project meetings and events and who act in an external advisory capacity to provide subjective advice to the consortium as required, to share knowledge and to assist in adding value and visibility to the ProPAT project. This is a voluntary Committee and its members will not have any decision-making capacity in the project.

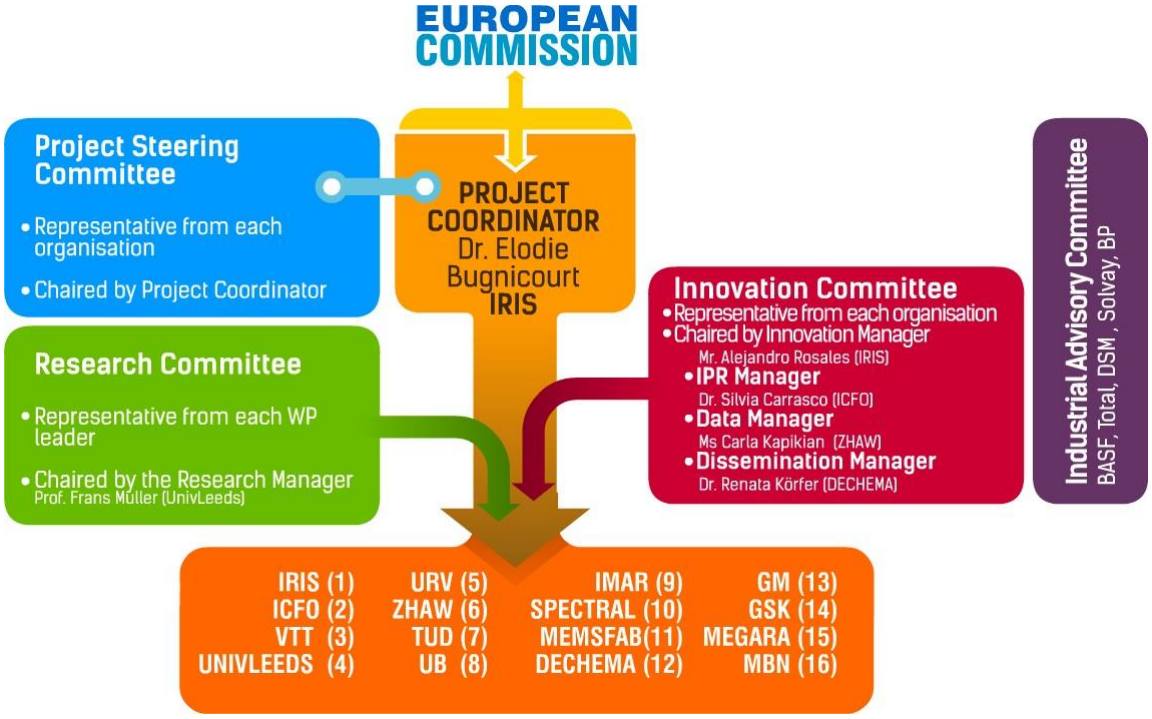


Figure 11- Overview of ProPAT Management Structure

<u>Project Coordinator</u>

The Project Coordinator is responsible for the day to day running of the project, and for ensuring that the work programme adheres to its schedule, fulfils its objectives, and the project meets the needs and expectations of all the partners. The Coordinator will be active in steering the evolution of the project by closely controlling the work that has been carried out by each partner and will be responsible for ensuring that **the project management procedures are being followed** and that all deadlines and obligations are met. The project coordination role requires a profile that is sufficiently experienced, qualified and capacitated to supervise the execution of work programme and delegation of the work packages (WPs), to act as the communication and administrative interface with the EC, to carry out the reporting and overall administrative management of the project, to motivate the team and encourage proactive participation, to facilitate collective problem solving and the taking of timely corrective actions.

Dr. Elodie Bugnicourt from IRIS will perform the role of Project Coordinator. In terms of her competence to successfully execute this role, Dr. Bugnicourt brings a wealth of project management experience from both private contract research and funded EC projects. In the context of FP7, she is now bringing her industry knowledge and expertise to the coordination two “Research for SMEs” projects: DURAWOOD (Grant Agreement No. 232296); and, OPTIJECT (FP7 GA nº 262623), as well as an NMP project OliPHA (FP7 GA nº280604): Among others, she was also technical manager and heavily involved in the General Steering Committee for the “Research for SME Associations” project, WHEYLAYER (Grant Agreement No. 218340-2) and follow up Demonstration Activity project WHEYLAYER2 (FP7 GA nº 315743). She is very familiar with all the modalities and administrative procedures involved in the management of EC research projects, from the negotiation process, preparing and maintaining the Consortium Agreement,

monitoring work, reporting requirements, the use of tools for communication with the EC, etc. Prior to this, she has successfully managed R&D projects in the chemical industry, and in particular in the development and study of novel nanocomposites, the formulation of bio-sourced resins and of flame retardant resins. She has shown herself to be a motivational leader, a fair and intelligent facilitator, with a sound understanding not only of materials science but also of process and associated industries.

Page

Project Steering Committee

To ensure that the project is implemented according to the needs and requirements of all the project partners, and that each **can exercise a role in the strategic management of the project,** a Project Steering Committee (PSC) is established consisting of a key representative of each of the beneficiaries. This PSC will meet every 6 months and will be facilitated through the chair of the Project Coordinator. Through this Committee the partners will closely monitor and control the work performed, and following the principles of **"quality, time, cost"**. Each representative will assume responsibility for mobilising and coordinating the active and coherent participation of the other personnel from their organisation who are involved in the ProPAT project, and will ensure that reporting and cost statements from their organisation are submitted on time and correctly. In terms of **decision making mechanisms**, when strategic decisions need to be taken, such as action to solve conflicts, action to mitigate critical risks to project implementation, action to manage changes in the work plan, etc., it will be the responsibility of the Project Coordinator to seek consensus with the Project Steering Committee, whereby each beneficiary will be **assigned one vote**. Decisions will be taken by simple majority. In the event of a tie situation, the Coordinator will take the final decision.

Research Committee

The Research Committee is established take care of scientific and technical (S&T) issues. This Committee will consist of each of the RTD and DEMO Work Packages and Task Leaders. The S&T Committee will meet every 3 months under the chair of the **Research Manager,** Prof Frans Muller (UNIVLEEDS). He will be responsible for supporting the Project Coordinator in ensuring that the S&T objectives and activities of the project are successfully accomplished in a timely manner, taking into account the interdisciplinary approach and the interdependence between tasks. It is expected that the appointment of the Research Manager to oversee the absolute S&T matters pertaining to the work plan will increase the potential for **early identification** of any scientific or technical risks or problems that require effective and timely management. In relation to decision making on **S&T issues of strategic importance to the project**, the Research Manager will alert the Project Coordinator and an emergency meeting of the Project Steering Committee will be called so that a decision can be arrived at.

Innovation Committee

The Innovation Committee consists of a qualified representative from each beneficiary. This Committee assumes the important role of Innovation Management in the project. Within this context this committee will manage the implementation of the innovation cycle, and the feedback linkages to the research. It will evaluate the knowledge as it is generated in line with its market impacts. It will also ensure that the project results are **effectively protected and exploited to the full for future industrial growth and competitiveness**. Key strategic issues, such as IPR protection mechanisms, IPR sharing and routes to exploitation will be discussed extensively during meetings of this Committee. It will also ensure that the project Dissemination Plan and Business Plan are further planned and developed based on the draft plans provided in Section 2.2.2.1. The Innovation Committee will also ensure that the data management plan (discussed in the same Section 2.2.2.1) is seamlessly executed. Moreover, the Innovation Committee will **decide on all matters with regard to IPR and exploitation, with each beneficiary being assigned a vote.** Decisions will be taken by simple majority. In the event of a tie situation the Project Coordinator will have the final vote. The Committee will be chaired by the Innovation Manager, Mr. Alejandro Rosales from IRIS who will lead the partners in effective management of the innovation cycle of the project (this role is discussed further down in the context of Innovation Management), as well as being a driver for the industrial partners in defining post-project commercial strategy. Mr. Rosales has ample expertise to perform this role as he has been the Chair of the Innovation Unit in IRIS for the past 2 years. He will be supported by a number of additional key management figures:

- ***IPR Manager***- Responsible for ensuring the project results are protected where possible and that the knowledge management procedures are implemented. This position will be fulfilled by Dr. Silvia Carrasco (ICFO), who has vast experience in relation to knowledge management and patent applications.
- ***Data Manager***- The Data Manager, Ms Carla Kapikian (ZHAW), will be responsible for leading the partners in drawing up, revising and implementing the Data Management Plan, which will provide an analysis of the main elements of the data management policy that will be used by the applicants with regard to all the datasets that will be generated by the project. She has a Master in Business Administration from MIT and is experienced in data management issues both from the private sector (banks and industry) and from the research side since2008.
- ***Dissemination Manager***- This position will be held by Dr. Renata Körfer (DECHEMA), who has considerable expertise in communication of results to industry, the general public, policy makers, the research community and other stakeholders, as well as vast reach to the process industries in Europe. She will be responsible for leading the consortium in further planning and implementing the project Communication and Dissemination activities, as well as ensuring the project reaches out to different audiences in the most effective way, and will also ensure the confidentiality of the results is safeguarded until the IP is protected.

Management Procedures

This proposed project represents a carefully planned series of inter-related activities that will require a set of resources (time, money, people, materials, energy, space, provisions, communication, quality, risk, etc.) to meet its pre-defined objectives. Robust Management procedures are required to ensure that the project is successfully managed. As such, the project will be managed according to proven standard project management and ISO 9001 principals, and founded on sound project planning and subsequent monitoring and controlling of the execution of the workplan. Project Management will reply on 3 key outputs from the project planning process: 1) the Project Definition (Section 2.1 of this documentl); 2) the Project Workplan (include in Part Section 3.1); and 3) Project management procedures (which are discussed below).

Technical Workplan Review and Update Procedure

The technical workplan (WP1-WP8) will be reviewed by the Research Committee on a monthly basis (via monthly Technical Reports that will be submitted by each WP and Task leader on the status of their tasks). In parallel, the schedule will be reviewed, whereby completed actions and tasks will be identified and registered as completed and any activities that should be completed, but are not, will be identified using the project management tool. This will ensure tight monthly monitoring of the technical work. The budget situation will be reported every quarter, whereby all partners will submit the status of their expenditure to the Coordinator, to enable a review of project performance against the budget to be performed. To this end, each quarter, after the workplan has been updated to show the current reality, the tool will reschedule the work to ensure the project will be completed within the original **effort, cost, and duration**. In addition, the **critical path** of the project will be evaluated during the quarterly meetings of the Research Committee, whereby the Project Milestones (Part A, Section 1.3) will be used as a valuable tool for determining whether or not the project is on schedule.

Any potential adjustments to the workplan will be discussed and proposed to the Project Steering Committee (PSC). Risks will be flagged and managed under the Risk Management Procedure.

Issues Management Procedure

Issues can be surfaced through verbal or written means, but eventually an issue must be formally documented using an Issue Submission Form, which is then entered into the Issues Log. The Project Coordinator will assign technical issues to a responsible from the Research Committee for resolution, whereby the impact to the project in terms of budget, schedule, and scope will be estimated. Issues of a non-technical nature will be dealt with by the Project Coordinator or by the most qualified representative from the PSC. The resolution or course of action will be documented on the Issue Submission Form, as well as documented briefly on the Issues Log. If resolving an issue involves changing the scope of the project, the issue will be closed and the Scope Change Management procedure will be used instead to manage the resolution.

Scope Management Procedure

While it is considered highly unlikely that any changes will be required to the scope of the project, for a complete set of Management Procedures it is important that scope change is included. In the unlikely event that a scope change is requested (only to be recommended in extraordinary circumstances and maybe arising through the resolution of an important issue or a high-level risk) a Scope Change Request form will be completed and brought before the PSC for discussion. If deemed vital, the request will be entered into the Scope Change Log and it will be assigned to the Research Manager for investigation. If the scope change involves an addition in scope that will not significantly impact on project cost, effort, and duration and the project will still be completed within the agreed upon cost, effort, and duration, the PSC may approve the Scope Change Request. However, if the scope change involves a shift in the scope of the project, the EC will be informed.

Risk Management Procedure

During the definition of this project, a complete assessment of project risks has been performed and a Risk Checklist has been documented (Table 3.2b), whereby risks have been listed, a risk level assigned and a mitigation strategy has been proposed. During the project implementation stage the Risk Checklist will be discussed and updated based on close monitoring of the evolution of the work plan (following an analysis the monthly technical reports, or following the quarterly budget and critical project path review). To this end, the Project Coordinator, by working closely with the Research Manager and the Innovation Manager (depending on the nature of the risk, as discussed below under *Critical Risks and Mitigation Measures*), will ensure periodic evaluation of risks throughout the project based on current circumstances. Risk Management will be included on the agenda of all meetings of Research Committee, and as appropriate in the case of the Innovation Committee and PSC meetings. An appropriate response plan will be created for each risk in keeping with its level to ensure the risk is managed successfully. This plan will include steps to manage the risk, people assigned, completion dates, and periodic dates to monitor progress. Activities associated with the risk plans will be moved to the project workplan and monitored accordingly.

Conflict Management Procedures

It is important to outline that the project ethos will centre on common understanding and vision, transparency, open dialogue and active participation. Project participants will be encouraged to raise any issues of concern or worry during meetings, or in private with the Project Coordinator if so preferred. It is considered that such a project ethos will contribute to conflict prevention. However, robust conflict management will continuously monitor for conflicts, to ensure early identification of conflicts, an analysis of their stage and the implementation of a resolution process in keeping with each stage. The conflict stages and corresponding resolution strategy are outlined in Table 10. The Project Coordinator assumes the responsibility to ensure that all conflicts are dealt with as soon as they arise in order to minimise their negative effect and progression to the next stage. All conflicts that cannot be resolved between the parties concerned will be brought before the PSC for a resolution or decision.

Table 10 - Conflict stages and resolution strategy

Stage No.	Conflict stage	Resolution strategy
1	Potential for conflict exists	Open dialogue and conciliation between the parties concerned
2	Latent conflict where a competitive situation could easily spill over into conflict	Open dialogue and conciliation between the parties concerned
3	Open conflict	a) Negotiation first among PSC (whereby decision-making procedures for PSC apply); b) Mediation next; c) Arbitration as a last resort
4	Aftermath conflict	a) Negotiation first among PSC (whereby decision-making procedures for PSC apply); b) Mediation next; c) Arbitration as a last resort

Communication Management

It is deemed appropriate that a Communication Plan is in place that guides the overall communication process for the project, the core aspects of which are discussed here. Formal project meetings will be scheduled. Agendas for all on-site and virtual meetings will be required. Discussions, decisions and actions will be recorded in minutes, which will be circulated to all partners, edited and agreed. The Coordinator will use the 'Kick-off' (KO) meeting to establish positive working practices in the consortium and to set the tone for all future conduct. Following this, the Project Steering Committee (PSC) will meet every 6 months to review the general progress of the work and to discuss all matters that are common

to all project beneficiaries. Emergency/extraordinary meetings of the PSC will be convoked in the event of identified risks, conflicts, problems, etc. The Innovation Committee will also meet every 6 months (after the PSC meetings) to discuss knowledge management issues, and plan for the exploitation and dissemination of the project results. The Research Committee will meet every 3 months (teleconferencing facilities may be used if considered appropriate) to enable the WP and Task leaders to review together the technical issues of the project. The technical progress and results of each of the active tasks will be presented, along with the actions required for the upcoming 3 month work period. Technical working groups may meet amongst themselves more regularly if required, and especially during periods of particularly high task activity or prior to major project milestones and deliverables. Virtual tools will be used in as much as possible for these extra meetings. Use of e-mail, Skype, videoconferencing tools, etc. among partners will be beneficial for the smooth progression of the project and to ensure sufficient regular communication. Emails will serve as communication records in themselves, and all conversations held will ensure the responsibility to report via the Project Coordinator to the consortium on all significant actions/conclusions and immediately should any problems arise.

Data Management procedures

These procedures, in keeping with the Data Management Plan (DMP) discussed in Section 2.2.2.1, will determine how research data will be handled during a research project, and even after the project is completed, describing what data will be collected, processed or generated and following what methodology and standards, whether and how this data will be shared and/or made open, and how it will be curated and preserved.

Quality Management procedures

Project quality monitoring will be ensured by a process of self-assessment of the project through the application of a Quality Management System (QMS) coherent with ISO 9001:2008. ISO 9001 defends a process-based approach for the development, implementation and efficiency of a QMS, whereby it is necessary to identify each activity (process), the required resources and inputs, specifications and results obtained, the responsibilities and the interaction between activities. This has all been ensured through the DoA. Such an approach enables continuous monitoring and control over the general process to assure the quality of the final product. The QMS will be implemented at the start of the project and will continually review the project management structure and plans, the specific quality objectives and quality indicators for each development stage of the project, the quality assurance and control activities (what is controlled, who controls, when), the details of personnel with respective responsibilities and assignments for each partners, the information exchange and dissemination, control of information exchange and reporting, and the documentation management and control including identification of critical points and risks assessment.

Appropriateness to the complexity and scale of the project

This organisational structure and decision making committees are considered appropriate for the management of the ProPAT research and innovation project, in view of the size and geographic disparity of the consortium, as well as the research ambition and expected innovation impact of the project results. In short, they will ensure:

1. effective management of the day-to-day running of the project and its resources and appropriate interfacing with the EC (Project Coordinator);
2. strategic management of the project (Project Steering Committee);
3. sound management of the overall research and it potential risks (Research Committee);
4. coherent and timely management of the innovation cycle of the projects towards success in market-oriented outputs (Innovation Committee).

It is important that dedicated committees with key roles and responsibilities are established to ensure that there is dedicated and qualified management and skills-sets mobilised at the different levels, as opposed to a centralised, one-dimensional management approach, which would be highly unlikely to deliver the necessary versatility. To this end, there is an integral approach to the management of the project, ensuring that all participants have a stake in management (via the Project Steering Committee) to ensure that the project achieves its research objectives (Research Committee), as well as the effective management of the knowledge and data generated, Intellectual Property Rights (IPRs), as well as the other innovation-

related activities required to deliver market exploitable results and to map out the route for further research in order to build on the foreground (via the Innovation Committee).

Innovation management

As a key objective of publicly-funded research is that it should lead to the exploitation of results, which goes one step further than the mere production and dissemination of new scientific knowledge[63], **effective management of the innovation cycle** is a vital component for ensuring success in market-oriented research projects.

The ProPAT innovation cycle will be based on a **continuous iterative process** whereby, the research, which is ongoing throughout the project, will be closely linked to feedback incorporation and exploration of market opportunities, involving a series of feedback loops, such as incorporating feedback from designers, engineers and users, as well as incorporating feedback and exploring market opportunities in parallel with other key steps of the innovation cycle (e.g. research, prototyping, industrial demonstration etc.). In addition, protecting and managing Intellectual Property Rights will be embedded in the innovation cycle as a continuous process. Moreover, as ProPAT is a complex development that implies trials on the client's side, these trials will commence towards the middle of the innovation cycle rather than closer to the end to enable user feedback to play a prominent role in the innovation cycle and allows for further advancement of the project outputs.

This entire processes needs to be well managed, through a robust innovation strategy, and indeed incorporating feedback and exploring market opportunities should start from the very beginning of the innovation cycle and not long after the beginning of research activities[70]. As the successful management of innovation requires an understanding of both the market and the technical problems, the **multidisciplinary ProPAT team** will be mobilised under the leadership of a competent **Innovation Manager** who is capable of fomenting an **integrated team approach**. This role will be performed by Mr. Alejandro Rosales who has solid experience chaining the Innovation Unit in IRIS.

Importantly, the ProPAT project management structure contemplates an Innovation Committee, consisting of a qualified member of staff from each of the participating ProPAT partners, who together will bring the necessary multidisciplinarity combining the most free thinkers from the research team, the market and business-focussed skills sets from the technological SMEs, and the senior actors from the end-user chain who will bring all the inputs of early adopters of first clients. Combining all these skill sets with knowledge management, IPR and patent expertise, ensures that we have a very capable innovation committee in place. Bi-annual meetings will allow the committee to **effectively manage the innovation cycle**, the core elements of which have been embedded as tasks into the Innovation-Related Activities work package to facilitate their formal management and execution. Moreover, the capable leadership of Mr. Rosales, who is a business focussed professional, and competent in leading teams to ensure "no thinking in terms of silos", and savvy enough to engage with external actors and the diverse stakeholder groups as required to ensure "openness for ideas from outside[70]" ensures that ProPAT has mobilised a solid and experienced approach to innovation management.

Critical risks and mitigation measures

Given that an inherent characteristic of research projects is technological uncertainty, which may result in project failure, and time and resources spent without any tangible return, these risks will be managed by the Research Committee under the leadership of the highly experiences and qualified **Research Manager**, Prof. Frans Muller (UNIVLEEDS). However, risks are not only technology-related, whereby they can be categorized as follows; market related risks, technology related risks, environment related risks, organization related risks[64]. Market-related risks and environment related risks will be closely analysed by the Innovation Committee, and organisational-related risks will be managed by the Project Coordinator supported by the Project Steering Committee, as part of the close monitoring of the evolution of the work and the mobilisation of the project resources.

During the preparation of the work plan considerable effort was dedicated to identifying the critical risks that could jeopardise the future or the integrity of the project as a whole and establishing risk-mitigation

[63] Innovation. How to convert research into commercial success story? Part 3: Innovation Management for Practitioners. European Commission. Directorate-General for Research and Innovation, 2013. Nanosciences, Nanotechnologies, Materials and New Production Technologies (NMP)

[64] Kasap et. al. (2007). Risk Management in R&D Projects. Management of Engineering and Technology, Portland, US. 5-9 Aug 2007. Print ISBN: 978-18908-4315-1

measures for each (see Critical Risks Table in Part A). During the project implementation phase, a continual process of monitoring during the management of this project (through the Risk Management procedures discussed above), will ensure that technical risks are quickly identified and brought before the relevant steering committee in order to discuss upon the most effective strategy for their resolution so that they do not jeopardise the future of the project, whereby agreement on all proposals from the Research Committee (in relation to technical risks) and the Innovation Committee (in relation to market- and environment-related risks) will be taken by the Project Steering Committee.

3.3. Consortium as a whole

Description of the consortium

The ProPAT consortium is made up of 16 partners from 8 European countries (Spain, Finland, United Kingdom, Ireland, Greece, Germany, Italy, Switzerland) and collectively represents a highly focused and synergistic balance of research and industry fields, expertise, skills, resources, and European reach, to constitute a complete value chain of actors that together are **capable of achieving the project objectives.** Without a European approach, the critical mass of expertise required for the multidisciplinary research team could not be gathered. Single technique specialists cannot, by definition, resolve multidisciplinary and multifaceted problems such as the delivery of a holistic approach to whole system optimisation.

There is solid **scientific presence** (ICFO, VTT, URV, UB, UNIVLEEDS, ZHAW, TUD, IMAR), combined with **strong industrial and commercial involvement** (IRIS, SPECTRAL, DECHEMA, MEMSFAB, GM, MBN, GSK, MEGARA) to ensure the research and innovation work and the resulting technological outputs are relevant for market needs. The presence of a highly qualified and experienced market actor (DECHEMA) with over 5500 members, who are scientists, engineers, companies, organisations and institutes completes are very high quality ProPAT consortium. Moreover, there is strong industrialisation capacity postproject with a clear route to market for the results, where the impacts of the ProPAT for increasing the competitiveness and sustainability of the European process industry will be realised. It is noteworthy that 5 for the ProPAT consortium partners are SPIRE members (DECHEMA, UNIVLEEDS, VTT, ZHAW, IRIS).

In order to evidence how each partner is relevant for the project and to **show the complementarity between the participants**, the partners are clustered according to their positioning on the value chain. The broad role of each cluster of partners in contributing to the project objectives, and, indeed, in contributing to the expected impacts of the project are illustrated in Figure 12, testament to how they are suited **and are committed to the tasks assigned to them.**

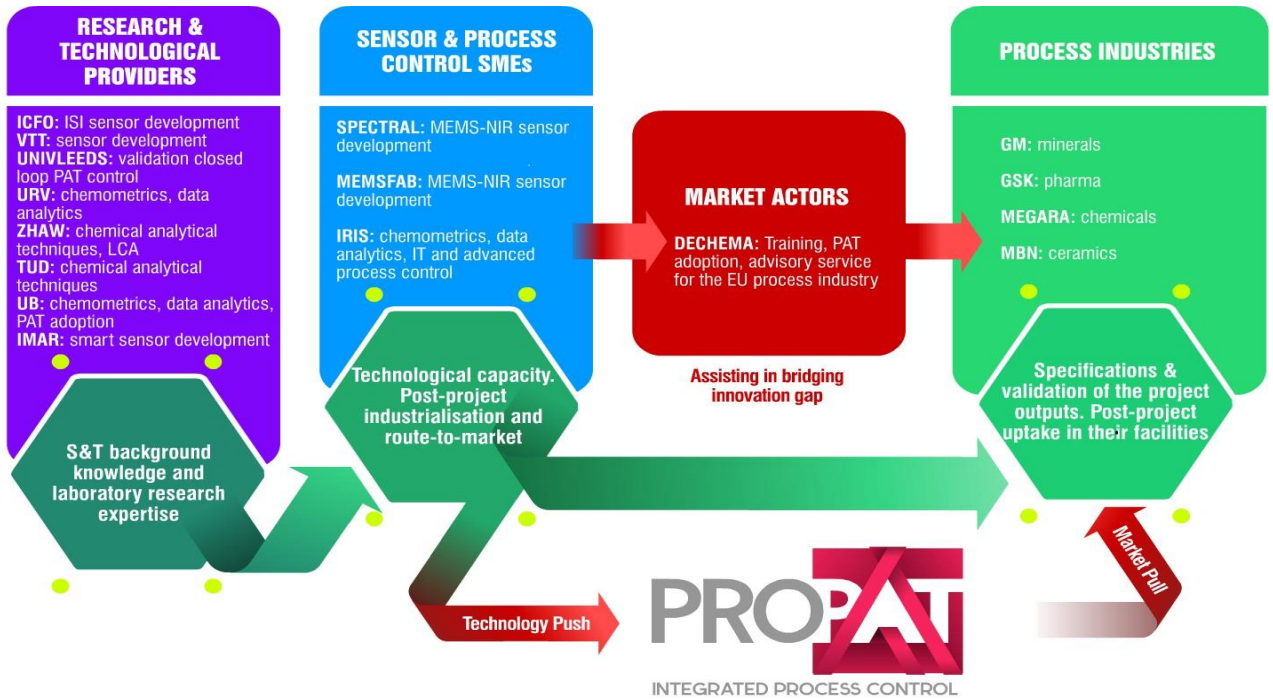


Figure 12- Overview of the consortium and how each contributes to ProPAT

In terms of the **relevance and capacity of the consortium to implement the work plan**, on the one hand, there is sufficient **research and technological background and expertise in the fields of:** sensor development (VTT, ICFO, IMAR, SPECTRAL), PAT adoption (UB, DECHEMA, IRIS), chemometrics, data analytics and advanced process control (IRIS, URV, UB), chemical analytical techniques (TUD, ZWAH), closed loop PAT validation (UNIVLEEDS), LCA (ZWAH), chemical processing (MEGARA), mineral processing (GM), ceramics (MBN) and pharma (GSK). There is **no overlap or conflict of interest** among the consortium and each participating organisation has a clearly defined role in the project, which will mitigate the risk of conflicts arising during the project. Indeed, our consortium covering the **whole value chain**, brings a series of key strengths due to some key features of its participants: DECHEMA: founder of the PAT methodology; GSK: 4th pharma company worldwide; UB: Reference in PAT. Organizer of the EuroPAT conference in 2014; ICFO: Research on graphene, plasmonic optics. Nature Publishing Index Top 200; IRIS: Participation as RTD and coordinator of more than 40 FP7 projects and active in offering high-end monitoring and control solutions to industry; SPECTRAL: innovative start-up in high end NIR-MEMS backed by TEKES and VTT; VTT: largest research institution in Scandinavia; UNIVLEEDS: participant in F3 Factory project that drew guidelines for process control topics; TUD: references for standards related to particle size; A core group of Suschem members who lobbied for the SPIRE programme creation.

In short, ProPAT unites sufficient reputable actors from the scientific fields, as well as from the industry value chain, in order to be able to realistically implement the ambitious work plan, to protect the knowledge and plan for its use and communication to a wide audience of different stakeholders, so that the true impacts of the results can be felt by European process industries and indeed the wider downstream discreet manufacturing industry in Europe, as the ProPATs are up-taken.

The industrial/commercial involvement in the project to ensure exploitation of the results

Strong industry involvement in the consortium, backed by robust knowledge management, knowledge transfer and market exploitation plans, will contribute to ensuring that the ProPAT project is successful transitioning novel PAT analysers and the latest low cost innovations in sensor technology to the industry and their integration into flexible control platform for providing an integrated process control solution to the European process industries. It is noteworthy that this consortium is industry led, through IRIS, who have vast experience in leading European applied R&D projects, and in particular projects that are geared towards market-oriented exploitation.

While the involvement of industry in R&D consortia increases the success rate of market-oriented exploitation[70], **vertical integration (i.e. including the whole value chain)** in R&D projects is a success factor for market-oriented exploitation.[70] To this end, ProPAT includes technological SMEs with research performing capacity who can contribute to **technology push** (the SME Sensor, instrumentation, and process control SMEs-IRIS, SPECTRAL, MEMSFAB), who are not only capacitated to contribute to the R&D effort, especially in the context of **scale-up, integration and demonstration of the results during the project,** but who can absorb the knowledge generated to protect it, in order to **quickly industrialise the outputs post-project towards commercialisation,** to serve strong industry demand that will be stimulated as a result of **the market pull** generated by the participating process industries (GSK, GM, MEGARA, MBN), who during the project will contribute end-user knowledge and experience to assist in mainstreaming the results of the project and in validating them in the industry. Importantly, the presence of large enterprises can make a difference in successful market oriented exploitation as long as the research outcomes are in line with their 'expectations'[70]. ProPAT has secured the interest and involvement of these large enterprises at the early stages of the preparation of this bid, and they all understand its scope and have been very committed in collaborating with the rest of the partners to contribute their vast knowledge of the processing environment, current technologies in use, as well as communicating their expectations in terms of research outputs which have helped to shape this bid.

Finally, the involvement of a non-profit making society for chemical engineering and biotechnology (DECHEMA) who is dedicated to **transferring research results to industry**, as well as **providing training and educational programmes**, is an invaluable market actor to complete the consortium by contributing to close the innovation gap that frequently prevents industry adoption the results and outputs of R&D, by facilitating the advice and training needed for the successful implementation of the PAT paradigm in the European process industry.

Other countries: The consortium includes a Swiss partner, the Zurich University of Applied Sciences. This partner has been included in the consortium due to their vast expertise in chemical analytical techniques as well as in LCA, which is considered essential for carrying out the WP2 and WP8 in the project. Erni Peter, director at the Eurosearch Head Office in Bern, Switzerland said that "In the event that Switzerland will be treated as a third country in calls for Horizon 2020 project proposals, funds for contributions to the EU framework programme will instead be paid directly to researchers in Switzerland, by the State Secretariat for Education, Research and Innovation of Switzerland.". The budget could also accommodate the Swiss partner while keeping below the maximum of € 6M stipulated in the call topic in case the situation evolves.

3.4. Planned use of resources

Table 3.4b: 'Other direct cost' items (travel, equipment, other goods and services, large research infrastructure)

1 - IRIS	Cost (€)	Justification
Travel	80000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting) and onsite installations in UNIVLEEDS and 4 industrial facilities (5 x 2 weeks, 2K € per week, 2 people); participation in conferences, fairs (5x2K €); travel for the industrial advisory board to attend specific meetings (€30 K); Travels to subcontractor to integrate software (€3000)
Equipment	3400	Equipment to integrate software to commercial granulometers (laptops, usb hubs, cables)
Other goods & services	142720	IPR protection and fees for open access journal (12K); organisation of workshops (3 x 2K €), printing of communication materials (2K €); opto-mechanical consumables for 4 prototypes- (4 x 15K €), software licences for designing the hardware and the GCSS (2 x 3K €); audit certificate (3K €); €14000 (prototyping and building the self-cleaning interface) for GSK; rental of particle sizer for GM and MBN use cases.
Total	226120	
2 - ICFO	**Cost (€)**	**Justification**
Travel	17000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting)
Equipment	62343.3	Equipment and consumables for 6 prototypes (computers, electronics, power LED, beam splitter, filters, CMOS image sensor kits, etc.)
Other goods & services	7000	Cost IPR protection and publication fees for open access journal (7000€)
Total	86343.3	
3 - VTT	**Cost (€)**	**Justification**
Travel	23000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting, 2 persons normally)
Equipment	25000	Silicon Wafers and MEMS process masks
Other goods & services	143549	Cost of IPR protection and publications fees for open access journal (7000€), audit cost (3000€), research facility costs (99549€), use of special equipment (34000€). Cost for hosting training session (3K €)
Total	191549	
4 - UNIVLEEDS	**Cost (€)**	**Justification**
Travel	28000	Travel costs to project and technical meetings (28 meetings in 4 years, 1000€ per meeting)
Equipment	20000	Laptops, NIR probe with optical fibres, and pump for process simulation
Other goods & services	171067	Processing and control consumables, cost of IPR protection and publication fees for open access journal and auditing cost (not over headed at 25%) and hosting of workshop. Piping, glass blowing, control, fabrication, chemicals, general laboratory consumables for building replica of GSK process
Total	219067	
5 - URV	**Cost (€)**	**Justification**
Travel	17000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting)
Equipment	6000	3 computer PCs (complete: CPU, monitor, Operative System, mouse...)
Other goods & services	11000	Cost of IPR protection and publication fees for open access journal (7000€); software licenses (4000€)
Total	34000	
7 - TUD	**Cost (€)**	**Justification**
Travel	20000	Travel costs to project meetings and visits to standards events (20 travels, 1000€ per travel)
Equipment		
Other goods & services	7000	Cost of IPR protection and publication fees for open access journal (7000€)
Total	27000	
8 - UB	**Cost (€)**	**Justification**
Travel	17000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting)
Equipment		
Other goods & services	21000	Cost of IPR protection and publication fees for open access journal (7000€); MATLAB software license (this is the environment used to produce the ProPAT software) (5000€); Conference registration (500€/senior research, 250 €/predoc); 1 meeting per year in specialized conferences related to PAT (euroPACT, APACT, EuPAT,...)(4000€); Accommodation costs for prof. Marcel Maeder, from Univ. of Newcastle, leading expert in hard-modeling to advise in task related to modelling of processes involving chemical reactions processes and for taining of industrial staff (5000€)
Total	38000	
9 - IMAR	**Cost (€)**	**Justification**
Travel	17000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting)

Equipment	10000	Sensor calibration Software and hardware suites, Sensor hardware housing
Other goods & services	27000	Cost of IPR protection and publication fees for open access journal (7000€); materials & consumables; Sensor & test environments including calibration beds. Development boards for sensor integration, Batteries & replacement parts (20K €)
Total	54000	
10 - SPECTRAL	**Cost (€)**	**Justification**
Travel	30000	Travel costs to project meetings (17 meetings in 4 years, 1500€ per meeting and 1-2 people travelling)
Equipment		
Other goods & services	40000	Cost of IPR protection and publication fees for open access journal (10000€); Materials and consumables: components for NIR sensors, optomechanics for test rings (30K €)
Total	70000	
11 - MEMSFAB	**Cost (€)**	**Justification**
Travel	25500	Travel costs to project meetings (17 meetings in 4 years, 1500€ per meeting)
Equipment	60000	Cleanroom equipment use for fabricating three batches of wafers
Other goods & services	10000	Materials and consumables: silicon wafers and other materials needed in wafer processing
Total	95500	
12 - DECHEMA	**Cost (€)**	**Justification**
Travel	20000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting); Travel to conferences, workshops (3K €);
Equipment		
Other goods & services	17000	Cost of IPR protection and publication fees for open access journal (7000€); consumables and supplies (3K €); license to access the standard sources (2K €); costs for organising and hosting training sessions (6K €)
Total	37000	
13 - GM	**Cost (€)**	**Justification**
Travel	17000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting)
Equipment	35000	Cost of modification to accommodate the PAT platform as well as consumables
Other goods & services	8000	Cost of IPR protection and publication fees for open access journal (3000€); consumables for building support structures and performing process adaptations (€5000)
Total	60000	
14 - GSK	**Cost (€)**	**Justification**
Travel	21000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting), as well as for attendance to 2 conferences (2x 1500 €)
Equipment	22000	Consumables for tests and installation of PAT system
Other goods & services	7000	Cost of IPR protection and publication fees for open access journal (7000€)
Total	50000	
15 - MEGARA	**Cost (€)**	**Justification**
Travel	17000	Travel costs to project meetings (17 meetings in 4 years, 1000€ per meeting)
Equipment		
Other goods & services	21000	Cost of IPR protection and publication fees for open access journal (7000€); consumables for tests and installation of PAR system (14K €)
Total	38000	
16 - MBN	**Cost (€)**	**Justification**
Travel	8000	Travel costs to project meetings (8 meetings in 4 years, 1000€ per meeting)
Equipment		
Other goods & services	43000	Spare parts/mechanical parts for milling machines, valve and connections, raw materials, Cost of IPR protection; consumables for support structures and process adaptations (€5000)
Total	51000	

2/VTT	**Cost (€)**	**Justification**
Large research infrastructure	99549	Research facility comprises of equipment located in Micronova M2 at Tietotie 3 in Espoo: Clean room M1; Clean room M2; SubTech lab, 155; Micropackaging lab, 1021; Metal CMP laboratory, 109; Chemical laboratory, 102; Optical laboratories 120 and 121.

4. Members of the Consortium

4.1. Participants

Participant no.	1	**Name (short name)/ Country**	IRIS Technology Solutions SL
Description of the legal entity			
IRIS was established in Barcelona in 2007 as an advanced engineering and R&D company, specialising in the field of process engineering. IRIS has expertise in the following fields of engineering: Production Processes (with a strong specialisation in novel technologies); Smart Systems; and Optical Systems, combined with a knowledge and research capacity in food technology, materials science and environmental science. IRIS applies its knowledge and expertise to develop solutions for industry in the fields of **advanced monitoring and control,** surface activation, decontamination and biocompounds, as well as to provide contract research services to the food, agriculture, chemical and pharmaceutical industries, among others. The company has a policy of continuous investment for growth, and currently employs 48 staff based in two sites (Barcelona and Dublin) which are equipped with engineering workshops (3), food testing and wet chemistry laboratory, optics labs (2), and an electronics/telecommunications lab. IRIS is **ISO 9001:2008 certified** and also is certified under **UNE-EN-ISO 166002: Management Systems R&D&i**.			
Main tasks			
IRIS will act as Project Coordinators and they have extensive expertise at this level, and in particular in the context of industry-driven FP7 projects, and private contacts with the industry. In addition, they will lead WP5, whereby, they will be responsible for the development of the Global System Control for successfully integrating the PAT developed during the project in the participating dairy processing facilities. As a forerunner to this IRIS will also lead Task 1.5, in order to draw up an integration strategy in collaboration with the consortium. IRIS will lead WP5 Development of the Global Control System Platform. IRIS will play an active role in validating the integrated PATs in the industry (WP7). IRIS will also lead WP10, especially in terms of bringing their skills in IPR management, commercial branding, reaching industry, the market and consumers and also contribute to knowledge transfer and training activities in WP9.			
How the legal entity profile matches the tasks in the project			
IRIS is currently developing a global process control platform for integrating PATs in the industry (PATBox™), the knowledge from which it brings as background to the project. In addition, the Smart Systems group at IRIS unites expertise in sensor readings, process logic and control, and modelling, coupled with knowledge predictive and adaptive learning capable of describing and analysing the given situation and taking smart actions based on the available data. The Group is currently working on a number of projects related to WSNs where the above mentioned is applied. With its team and resources, including telecom and electronics engineers, system engineering, application programming and web development etc. the group can carry out full system design and engineering, including electronic design (PCB) and assembly, firmware and software development, electrical harnesses and interconnections, mechanical design for enclosure and prototype integration. Moreover IRIS has optical engineers, mechanical and mechatronics engineers, specialised in integration work, as well as installation, validation and industrialisation of prototypes and equipment in			

food and pharma production plants. In terms of project management expertise, IRIS is a highly experienced coordinator of over 30 projects in FP7 to date.
In terms of programming and advanced ITs capabilities, IRIS has a specialized staff (**software and** telecommunication engineers with programming experience) that work on the basis of the Agile **methodology** and philosophy a (scrum **and kanban**) for developing complex applications that include hardware interfaces, **SQL and No-SQL** databases **(MySQL, PostgresDB, Couchbase among others)** and advanced and user-friendly graphic user interfaces **not only for computers (using Windows Presentation Foundation) but also for smartphones (Android, iOS) or web-based access (HTML 5)**. Our software engineers are skilled in a wide variety of languages and operating systems as well as low-level programming (microcontrollers).

CV of the persons who will be primarily responsible for carrying out the work

Dr. Ing. ELODIE BUGNICOURT. Masters in Material Science & Engineering (2002), PhD in Polymers and Composite Materials (2005), both from National Institute of Applied Science (INSA), Lyon, France. Leader of the Eco-Materials business unit in IRIS since 2009. Coordinated several European projects in the fields of bio-based packaging materials development, process monitoring, surface treatment. Previous industry experience: R&D in chemical industry especially related to reactive polymer formulations: i) Arkema (ex. Cray Valley, 2002, Spain) in the field of polyester powder coating; Wacker Chemie (2002-2005, Germany and France), Silicone division that funded her PhD thesis: development of **nanocomposites** based on an epoxy matrix and silica; iii) Ashland (2006-2009, located sequentially in US, Spain and France) in the Composite Polymers division, focussing on bio-sourced resins, flame retardant resins, SMC resins, PU and epoxy structural adhesives. Led and participated in many R&D projects aimed at developing new formulations, understanding mechanisms and structure-properties relationships, developing new analytical methods and modelling (for improved quality management); as well as collaborations with academic labs and other players in the supply and value chain. Author of several papers in peer-reviewed and technical journals, and has presented her work in numerous international conferences.

PAU PUIGDOLLERS DE BALLE. BSc in Environmental Science from the Universitat Autònoma de Barcelona. MSc in Terrestrial Ecology (Research Thesis: "Effects of biochar from fast pyrolysis on soil water availability and basal respiration", 2010) from the same university and MSc in Renewable Energy Engineering and Management from the Universitat de Barcelona. Prior to joining IRIS he held positions as technical sales person on Servicios Normativos consultancy group, working with environmental and quality projects, and quality field inspector on waste treatment plants for Ingeniería IDOM International, performing quality audits and reports. He has also held different research positions at the CSIRO (Commonwealth Scientific and Industrial Research Organization), in Australia and in the CREAF (Centre de recerca ecològica i aplicacions forestals) in Barcelona. He has also experience as environmental consultant at Eco Intelligent Growth, working with renewable energies, life cycle assessment, C2C certifications and green building. He has extensive project management expertise, whereby he has successfully managed to a positive conclusion on 4 EU R&D projects in the fields of advanced monitoring and decision support system development.

MSc. ALEJANDRO ROSALES has a M. Sc. in Physics from the Universidad de La Habana (1986). His background comprises theoretical and technical skills in several scientific and technological areas: optics, lasers, high vacuum, colorimetry, UV-VIS-NIR-MIR spectroscopy, chemometrics and design of measurement instruments. He has extensive practical experience in programming, electronics and mechanical CAD design as well as R&D methodology and management. His career has included positions as researcher applications engineer, design engineer, business developer and R&D manager. In 2000s, he worked in the FTIR spectroscopy field applied to quality control in the agrofood (wine) industry, whereby he developed two new measurement instruments for wine analysis. His career as industrial physicist has been focused in applying his wide expertise to provide solutions for industrial problems. Since 2011, he has been the Technical Manager in IRIS, leading a staff of 20 engineers and researchers involved in R&D projects aimed at developing novel high-end solutions for industry. Currently, he is the Science & Technology Manager in IRIS.

Dr. CRISTINA SERRANO is currently Project Manager in the Eco-Materials business unit in IRIS where she coordinates projects in the fields of sustainable packaging and surface treatment. She received her PhD (2011) at the University Complutense of Madrid (Spain) working at the Institute of Science and Technology of Polymers (2006-2011) on the development and characterization of new polymeric nanocomposites for self-sterilization of food packaging materials, based on the incorporation of non-doped and doped-TiO_2. She benefited from short pre-doctoral stays in PLAPIQUI (Argentina, 2008) to study the rheological behaviour of thermoplastics and composites, and in the Max Planck Institute for Polymer Research (Germany, 2011) where her studies were focused on obtaining nanostructured surfaces by plasma etching. After her PhD degree she worked as a post-Doc (2011-2013) at the Max Planck Institute for Polymer Research, Germany, on several R&D projects in the field of bio-inspired adhesives and stimulus sensitive polymers involving synthesis and modification of polymers and polymer´s characterization. She is co-author of fourteen scientific articles and has presented her work in national and international conferences.

JAIME PEREIRAS. Masters Degree in Mechatronics (2011) and a Bachelor's Degree in Electronics (2009), both at University of Oviedo. His professional background comprises technical experience in the areas of automatics, electronics, mechanics and manufacturing processes, with special emphasis on signal analysis and control engineering. He has previous experience at developing special machines for assembly lines and development of applications for monitoring process control. His software expertise includes SolidWorks, Matlab/Simulink, LabView, OrCAD Cadence/PSpice/Layout, among others.

DR. ANDRAS GEFFERTH. Ph.D. in Computer Science (2005) and M.Sc. in Computer Science (1998) from Budapest University of Technology and Economics. Besides computer science his background includes mathematical modelling and stochastic processes. Industry experience at Morgan Stanley as a quantitative researcher working on pricing and risk models for various financial derivatives. Research fellow at the Budapest University of Technology and Economics participating in several R&D projects in the fields of video transmission, data network traffic identification, data network traffic modelling and analysis. Teaching and supervision of M.Sc. students, giving lectures and supervising laboratory work. Expertise in various programming and scripting languages-C++, Matlab, Tcsh, Awk and Python. Co-author of several peer reviewed publications.

MIROSLAV SPAJIC. B.Sc. in Programming, Uni. Belgrade. M.Sc. Programming, Megatrend University, School for Computer Sciences. Senior Programmer in IRIS since March 2009. Main fields of interest are Programming, DB administration, SAP, Networking and multimedia. Industry expertise: 1996 to 2008- Senior Programmer at "TE Nikola Tesla" Power Plant: programming; DB admin, creation, scripting, automatic backups and maintaining of database security; Networking. SAP (System installation and configuration, Data migration, QHQ Client, Customizing etc.) and SAP Training for System Administrators in EPS/ICAS. 1994-1996- design and video editing in Studio MAG, MSC Studio, TV Mix Bourgas – Bulgaria and Studio Film Belgrade: non-linear audio-video editing, radio and TV advertisements and movie production. 2008-2009- Snr. Prog. At Sol90Labs, Barcelona, developing the program "Xalok" (editorial software for newspapers) in C++, XML, XSLT, ASP and JS; Network and DB admin of MS SQL databases.

IVAN PAQUICO RODRIGUEZ. Telecommunications Degree in UPC (Barcelona) and a Joint Degree in Multimedia Development in LUAS (Finland). His final thesis, 'Augmented Reality Chess and Application to Detect Chess Moves', implemented by using Computer Vision, Augmented Reality and Artificial Intelligence techniques, has won several awards, such as the best final thesis of the UPC and the best final thesis of Spain (received from Vodafone Spain and Universia foundation). He is an official iOS and Android Developer since 2010, creating a social network for iPhone called: 'BeThere' developed by iOS SDK from Apple and used Couchbase technology for the database. He also has experience in different programming languages such as C, C++, Objective C or Java, 8 bits embedded and real time systems. He has expertise in multimedia methodologies ranging from Virtual Reality programming techniques to 3D/2D modelling software such as OpenGL, Quest3D, Autodesk Maya, Autodesk 3DS Max or Adobe Photoshop.

DR. TIMOTHY KEHOE. PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast. Designed an Echelle spectrometer while based at Andor Technology Ltd. Worked at Tyndall National Institute, Cork, and Catalan Institute of Nanotechnology, Barcelona, developing diffractive optical and acousto-optical metrology techniques for nanotechnology applications, in particular for characterisation of nanoscale structures fabricated by nanoimprint lithography and directed self-assembly. Other research activities: quantification of ordering in selfassembled structures by image analysis techniques and optical sensors using hydrogel materials.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Development of a commercial system for real-time monitoring of moisture, CO2, temperature and filling level of grains in industrial silos. It comprises a network of multi-sensor probes and a data management subsystem. B2B project (private contract research).
2. System for real-time on-line multipoint monitoring the olive oil extraction (centrifuge) in order to optimize the yield and reducing wastes. B2B project (private contract research).
3. A platform for in-field multi-point monitoring of salinity in rice paddies in order to optimize the water consumption (AgriBox, in IRIS products line)
4. A database-based system for decision support to ensure quality of ready-to-eat foods production (from FP7 project Startec: Decision Support Tools to ensure safe, tasty and nutritious Advanced Ready-To-Eat foods for healthy and vulnerable Consumers, Grant Agreement no: 289262.
5. VIS and NIR-based on-line QC control in plastic injection process and DDS using chemometrics, as well as high temperature and pressure reflectrance and transmittance probes relevant to ProPAT (from FP7 OPTIJECT: A novel NIR-based instrument for in-line monitoring during injection moulding , GA nº 262623)

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. OPTI-CLEAN-Effective and reliable optical system (NIR-Chemical Imaging) for cleaning validation in pharmaceuticals manufacture- Grant Agreement no: 285836
2. PARTICLE-PRO-Development of a hybrid process analytical technology to characterise the physical and chemical properties of particles- Grant Agreement no: 312050
3. CRYSTAL-VIS. A process analytical technology for characterising the physical properties of crystals. Grant Agreement no: 605814
4. PUREFORMULA-Development of a novel control technology for ensuring high quality Powdered Infant Formula. Grant Agreement no: 605134
5. COMPO-BALL- Novel on-line composting monitoring system. Grant Agreement no.: 243625

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Spectroscopic Laboratory equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS
(PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).
Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others.
IRIS' optical engineers work with ZEEMAX software for optimally designing of optical systems and to setup the designs by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

Participant no.	2	**Name (short name)/Country**	Institute of Photonic Sciences (ICFO)/SPAIN
Description of the legal entity			

ICFO - The Institute of Photonic Sciences (www.icfo.eu) was created in 2002 by the regional government of Catalonia and the Technical University of Catalonia. The Institute was launched with the mission to become a world-leading research center in Photonics. ICFO currently hosts more than 250 researchers organized in 23 research groups working in 60 state-of-the-art research laboratories, equipped with the latest experimental facilities and supported by a range of cutting-edge facilities for nanofabrication, characterization, imaging and engineering. The Institute is located in a specially designed, 14.000 m2-building situated in the Mediterranean Technology Park in the metropolitan area of Barcelona. ICFO currently hosts 12 ICREA Research professors, 14 European Research Council projects and 6 Fundacio Cellex Barcelona Nest Fellows. ICFO researchers have already published some 1000 papers in the top topical journals and more than 30 papers by ICFO researchers have been published in journals of the Nature & Science groups during the last few years. The institute was given athe top worldwide position in the Physics category in the "Mapping Scientific Excellence" ranking published by the Max-Planck Society and ETH in 2013. On the industrial side, ICFO participates actively in the European Technological Platform Photonics21 and is also very proactive in fostering entrepreneurial activities and spin-off creation. The center participates in incubator activities and seeks to attract venture capital investment. ICFO hosts an active Corporate Liaison Program that aims at creating collaborations and links between industry and ICFO researchers. To date, ICFO has helped create 5 start-up companies.

The Optoelectronics group at ICFO, lead by the PI, studies and develops new advanced materials and devices for the photonics industry. These include ultrathin materials, such as metals, graphene and phase change compounds for transparent electrode and optical switching, nano-structured glass surfaces for antifingerprint, antimicrobial and microfluidic applications, and low cost integrated optical sensors for cell, micro-organism counting, sorting, and DNA sequencing. The group aims at bridging the academic and industrial worlds by pursuing fundamental ideas with great potential for revolutionizing future photonic products. It thus works closely with local and international industrial organizations to make sure that the society at large benefits from its research effort in the future, including major industries, in Spain, Europe and USA, including Corning Inc., Oclaro Inc., ABB, Ficosa International, IBM, SEAT/Wolkswagen, BASF, Prysmian, Carl Zeiss, Alter Technology, HP and the European Space Agency.

Main tasks

ICFO will be focus on tasks 2.1 nad 2.2, i.e. the development of a new prototype of Granulometer for the ProPAT project (ProG) adapted to specifications defined in WP1, based on previously tested Image Scatterometers prototypes made at ICFO, and the integration of this ProG prototype for real-time and in-line measurements during granulation in a real industrial environment.

ICFO will carry out the development of four units of a beta version of the novel ProG, focussing on the hardware

development including optics, electronics and mechanical parts, with the necessary modifications to each end-user requriments. These novel sensor concept will allow to achieve a low-cost and compact Granulometer for in-line measurements.

How the legal entity profile matches the tasks in the project

Among the current PI and his group's activities listed above, the most relevant for the proposed project concerns the development of lens-free microscopy based optical sensors. These rely on a CCD and CMOS detection and can be used for different applications, ranging from scattering measurements to material defects, micro-organism, cell analysis and counting, and DNA detection and sequencing. The group and the PI has produced 1 granted patent and 3 patent applications in this area, all relevant and part of the background IP for the project.

CV of the persons who will be primarily responsible for carrying out the work

VALERIO PRUNERI (male), the PI, is an ICREA Industrial Professor, Corning Inc. chair and group leader at ICFO. Prior to ICFO, from 2000 to 2006, he had worked as head of advanced research in industry for Avanex Corp. (Italian branch), Corning OTI and Pirelli Cables and Systems. From 1993 to 2000 he was with the Optoelectronics Research Centre, University of Southampton. He received his PhD in Laser Physics/Optoelectronics from the University of Southampton and his Master ("Laurea") in Nuclear Engineering "cum laude" from the Politecnico di Milano. Since 2010 he has spent visiting periods at Corning Inc. for more than a year. He has about 30 granted or pending patent families, 50 invited talks at major international conferences, and 250 refereed journal/conference papers. He has served on about 40 international conference committees; in particular, he will be Technical co-chair of the Conference on Lasers and Electro-optics (CLEO) Europe 2015 and has been technical/general chair of the European Conference on Integrated Optics (ECIO) 2012. He is an international R&D advisor of ACREO fiber optic centre in Sweden, member of the advisory board of VLC Photonics, technical advisor of Medlumics SL, chair of the photonics panel of the National Evaluation Agency in ICT and regular consultant for the EU commission and photonics industries. For his research and technology transfer effort, he received the Philip Morris Prize for Scientific and Technological Research, the Pirelli Research Fellowship, the IBM Faculty Award, the Duran Farell Prize for Technological Research and the Corning Inc. Chair.

DAVIDE JANNER is a senior researcher in the Optoelectronics group. He received his PhD in Physics from the Politecnico di Milano and his Master in Physics from Università degli Studi of Milano. His research interests cover both theoretical and experimental aspects in optical sensors based on micro-nano-structured fibers, photonic crystals, surface nano-structuring and optofluidics. He published more than 30 papers in international peer reviewed journals and gave 5 invited talks at major international conferences. He is co-author of 3 issued patents. His main contributions to the project are on optical modelling, optical design and optical characterization.

MARC JOFRE received the M.Sc. degree in electrical engineering (Telecommunication engineering) from the Technical
University of Catalonia (UPC), Barcelona, in 2008, and the PhD in Photonics from ICFO - The Institute of Photonic Sciences, in 2013. He acquired experience in optical engineering and system measurements by working on novel transmitters for quantum communication. His research interest includes optical communication and bio- sensors. His main contributions to the project include optical, electronic integration and engineering, in particular design and realization of the image sensor system.

PEDRO ANTONIO MARTÍNEZ CORDERO is research engineer at ICFO. He has more than 5 years of experience in the field of optoelectronics systems being previously head of R+D photonics department in a private company specialized in radio frequency and photonic devices. He has a large expertise in imaging systems, optical-electronic integration and image processing. He will be responsible for tasks related to software and hardware at the design and development stages. He will apply his expertise in software development focus on holographic image processing and statistics algorithms for analyse and counting particles.

DR. SILVIA CARRASCO. FEMALE. Knowledge and Technology Transfer Head (KTT), ICFO. She also serves as vicepresident of the Spanish mirror of the European Platform Photonics21 since February 2009 and is also a Member of the Board of SECPhO, the Southern European Cluster in Photonics and Optics since 2013. Carrasco received a BSc in Physics in 1999, a PhD with highest honors in 2003, an MBA in 2008 and was awarded a Fulbright Fellowship. She has enjoyed research stays at several institutions in the USA, including the Centre for Research and Education in Optics and Lasers (currently College of Optics), University of Central Florida, the Photonics Center, Boston University, and Harvard University, Boston. In her capacity as ICFO KTT director, she has a broad experience in IPR management. In particular she has built at ICFO a Corporate Liaison Program that serves as a bridge for ICFO and all types of industries and corporations to achieve common goals. Her experience in fostering collaborative projects between ICFO and all types of corporations includes the establishment of worldwide relationships with photonic manufactures, photonics distributors, and photonic users in different sectors. Since her arrival to ICFO in 2006 to manage the KTT unit, she has built a patent portfolio at ICFO of more than 50 patent families. She is also the founder of the "From Science to Business" Program that ICFO launched in 2008 jointly with ESADE

Business School to promote the entrepreneurial spirit among young scientists. Five start-up companies have been incubated and launched at ICFO based on property know-how: Radiant

Light, Signadyne, Cosingo, ProCareLight and HemoPhotonics.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Apparatus for measuring light scattering, V. Pruneri et al., patent application EP14382043 (2014)
2. Optically transparent support for bubble-free specimens, V. Pruneri et al., patent application P201331446 (2013)
3. Compact system for optical measurements using a discretized plate, V. Pruneri et al., patent application ES201330270 (2013)
4. Durable, superhydrophobic, antireflection and low haze glass surfaces using scalable metal dewetting nanostructuring, D. Infante, K. W. Koch, P. Mazumder, L. Tian, A. Carrilero, D. Tulli, D. Baker, V. Pruneri Nano Res. 6, 429-440 (2013)
5. Photonic crystal fiber sensor array based on modes overlapping, G. A. Cárdenas-Sevilla, V. Finazzi, J. Villatoro, V. Pruneri, Opt. Express 19, 7596-7602 (2011)

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. Multi sensor technology for management of food processes (MUSE-TECH)
2. Nano-structured optical surfaces for microelectronics, consumer electronics and biosensors (OPTICAL TOUCH) – La Caixa bank
3. Micro-nano-photonic devices for energy efficiency (MINDER) – TEC 14832 – Spanish Ministry of innovation 4. Superolephobic and antifingerprint nano-structured transparent surfaces (NANOSURF) – Corning Inc.

5. Microfluidic devices via femtosend laser writing (MINILAB) - Cellex Foundation

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

The Optoelectronics group has available three state-of-the-art laboratories, one of which fully dedicated to activities related to the project, i.e. lens-free microscopy for sensing in the material processing, biology and water industries. In addition to three own laboratories, the group can count on several central facilities. Particularly relevant for the projects are the nano-photonics and bio facilities. The Nanophotonics Fabrication Lab (NPL), which is devoted to the fabrication and characterization of nanostructured photonic devices. This laboratory includes state of the art nanofabrication tools and techniques for a wide range of optical applications, ranging from biosensing to optical communications, and spanning a broad range of materials, including polymers, semiconductors, metals, or crystals,among others. ICFO holds a total of 215m2 of space dedicated to BioFacilities (within a total of approximately 500m2 dedicated to life sciences related photonics research). These facilities are composed of BioSafety Level 1 (BSL1) and BSL2 labs that work with both pathogenic and non-pathogenic material. All the biofacilities (that are composed of Wet labs, Cell Culture areas, Bacteria and Model Organism Labs and all the associated supporting facilities) are located “door to door” with ICFO’s

Super-resolution Light Nanoscopy and Microscopy facility (SLN@ICFO). ICFO’s Biological Laboratories (BSL1 and BSL2) follow the European, National and Institutional regulations organized by the ICFO-Biosecurity Committee (IBC) and managed under an implemented Biosecurity Program. ICFO holds also a powerful programme on a variety of advanced and super-resolution light microscopy techniques and functional imaging with applications that range from single molecule to medical imaging. The Super-resolution Light Microscopy & Nanoscopy Lab at ICFO (SLN@ICFO) has as a mission to perform activities in three large pillars: user-oriented access, high quality comprehensive training, and research and development in the most advanced imaging techniques. The current equipment of SLN@BCN comprises custom-built as well as the latest commercially available microscopes and imaging devices. The currently available microscopy equipment at the core labs include a CW-STED microscope (LEICA) that has been modified to enhance its imaging capabilities with other nonlinear imaging techniques, a confocal microscope (NIKON E-TI) modified to work as a multimodal platform including several nonlinear imaging techniques., two home-made multiphoton microscopes (TPEF, SHG, THG, PSHG), a Raman microscope (Renishaw) and a home built multimodal SPIM microscope. Within this imaging facility ICFO has a library of antibodies for super-resolution available to the facility users.

Participant no.	3	**Name (short name)/Country**	VTT Technical Research Centre of Finland (VTT)/ Finland

Description of the legal entity
VTT Technical Research Centre of Finland (www.vtt.fi) is the largest multidisciplinary contract research organization in Northern Europe, with approx. 3100 staff and turnover of 292 million €. VTT is a part of the Finnish innovation system under the domain of the Ministry of Employment and the Economy, but gains a significant fraction of its income from contract research from the private sector. VTT is impartial non-profit expert organisation acting independently in respect of its customers. The objectives of VTT are to create high level scientific and techno-economic knowledge and knowhow and to generate technology and innovations for industry and society.
Main tasks
VTT's main role in this project is to MEMS process platforms development in WP3.3 and sensor test rig development in
WP3.4.
How the legal entity profile matches the tasks in the project
VTT is well known worldwide for innovative and state-of-the-art MEMS devices. We design, model, simulate, fabricate and characterize silicon-based components and readout electronics, focussing on surface- and SOI MEMS technologies. Core competences include also opto-electromechanical design, integration & execution of high complexity MEMS fabrication processes for various MOEMS devices, such as monolithic spectrometers and optical filter structures. We have successfully established commercial production pilot line of carbon dioxide sensors based on MOEMS FPI platform designed for the IR range. VTT has a vast experience of working in various EU projects as a partner and as a coordinator. In 2011 alone, VTT was involved in 322 EU-funded research projects. VTT is ranked among the leading European research and technology organisations and is committed to responsible collaboration for achieving the EU H2020 goals.
CV of the persons who will be primarily responsible for carrying out the work
DR. ANNA RISSANEN. Female. Research Team Leader of the VTT MOEMS and BioMEMS Instruments team and responsible for project management of this project. She has more than ten years of expertise in MEMS development, including optical and electrical design of MOEMS structures and process integration/development for surface micro machined Fabry-Perot interferometers. Her core competence is in the opto-electromechanical design, integration & execution of high complexity MEMS fabrication processes for optical sensors, micro-spectrometers and optical filter structures for MEMS hyperspectral imagers. **MIKKO TUOHINIEMI** has a M.Sc in Engineering Physics from the Aalto Univ. (formerly Helsinki Univ. of Technology). He will be responsible for the MEMS design and process integration in WP3. He worked as a MEMS designer 20032008 with Murata Electronics Oy (formerly VTI Technologies Oy) and has been with VTT in MEMS process development since then. Currently he is a senior scientist in the team for MOEMS and BioMEMS Instruments.
List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content
1. Antila, J., Tuohiniemi, M., Rissanen, A., Kantojärvi, U., Lahti, M., Viherkanto, K., Kaarre, M. and Malinen, J. 2014. MEMS- and MOEMS-Based Near-Infrared Spectrometers. Encyclopedia of Analytical Chemistry. 1–36.) 2. Blomberg, Martti; Torkkeli, Altti; Lehto, Ari; Helenelund, Christer; Viitasalo, Mauri; Electrically tuneable micromachined fabry-perot interferometer in gas analysis; Physica Scripta. Vol. T69 (1997), 119 - 121 3. Tuohiniemi, Mikko; Näsilä, Antti; Mäkynen, Jussi ; Characterization of the tuning performance of a micro-machined Fabry-Pérot interferometer for thermal infrared; Journal of Micromechanics and Microengineering. IOP Publishing. Vol. 23 (2013) No: 7, Article no. 075011 4. Tuohiniemi, Mikko; Blomberg, Martti; Akujärvi, Altti; Antila, Jarkko; Saari, Heikki; Optical transmission performance of a surface-micromachined Fabry-Pérot interferometer for thermal infrared ; Journal of Micromechanics and Microengineering. Vol. 22 (2012) No: 11, 115004

5. Rissanen, Anna; Kantojärvi, Uula; Blomberg, Martti; Antila, Jarkko; Eränen, Simo; Monolithically integrated microspectrometer-on-chip based on tunable visible light MEMS FPI; Sensors and Actuators, A: Physical . Elsevier.
Vol. 182 (2012) No: August, 130 – 135

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. IQFUEL: development of MEMS processing technology for NIR FPI for fuel quality sensing

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

VTT has two major cleanroom facilities for fabricating nanoelectronic devices and microsystems; at Micronova in Espoo and Micromodule Centre in Oulu. The former is an extensive wafer fab where silicon and glass wafers are processed into devices, the latter concentrates on system integration. Main cleanrooms cover a total area of 2 600 m². The cleanroom classification ranges from Class 10 (ISO 4) to 1000 (ISO 6). Micronova cleanrooms are defined as a National Research Infrastructure by the Ministry of Education, maintained by VTT and Helsinki University of Technology.

Participant no.	4	**Name (short name)/Country**	University of Leeds (UNIVLEEDS) / UK

Description of the legal entity

University / Higher Education Institute. Established in 2008, the Institute of Process Research and Development (iPRD) of the University of Leeds offers a unique set of facilities and expertise for product and process development. iPRD is comprised of experts from the fields of chemistry and chemical engineering who work closely with the chemical industries to develop technologies that can deliver cost reductions, quality benefits, increased productivity and reduce waste and energy utilisation in product manufacture. The team has over fifty year's combined experience of working in the fine chemical and pharmaceutical sectors and are able to offer companies focused contract-based services for problem solving, process understanding, development of new process technologies, small-scale manufacture, training and consultancy. iPRD works with companies of all sizes, including SMEs from the region. They are also actively

involved in regional and European funded projects.

Main tasks

UNIVLEEDS will contribute to WP1 supporting the definition of the specifications for the processes and sensors to be used in WP6 and to WP5 in supporting the specification and development of the control hardware, however WP6 is the main focus of UNIVLEEDS activity in ProPAT. In WP6, three types of processes will be implemented at an industrially relevant pilot scale: (Task 6.2) high temperature polymerisations, (Task 6.3) Wet and Dry milling processes and (Task 6.4) granulation processes. In each of these tasks the following activities will be undertaken: (i) process familiarisation in laboratory scale equipment, (ii) the design and construction of processing facilities at an appropriate pilot scale with locally designed control. In Task 6.1, the local control system will be coupled to the ProPAT control system and sensors. These will be tested initially with model systems and subsequently with industrial materials & processes in order to collect sensor calibration data under realistic processing conditions This data will improve the calibration models. Full automation of the processing equipment will be completed by use of the updated calibration models. The fully automated system will be tested using model systems, and on successful completion of these tests, industrial processes will be run at pilot scale in “closed loop” automation mode in order to demonstrate (a) validity of the ProPAT developed technology and control, (b) the robustness of the sensor calibration at scale, (c) the stability of the sense across multiple batches and for prolonged running times and (d) the environmental and economical benefits of the sensors (waste/energy reduction, selectivity and yield increases, cycle time reductions).
UNIVLEEDS will also contribute to WP7 (knowledge transfer) to ensure the operational learning from WP6 is effectively embedded in the industrial pilot studies executed in WP7.

How the legal entity profile matches the tasks in the project

UNIVLEEDS has the facilities and expertise to design, construct and automate batch and continuous chemical and particulate equipment required for this project. **Muller** has 17 years of industrial experience in process development and scale up and manufacturing. Since arriving at the university in 2011, he has developed a formulation process and scaled it to 20L scale manufacturing, and together with Alfa Laval designed and built an automated 5L scale slurry hydrogenation rig. He also has significant experience in software development. **Bourne**’s expertise lies in the development and operation of fully automated, self-optimising continuous processing systems coupled to inline analysis. **Hunter** has significant research experience in the use of sensors as online and in situ process control systems for suspensions of solids in liquids. **Hassanpour** has a wide range of expertise in the field of particulate processing. His interests lie in understanding the relationship between particle properties and bulk behaviour of particles and coupling that understanding to particle processes like size enlargement and size reduction of particles. He is an expert in the application of discrete element method (DEM) using combined continuum (CFD) and DEM techniques and coupling this to detailed experimentation. Our expertise in continuous processing is particularly well suited for testing control systems and sensors for longer duration.

CV of the persons who will be primarily responsible for carrying out the work

PROFESSOR DR. IR. FRANS MULLER PhD Cantab CEng FIChemE (*Male*) holds a joint chair in Chemical Process Engineering in the School of Process Environmental and Material Science (SPEME, ranked 3rd in its subject (RAE2008)) and the School of Chemistry (ranked 8th in its subject (RAE2008)). He works on the interface between chemical engineering and chemistry and his research interests lie in the dynamics of manufacturing processes and understanding of the interaction between process and equipment. His work is currently supported through a £1M strategic investment of the University of Leeds as part of the establishment of the Institute of Process Research and Development (iPRD) and directly from industry (AstraZeneca (AZ), Syngenta, Alfa Laval and Aesica). Before joining iPRD in August 2011, he spend 17 years in industry (AstraZeneca (AZ), Syngenta) as a process engineering scientist working on the design and development of industrial processes & equipment. His group includes 4 PhD students funded by Leeds and industry (AZ, Syngenta).

He was the lead investigator for AZ on a large EU consortium grant (F3 Factory, 19M€ EU funding, 25 partners involved in continuous mobile modular manufacture, www.f3factory.com), representing AZ's 5 academic partners on the consortium management team and leading the AZ project team. At Syngenta he developed and in 2000 subsequently commercialised a batch processing simulation program, known commercially as DynoChem, to a SME software company (Scale-up Systems Ltd, http://www.scale-up.com). Now over 450 DynoChem licences are used on ~100 sites in 23 countries by 13 of the top 15 pharmaceutical companies. Scale-up Systems employs 20 people in 5 countries and has an annual turnover of over €2M2. His research has resulted in over 26 archival journal and reviewed conference publications as well as 9 invited lectures and seminars.

Dr ALI HASSANPOUR (*Male*) is a lecturer at the Institute of Particle Science and Engineering (IPSE) of University of Leeds. He is also an active member of management committee for Particle Technology Special Interest Group of

IChemE and Process Engineering Formulation Design and Manufacturing focus group of Academy of Pharmaceutical Sciences (APS). Ali's research has mainly focused on experimental work and modelling in various areas of particle science and engineering. This includes (i) the characterisation of mechanical properties of particles and their correlation with bulk behaviour; (ii) size enlargement and size reduction of particles; (iii) tribo-electrification and electrostatic effects on particles (iv) modelling bulk behaviour of particles using discrete element method (DEM) and particle-fluid interactions in turbulent flows using combined continuum (CFD) and DEM techniques. His research has resulted in over 80 archival journal and reviewed conference publications as well as 12 invited lectures and seminars.

DR RICHARD BOURNE (Male) is a tenure track fellow in (SPEME) at the University of Leeds. He works on the interface of Chemistry and Chemical Engineering and is interested chemical processing, online analysis and automation. Of particular relevance to this project, RAB has demonstrated self-optimising flow reactors for rapid optimisation of chemical processes using inline gas chromatography, liquid chromatography and infrared spectroscopy. Prior to this position he worked at the University as Nottingham with Prof. Martyn Poliakoff, CBE, FRS and Prof. Michael W. George developing self-optimising continuous systems. He currently has an AstraZeneca funded industrial CASE award, HEIF funding and industry funding, total value ca. £100k. He has published 23 peer-reviewed journal and conference articles. **DR TIMOTHY HUNTER** (*Male*) is a lecturer in IPSE, at the University of Leeds. He has a general research interest in multiphase suspension and slurry processing, as well as interfacial colloidal and soft matter systems. He has current active collaborations with P & G, Sellafield Ltd, MMI Engineering, NNL Ltd, Unilever and Nestle. Of specific relevance to this project is his significant research experience looking at the application of commercial backscatter sensors for use as online and in situ process control systems, to monitor suspensions in tanks and pipeflow environments. Additionally, he has looked at novel applications of multiangle ellipsometers to measure the interfacial structures and contact angles of colloidal monolayers. He is currently Co-I (Co-Academic lead) on a ~£1.2M Technology Strategy Board grant with Sellafield Ltd and MMI Engineering, and has +25 peer reviewed journal and conference articles with >330 citations.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Muller F.L., Gray K.A., Robinson G.E. (2011). Understanding Scale Up and Quality Risks on the interface between Primary and Secondary Development. Pharmaceutical Process Scale-Up. 3rd edition edited by M. Levin. Informa Healthcare NY, Pharmaceutical Process Scale-Up, Third Edition, chapter 3. Feb 2011.
2. DynoChem (http://www.scale-up.com/) process modelling system. Muller was the inventor of this batch simulation program when at Syngenta AG. He oversaw the successful licensing of DynoChem to scale up systems. It is now used by 3 of the top 15 pharmaceutical companies.
3. Hassanpour, A., Kwan, C.C. Ng, B.H., Rahmanian, N., Ding, Y.L., Antony, S.J., Jia, X.D. Ghadiri, M., Effect of granulation scale-up on the structure and strength of granules, Powder Technology, 189 (2) 304-312 (2009).
4. Hunter, T.N., Darlison, L., Peakall, J., Biggs, S. "Using a multi-frequency acoustic backscatter system as an in situ high concentration dispersion monitor", Chemical Engineering Science 80, 409–418 (2012).
5. R. A. Skilton, A. J. Parrott, M. W. George, M. Poliakoff and R. A. Bourne, Real-Time Feedback Control Using Online Attenuated Total Reflection Fourier Transform Infrared (ATR FT-IR) Spectroscopy for Continuous Flow Optimization and Process Knowledge, Applied Spectroscopy, 2013, 67, 1127-1131.

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. F3 Factory project (EU FP7: 228867)
2. TSB *Developing the Nuclear Supply Chain* (2013-16) 'Measurement and modelling of sludge transfer processes'
   (Total project ~£1.18M of which University of Leeds received £440k). Project No 101433
3. TSB / Procter & Gamble: Creating high value compact products to increase total UK's total supply chain reach (TS/L00351/1)
4. Industrially funded by Alfa Laval: Slurry Hydrogenation case studies in flow (2013)
5. Various Company supported PhD projects: EPSRC/Johnson Matthey Industrial Case Award: Effect of Milling on the Catalytic Characteristics of Gamma Alumina;
   EPSRC/Astra Zeneca Industrial Case Award: Rate based design of experiments ;
   EPSRC/Astra Zeneca Industrial Case Award: Milling of Pharmaceutical Powders;
   EPSRC/Astra Zeneca Industrial

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

The Institute of Process Research and Development (iPRD) at the University of Leeds is jointly supported by the Schools of Chemistry and Chemical Engineering (SPEME) and has a process R&D and scale up facility designed to industrial standards. This was funded by a £4.5 M grant from Yorkshire Forward and the ERDF in 2008 and consists of 650m^2 lab and office space. The lab's centre-piece is a state of the art, 20L and 50L reactor facility, which is ATEX compliant (i.e. safe use of flammable solvents), that enables industrially relevant kilo-scale manufacture, development and demonstration of process and technology innovation. The reactors are fitted with process analytical technology (PAT) capable of all standard batch reaction operations such as charging liquids and solids, heating, cooling and temperature control, distilling (atmospheric and under vac), extraction and phase separation, crystallisation and filtration. Typically the 20L vessel is used as reactor with the 50L vessel being used for work up.

In addition to the batch processing facilities iPRD operates a number of continuous reactors: (i) a Alfa Laval ART reactor (120 mL single pass heat exchanger now marketed by Ehrfelt), (ii) a Coflore reactor (1L reaction volume in 10 stirred units in series, AM technology) and (iii) a CSTR in series reaction system (Parr, 100 Bar). These equipment can be operated under pressure, and we have successfully conducted the scale up of a range of chemistries, including continuous hydrogenations and high temperature chemistry (140C). In collaboration with Alfalaval we have designed a modular slurry hydrogenation facility which Alfa Laval has donated iPRD. The modular approach builds on the F3 factories concept of plug and produce, and we will use the existing modules to support the work in WP6.

Our state of the art facility is operated by world class academic and support staff that at all levels combine relevant industrial experience with a strong research capability. This skill base provide us

with industrial quality project management that ensures delivery of products without compromising safety. In terms of academic capability, iPRD has significant expertise in all aspects of process development: from synthesis design, via scale up and reaction engineering, crystallisation, particle engineering to product design and manufacture and the control associated with these processes.
The Institute of Particle Science and Engineering (IPSE) is one of 3 major research institutes in SPEME and works in close collaboration with iPRD through a number of joint research projects. Research at IPSE is predominantly focused on the measurement, modelling and processing of particulate solids and multiphase systems in various sectors, e.g. pharmaceutical, food, detergent, agricultural products and nuclear engineering. The facilities at IPSE that are relevant to this project include batch granulators (medium and high) and various types of particle mills such as ball mill, bead mill, planetary ball mill, disk mill and air jet mill. Also numerous facilities are available at IPSE for characterisation of physical and mechanical properties of particles (e.g. Atomic Force Microscopy, Electron Microscopy, x-ray microtomography, nano-indentation, dry and wet particle sizers, etc.).

Participant no.	5	**Name (short name)/Country**	Universitat Rovira i Virgili (URV)/Spain

Description of the legal entity

Universitat Rovira i Virgili (URV) was created in 1991 as a public institution orientated to place knowledge at the service of society so as to contribute to the social and economic development of the region. The URV trains professionals under the precepts of the European harmonization, a key factor in order to open the doors to a professional world without frontiers, by providing 52 programs of study in a wide variety of knowledge areas for over 13.000 students. In 2011, URV was ranked the third University in Spain in terms of scientific productivity.

The staff involved in the project belongs to the “Chemometrics, Qualimetrics and Nanosensors” research group, at the department of Analytical Chemistry and Organic Chemistry at the Faculty of Chemistry of the URV. The Faculty of Chemistry is one of the pioneering centres at the URV and, in recent years, it has been consolidated at all levels and is now a recognized centre for quality in teaching, research and knowledge transfer. According to the Shanghai ranking, URV has ranked among the 200 world's leading universities in the field of Chemistry.

Main tasks

Calculation, validation, improvement and transfer of multivariate calibration models. Software development in Matlab language of the above mentioned tasks.

How the legal entity profile matches the tasks in the project

This partner has extensive knowledge in multivariate modelling such as the one required in this project. The two main members of the group have been teaching chemometrics at the university since 1994 and have directed several doctoral thesis about this subject. Their research is focussed in the development and application of multivariate and multiway chemometrics techniques to solve classification and prediction problems. Matlab subroutines for chemometric analysis developed in this group over the years can serve as a basis for developing some of the software required in this project.

CV of the persons who will be primarily responsible for carrying out the work

Dr. Joan Ferré (male) is Associate Professor at the Department of Analytical Chemistry and Organic Chemistry of the Universitat Rovira i Virgili. His main expertise is chemometrics (especially experimental design, multivariate classification and multivariate calibration) and Matlab programming. He is author of about 50 scientific papers in international journals and has supervised 5 doctoral theses. He has participated in seven competitive Spanish research projects and three European projects. Knowledge-transfer to companies is carried out with regular participation in in-company chemometrics training courses and R+D contracts.

Dr. Ricard Boqué (male) is Associate Professor at the Department of Analytical Chemistry and Organic Chemistry of the Universitat Rovira i Virgili, and has an extended experience (both research and teaching) in the area of analytical chemistry, and specifically in chemometrics and qualimetrics.

He has published about 70 scientific papers in international journals. He has participated in three European projects. His main research interests include qualimetrics and multivariate calibration and classification methods.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. R. Boqué and A.K. Smilde. Monitoring and diagnosing batch processes with multiway covariates regression models. AIChE Journal, 47 (1999) 1504-1520.
2. Santiago Macho, Ricard Boqué, Mª Soledad Larrechi, F. Xavier Rius. Multivariate determination of several compositional parameters related to the content of hydrocarbon in naphta by MIR spectroscopy. The Analyst, 124 (1999) 1827-1831.
3. Néstor F. Pérez, Ricard Boqué, Joan Ferré. Establishment of multivariate specifications for food commodities with Discriminant Partial Least Squares. Talanta, 83 (2010) 475–481.
4. Eva Borràs, José Manuel Amigo, Frans van den Berg, Ricard Boqué, Olga Busto. Fast and robust discrimination of almonds (Prunus amygdalus) with respect to their bitterness by using near infrared and partial least squaresdiscriminant analysis. Food Chemistry, 153 (2014) 15-19.
5. A. C. Olivieri, N. M. Faber, J. Ferré, R. Boqué, J. H. Kalivas, H. Mark. Uncertainty estimation and figures of merit for multivariate calibration (IUPAC Technical Report) Pure and Applied Chemistry 78 (2006) 94-106

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. Robust process analytical methods for industrial practice and their validation (SMT4-CT95-2031).
2. European Network for the Intercomparision of Chemometric Software and Methods (SMT4-CT98-7521).
3. VICIM. Virtual Institute for Chemometrics and Industrial Metrology (G7RT-CT-2001-05067).
4. TRACE. Tracing Food Commodities in Europe (FP6-FOOD-2004-006942).
5. Chemometric maintenance of on-line NIR analyzer technology (contract, 2001-2014). REPSOL refinery, Spain.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

- Analytical instruments in our own laboratory, including NIR, MIR (one of them portable) and UV-Vis spectrophotometers, EEM spectrofluorimeter, headspace-mass spectrometer, GC-MS and LC-DAD.
- Programming capabilities with Matlab software, for development of chemometric data analysis algorithms.

Participant no.	6	**Name (short name)/Country**	Chahan YERETZIAN / Switzerland
Description of the legal entity			

Zurich University of Applied Sciences in Wädenswil (ZHAW) is one of the largest and most productive universities of applied sciences in Switzerland, with a strong presence both nationally and internationally. Among the ZHAW's distinguishing features are interdisciplinarity and a close connection to everyday working practice. About 9'700 students are enrolled at the ZHAW in a total of 26 Bachelor's degree programmes and 13 consecutive Master's programmes. The university also offers a broad range of continuing education courses, including over 30 Master's of Advanced Studies programmes. 30 institutes at the ZHAW network their resources to provide business and industry with customised services. Research at the ZHAW is led by three key principles:

Life Sciences – a key field of knowledge in the 21st century: Life Sciences are key fields of activity for the 21st century, demanding interdisciplinary thinking and action. In response to this challenge, the School of Life Sciences provides opportunities for networking – from biotechnological, chemical and food technology related issues to logistical, ecological and management aspects – and carries out complex projects using an interdisciplinary approach.

Infrastructure and human resources: Depending on the particular challenges of a project, researchers, research assistants and students from various disciplines work together in interdisciplinary teams. The ZHAW's equipment and apparatus are state-of-the-art and allow professional processing of application oriented research and development projects. Modern laboratories as well as testing and production facilities provide a solid basis for this.

Cooperation with industry and business: At the interface between theory and practice, the Institutes work closely with industry and business. Expertise and infrastructure are made available in accordance to clients' needs.

Main tasks

One key responsibility of the Zurich University of Applied Sciences is to provide and a range of analytical technologies and competencies for determine different product compositions or polymerization aspects. Furthermore, it will apply it competencies on LCA on the evaluation of processes and products within the frame of the project.

How the legal entity profile matches the tasks in the project

Their research group is specialized in the on-line, and time resolved analysis of volatile organic compounds at very high time resolution, mass-resolution and sensitivity.

CV of the persons who will be primarily responsible for carrying out the work

PROF. DR. MBA CHAHAN YERETZIAN. Received in 1991 his PhD in Chemistry from the University of
Bern/Switzerland for Experimental *Studies on Metallic Nano-Particles in Molecular Beams by Photo and Laser*
Spectroscopy. A post-doc at the University of California, Los Angeles (UCLA), from 1991 to 93 on Fullerenes, and an Alexander von Humboldt Junior Award for a 2-year position at the Technical University of Munich, from 1993 to 95 on ZEKE spectroscopy complemented his academic education. In 1995 he join the Nestlé Research Center in Lausanne, Switzerland, where he held various research and R&D management positions, mainly in the product categories Coffee and Culinary. During this period (2003-2004) he obtained a MBA degree from the University of Lausanne / Switzerland.

In 2007 he joined the Corporate Communication Department of Nestlé Nespresso, developing communication platforms on the science of coffee. Since 2008, he holds the position of head of the Analytical and Physical Technology Unit at the Institute of Chemistry and Biological Chemistry (ICBC) of the Zurich University of Applied Sciences. His current research is focused on the science and technology of coffee, the development of novel on-line analytics for volatiles, the application of state-of-the-art chromatography, and on applications of chemometry in flavour science and technology. In 2012 he was appointed Professor and Head of the center for Analytical and Physical Chemistry at the ZHAW, in Wädenswil. He has published over 80 peer reviewed papers, among others as first author in NATURE and SCIENCE. Prof. Yeretzian is an internationally renowned coffee expert. He collaborates and consults with a large number of coffee companies and initiated the first post-graduate degree in the "Science and Technology of Coffee". Since 2009 he is a Board Member of the Swiss Section of the Specialty Coffee Associations of Europe (Swiss-SCAE).

MSC. MATTHIAS STUCKI received his Master of Science Degree in Environmental Sciences from ETH Zurich in 2007 and is an expert in the field of Life Cycle Assessment (LCA). Between 2008 and 2012 he worked as a project leader at ESU-services Ltd. where he conducted a range of LCA projects in the various fields such as power generation, solar energy, bioenergy, geosynthetic materials, biogenic plastics, leisure activities, consumption patterns, sanitary systems etc. He published his findings in scientific journals such as the International Journal of LCA and submitted datasets to the worldwide ecoinvent database. Affiliated with the ecoinvent Centre, he started the Life Cycle Inventory Network in India and trained Indian engineers in conducting LCA projects. Since 2012 he holds a position as a Research Associate at the Institute of Natural Resource Sciences of the Zurich University of Applied Sciences, where he is responsible for research and teaching in the field of LCA. Currently, he is engaged in national and international projects that analyse the environmental impacts of food production (including dairy), sewage treatment, and power generation. He is a reviewer at the Journal of Industrial Ecology, member of the editorial board of the ecoinvent database, and an individual verifier at the International EPD® System.

DR. IVANA KROSLAKOVA. Female. Specialist in analytical chemistry having 10 years of experience in mass spectrometry. In 2006 she received her PhD degree at ETH Zürich for fundamental studies on matrix effects occurring in inductively coupled plasma using laser ablation as sampling technique. Since 2007 she works at ZHAW where she has built-up a Matrix Assisted Laser Desorption Ionization - Mass Spectrometry (MALDI –MS) laboratory and established R&D collaborations involving application of this technology. Her main engagement is applied research. She has participated on various research projects and possesses an expertise in analysis of wide spectrum of samples and materials including biological macromolecules, microorganisms' identification using MALDI-MS as well as synthetic polymers. As an SNF project leader she was responsible for characterization of natural polymers extracted from bark of soft wood species.

DR. SEBASTIAN OPITZ performed his doctoral thesis at the department of Chemical Ecology, Bielefeld University on the analysis of secondary plant metabolites (e.g. glucosinolates, iridoid glucosides and terpenoids), which have ecological functions in plant-insect interactions. He has a strong background and expertise in:

- The application of different chromatographic systems like GC-MS, UHPLC-TOF MS, HPLCMS/MS, HPLC-DAD, radio-HPLC, ion-chromatography, HPLC-SEC, HPLC-GPC.
- Semi-preparative chromatography, purification and identification (NMR) of metabolites.
- Experimental design of experiments, data management, statistics and behavioural bioassays.

He will be responsible for the all chromatographic analysis to be conducted in the frame of this project.

DR. PETER LIENEMANN obtained in 1986 his PhD degree from the ETH of Zurich on ICP-OES. From 1986 to 1989 he headed the group on X-Ray spectroscopy of the Swiss Federal Laboratories for Materials Science and Technology (EMPA). In 1989 he became the deputy head of the analytical laboratories of the EMPA, where is built up the interdisciplinary analytical laboratory on material-science, with a focus on metallic and ceramic materials, construction material and mineral and rocks. In 2009 he joined the Zürich University of Applied Science to head the group on elemental analysis. He is renowned expert in elemental analysis and is the authored, among others of significant textbook on elemental analysis (N. Kläntschi, **P. Lienemann**, P. Richner und H. Vonmont, Elementanalytik, Instrumenteller Nachweis und Bestimmung von Elementen und deren Verbindungen. SPEKTRUM ISBN 3-86025-134-1 (1996).

MS CARLA KAPIKIAN. FEMALE. earned in 1988 a Bachelor of Science Degree in Mathematics from the Massachusetts Institute of Technology (MIT) and in 1993 a Master in Business Administration (MBA) from the Harvard Graduate School of Business Administration, Boston, Massachusetts. She worked in a large range of small, medium large corporations (e.g. Andersen Consulting as Management Consultant, Chase Manhattan Bank as Assistant Vice President (1995-1996) and Assistant Treasurer (1993-1995), Young & Rubicam Brands as Associate Director and later Director (1997-2005) and at HRA Research as Director (2006 – 2008)) before moving in 2008 to Switzerland. Today she is working as Senior Academic Research and Development Assistant at the Luzern University of Applied Sciences and
Arts.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Franco BIASIOLI, Chahan YERETZIAN, Flavia GASPERI, Tilmann D. MÄRK, ***PTR-MS monitoring of VOC and BVOC in food science and technology***, Trends in Analytical Chemistry (TrAC), **30(7)**, 968-977 (2011)
2. Franco BIASIOLI, Chahan YERETZIAN, J. Dewulf, H. Van LANGENHOVE, Tilmann D. MÄRK, ***Direct injection mass spectrometry (DIMS): adding the time dimension to (B)VOC analysis***, Trends in Analytical Chemistry (TrAC),**30(7)**, 1003-1017 (2011).
3. Flurin WIELAND, Alexia N. GLOESS, Marco KELLER, Andreas WETZEL, Stefan SCHENKER, Chahan YERETZIAN, ***OnLine Monitoring of Coffee Roasting by Proton-Transfer-Reaction Time-of-Flight Mass-Spectrometry (PTRToF-MS): Towards a Real-Time Process Control for a Consistent Roast Profile,*** Analyt. Bioanalyt. Chem 402, 2531–2543 (2012)
4. Christian LINDINGER, Philippe POLLIEN, Santo ALI, Chahan YERETZIAN, Imre BLANK, Tilmann MÄRK, ***Unambiguous Identification of Volatile Organic Compounds by Proton-Transfer-Reaction Mass-Spectrometry (PTR-MS) Coupled with GC-MS***, Anal. Chem., 77, 4117-4124 (2005
5. Philippe POLLIEN, Christian LINDINGER, Chahan YERETZIAN, Imre BLANK, ***Proton Transfer Reaction Mass Spectrometry, a Tool for On-Line Monitoring of Acrylamide Formation in the Headspace of Maillard Reaction Systems and Processed Food***, Anal. Chem, **75(20)**, 5488-5494 (2003)
6. Jungbluth, N., Büsser, S., Frischknecht, R., Flury, K., & **Stucki, M**. (2012). Feasibility of environmental product information based on life cycle thinking and recommendations for Switzerland**. Journal of Cleaner Production**, 28(0), 187-197
7. Faist Emmenegger, M., **Stucki, M**., & Hermle, S. (2012). LCA of energetic biomass utilization: Actual projects and new developments, April 23, 2012, Berne, Switzerland. **The International Journal of Life Cycle Assessment**, 17(9).
8. Frischknecht, R., & **Stucki, M**. (2010). Scope-dependent modelling of electricity supply in life cycle assessments. **The International Journal of Life Cycle Assessment**, 15(8), 806-816.

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. Nose-space analysis: Chahan YERETZIAN, Philippe POLLIEN, Christian LINDINGER, Santo ALI, ***Individualization of Flavor Preferences: Toward a Consumer-centric and Individualized Aroma Science***, Comprehensive Reviews in Food Science and Food Safety (CRFSFS), 3(24), 1-8 (2004
2. Monitoring and control of coffee roasting
3. Analysis of air quality inside building and airplanes
4. Analysis of the extraction dynamics of coffee
5. On-line, time resolved analysis of formation of organic compounds during thermal treatment (flavour compounds, acrylamide…).
6. Doublet G., Jungbluth N., Flury K., **Stucki M.** and Schori S. (2013) Life cycle assessment of Romanian beef and dairy products. SENSE - Harmonised Environmental Sustanainability in the European food and drink chain, Seventh Framework Programme: Project no. 288974. Funded by EC. Deliverable D 2.1
7. Doublet G., Jungbluth N., Flury K., **Stucki M.** and Schori S. (2013) Life cycle assessment of orange juice. SENSE - Harmonised Environmental Sustanainability in the European food and drink chain, Seventh Framework Programme: Project no. 288974. Funded by EC. Deliverable D 2.1

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

- 5 fully equipped state-of-the-art laboratories
- Gas Chromatography - Mass Spectrometry (GC-MS)
- High Performance Thin Layer Chromatography (HPTLC)
- High Performance Size Exclusion Chromtography (HP-SEC)
- Matrix-Assisted Time-of-Flight Mass Spectrometry (MALDI TOF-MS)
- Proton-Transfer-Reaction Time-of-Flight Mass Spectrometry (PTR TOF-MS)
- Ultra Performance Liquid Chromatography - Mass Spectrometry (UPLC-MS)
- Elemental Analytics (XRF, ICP OES, AAS / AES)
- Optical Spectroscopy (FTIR, UV, Fluorescence)
- Confocales Raman spectroscopy
- Flow Injection Analysis (FIA)
- Software and Databases for Life Cycle Assessment, e.g. SimaPro 8 and ecoinvent v3.0

Participant no.	7	**Name (short name)/Country**	Technische Universitaet Dresden (TUD)/ Germany
Description of the legal entity			

The TU Dresden is one of eleven German universities that were identified by the German government as an 'excellence university'. TUD has about 36.500 students and almost 5319 employees, 507 professors among them, and, thus, is the largest university in Saxony, today. TU Dresden is strong in research, offering first-rate programmes with an overwhelming diversity, with close ties to culture, industry and society. As a modern full-status university with 14 departments it offers a wide academic range making it one of a very few in Germany. The TUD has extensive experience in project co-ordination and project management at national and international level.
The Mechanical Process Engineering department of TUD is widely acknowledged for its expertise in particle systems characterisation. Currently, there are approximately 12 scientists working on a variety particle technology projects. The department has a focus on nanoparticle suspensions and aerosol as well as on the development of particle sensors for submicron and micrometre particles. It further provides technology and metrology solutions to German industry and gives strategic input to ISO TC 24/SC4 and to the German DIN committees. The department includes the Particle Technology Centre – an excellently equipped laboratory for particle system characterisation, which has evolved to renowned place for teaching young and experienced researchers.

Main tasks

TUD will lead Task 2.3 Off line calibration of the sensor vs. standard measurements (granulometric sensor) and contribute to the other tasks in this WP in view of their outstanding knowledge regarding particle size measurements. They will also contribute to specifications in WP1, support for test at scale pilot scale WP6, industrial scale WP7 and then for the final comparison of ProPAT vs. existing quality control and recommendations for standardization (granulometric sensor) in WP8.

How the legal entity profile matches the tasks in the project

The mission of the Particle Technology Centre in Dresden is the process -relevant Particle Characterization including selection and optimization of the measuring methods and their validation matches WP2 "Set up and calibration of the novel granulometric analyser". Experience with the project leadership in several work items for international standardisation in CEN (EC-mandate) and ISO supports the WP8 "Demonstration"

CV of the persons who will be primarily responsible for carrying out the work

MICHAEL STINTZ, PROF. DR.-ING. HABIL. (male), Head of the Research Group Mechanical Process Engineering,
Institute for Process Engineering and Environmental Technology, Technische Universität Dresden.
Education and
Qualifications- 2005 Habilitation in the field of particle technology at the Faculty of Mechanical Engineering at the TU
Dresden, thesis: "Technologically relevant characterisation of particles and particle systems"; 1985 PhD thesis on the
"Combination of multi-channel analysis with different signal transducers"; 1977-1982 Study of Process Engineering, Technical University Dresden. Work experience- Since 2005, Head of the Research Group Mechanical Process Engineering, Institute for Process Engineering and Environmental Technology; 1993-2005: Senior associated scientist at the Chair of Mechanicla Process Engineering Institute for Process Engineering and Environmental Technology; since1985:Head of the Centre of Particle Technology (Zentrum für Partikeltechnik / Granulometrie) – an application oriented laboratory for the characterisation of particle systems.
Memberships
Since 2008-OECD Working Party on Manufactured Nanomaterials, Steering Group 8: "Exposure Measurement and Exposure Mitigation"
Since 2006- ISO Technical Committee TC 229, "Nanotechnology", Liaison officer TC24/SC4
Since 2006 - DIN Normenausschuss 062-08-17 AA "Nanotechnologien" des Normenausschusses Materialprüfung
Since 1996- ISO Technical Committee TC 24, Subcommittee SC4 "Particle Characterization", (since 2002 Convenor of WG1 "Representation of analysis data")
Since 1996- DIN Arbeitsausschuss 11.42.00 "Partikelmesstechnik" des Normenaus-schusses Bauwesen, (since 2001 Obmann des Arbeitsausschusses)
Since 1992- Assigned member in the scientific comittee "Particle Sizing" of the VDI-Gesellschaft Verfahrenstechnik und Chemieingenieurwesen GVC, since 2009 "Processnet" by GVC and DECHEMA
Latest international Standardization Project leaderships
ISO TC 229, "Nanotechnology", Project leader of WG2, PG10: ISO TS 12025: 2012, Nanotechnologies —
Quantification of nano-object release from powders by generation of aerosols",
ISO TC 24, SC4 "Particle Characterization", since 2002 Convenor of WG1: ISO 26824:2012 Particle characterization of particulate systems – Vocabulary

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. L. Hillemann, A. Zschoppe, R. Caldow, G. Sem, A. Wiedensohler, Ultrafine particle monitor for size-resolved number concentration measurements in atmospheric aerosols. J. Aerosol Sci., 68 (2014), 14-24. doi:10.1016/j.jaerosci.2013.10.007
2. R. Weser, S. Wöckel, B. Wessely, U. Hempel, Particle characterisation in highly concentrated dispersions using ultrasonic backscattering method. Ultrasonics, 53 (2013) 3, 706-716. doi:10.1016/j.ultras.2012.10.013
3. F. Babick, S. Gropp, U. Kätzel, M. Vorbau: Dynamic light scattering of dispersed fumed silica aggregates. Powder Technol., 217 (2012), 39-45. doi:10.1016/j.powtec.2011.10.064
4. F. Babick, K. Schießl, M. Stintz, Characterization of pyrogenic powders with conventional particle sizing technique: I. Prediction of measured size distributions. Part. Part. Syst. Charact., 29 (2012) 2, 104-115. doi:10.1002/ppsc.201000024
5. O. Ruscitti, R. Franke, H. Hahn, F. Babick, T. Richter, M. Stintz, Application of Particle Measurement Technology in Process Intensification. Chem. Eng. Technol., 31 (2008) 2, 270-277. doi:10.1002/ceat.200700465

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. NanoDefine, "Development of an integrated approach based on validated and standardized methods to support the implementation of the EC recommendation for a definition of nanomaterial' – one of the three leading research partners, responsible for WP4 "Screening of particle sizing methods"
2. NanoSTAIR, member of the IAB for support the standardization progress in the result of European research projects
3. Nanofutures (ETP) leader of Working Group "Standardization" – coordination with industrial safety research and regulatory issues.
4. The Value4Nano project has as its central aim the valorisation and elaboration of four key European **v**alue **c**hains which utilise nanomaterials. Especially VC3 - Manufacturing of powders made of functional alloys, ceramics and intermetallics matches with this project.
5. CoNanoMet – successfully finished European Project on Nanometrology development and standardization.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

The Particle Technology Centre is equipped with state of the art measurement and characterisation facilities for particle measurement in aerosols and suspensions, like SEM with cryo-preparation, SMPS, FMPS, CPC, DLS, LD and LS, analytical lab centrifuges, utrasonic spectroscopy, ultrasonic dispersion devices, powder aerosol generators, laminar flow bench. Metrology projects range from workplace exposure measurement to slurry quality assurance of chemical mechanical planarisation of wafers.

Participant no.	8	**Name (short name)/Country**	Universitat de Barcelona (UB)/ Spain

Description of the legal entity

The University of Barcelona (UB) was founded in 1450. Today it boasts of a student body of 63,020 and a research staff of 5,312 members. Degrees are offered in 73 different areas of teaching with numerous postgraduate and doctorate programs as well as continuing education courses. The University of Barcelona (UB) is the largest of the six universities of Barcelona and of the ten in Catalonia. The UB is ranked the first Spanish university, and the twenty third European institution, in scientific quality and productivity. The UB manages in average 150 European projects per year, for an amount of about 8.6 million euro a year. The University of Barcelona develops all its activities related to the research in Europe through its European Research Projects Office, a unit in charge of the promotion of EC programmes, technical assistance to the elaboration of proposals and management of projects with participation of researchers of the UB, coordinated by the Vice-rector for Innovation and Transfer of Knowledge.
The Fundació Bosch i Gimpera (FBG) is a third party of the Universitat de Barcelona (UB). This foundation is controlled by the UB, and is in charge of the administrative and financial management of the European projects in which the UB is involved. The FBG is not a subcontractor, since it belongs to the UB Group.

Main tasks

Main tasks involving the UB group will be related to provide process modelling tools (hard- and soft-modeling) to handle the response of NIR sensors (see WP3, task 3.6), to propose data fusion strategies combining spectroscopic responses with smart sensor responses or, eventually, scatterometer multivariate outputs (WP4, task 4.3) and to assess in the design of a decision support system for the Global Control System Software, based on the combination of the different responses obtained from sensors and platforms and acquired during the different stages of the process (WP5, task 5.2).

How the legal entity profile matches the tasks in the project

University of Barcelona is a large institution with a lot of human and material resources dedicated to research. As mentioned above, UB researchers are commonly involved in EU research projects, as leading recipients or collaborating partners, which prove their scientific qualification. Some outstanding instrumental facilities are also owned by the institution, such as the Centres Científics i Tecnològics de la UB (CCiTUB, http://www.ccit.ub.edu/EN/home.html), which provide a wealth of techniques that can be used for characterization of materials and processes if required. Within the UB, the Chemometrics group at the department of Analytical Chemistry of the faculty of chemistry is well known for the development of chemometric tools for process understanding (modelling and control), notably soft-modelling Multivariate Curve Resolution (MCR). In this sense, stable collaborations with related research groups, such as chemometrics at
IDAEA-CSIC, Spain) headed by Romà Tauler, leader in soft-modeling methodologies or Marcel Maeder, from the

University of Newcastle (Australia) leader in process hard-modeling are invaluable. Process analysis is now one of the core research lines of a national funded research project (see list of projects below).

CV of the persons who will be primarily responsible for carrying out the work

ANNA DE JUAN CAPDEVILA. Female. Associate professor of the Analytical Chemistry Department at the Universitat de Barcelona, will be the main responsible from this institution in the project. She is devoted to research in Chemometrics since more than 20 years. Since 2002 she is member of the Editorial Advisory Board of *Chemometrics and Intelligent Laboratory systems and since 2006 of Analytica Chimica Acta. Recently has acted as editor of the linear soft-modeling section for the reference work Comprehensive Chemometrics, Elsevier (2009).* In 2004 she received the 4th Chemometrics Elsevier Award together with Karl Booksh. She has published around 100 papers in international journals and books and has given more than 100 presentations in different international conferences, basically on multivariate curve resolution developments and related methods and on applications to process analysis, hyperspectral image analysis and general analytical applications. Recently she has an active commitment in the organization of PATrelated activities, such as the European Conference of Process Analytics and Control Technologies (EuroPACT) and specialized schools and activities of knowledge transfer.
The participation of a new PhD student fully devoted to develop tasks related to this project is envisioned.
Minor collaborations can also be done by:
SÍLVIA MAS GARCÍA, post-doctoral researcher since 2013, has experience in environmental process modelling (data fusion coupling different techniques and experiments) and has collaborated with foreign research institutions, like CEMAGREF (Montpellier, France) adapting hybrid methodologies for process modelling monitored by NIR.
SARA PIQUERAS SOLSONA is ending a PhD project on hyperspectral image analysis and has done research stages at the laboratory of Raman and IR spectroscopies in Lille (France) and at the Institute of Photonic Sciences (Jena, Germany). She has been working on handling image data sets coming from different platforms.
The collaboration of expert external researchers oriented to process analysis, such as **PROF. MARCEL MAEDER**, from the University of Newcastle (Australia), a worldwide recognized leader on hard-modelling process analysis, is also considered to enhance the value of the project.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Jaumot J, Igne B, Anderson CA, Drennen JK, De Juan A. Blending process modeling and control by multivariate curve resolution. Talanta, (2013) 117: 492-504.
2. A. de Juan, M. Maeder, S.C. Rutan and R. Tauler. Multivariate Curve Resolution (MCR) chapters in the reference work 'Comprehensive chemometrics' (eds. S. Brown, R. Tauler, B. Walczak), Elsevier (2009), pp. pp 249-260,325344,473-506.
3. A. de Juan, M. Maeder, M. Martínez, R. Tauler. Combining hard- and soft-modeling to solve kinetic problems. Chemometrics and Intelligent Laboratory Systems, 54 (2000) 123-141.
4. Plenary lecture in 5th Pan-European QbD & PAT Science conference (EuPAT 2012, Ghent, Belgium): de Juan, A.; Jaumot, J.; Mas, S.; Piqueras, S.; Duponchel, L; Igne, B; Anderson, C.; Tauler, R. 'Chemometric tools for process understanding'. Similar invited talks in process analysis conferences, such as EuroPACT 2008, APACT 2009.

5. Web page with free downloadable soft-modelling tools (MCR): www.mcrals.info, shared with the group at IDAEACSIC (Barcelona, Spain).

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. Coordinator of project: 'Development and application of chemometric tools for process and hyperspectral image analysis' (Spanish grant: CTQ2012-38616) 2013-2015.
2. Chair of the organizing committee of the 3rd European Conference on Process Analytics and Control Technologies (EuroPACT 2014) held in Barcelona, 6-9 may 2014.
3. Coordinator of the recognized research group: Development and transfer of multivariate resolution chemometric methodologies' (Catalan grant: SGR2009 0045). 2009-2013. Web page: www.chemometricsbcn.info
4. Coordinator of the thematic network of Chemometrics, formed by 21 Spanish chemometric groups. (Spanish grants: CTQ2010-09864-E, CTQ2007-28940-E/BQU). 2008-present.
5. Organizer of the school devoted to PAT knowledge transfer in Spain '*Herramientas quimiométricas para PAT'* chemometric tools for PAT) celebrated in Barcelona, 10-12 July 2012.

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Although performing measurements is not the main task of the UB partner, we want to mention again the facilities of the Centres Científics i Tecnològics de la UB (CCiTUB, http://www.ccit.ub.edu/EN/home.html), which could eventually be used to perform complementary measurements of physicochemical nature required by the project.

Participant no.	9	**Name (short name)/Country**	Institute of Technology Tralee (ITT)/Ireland

Description of the legal entity

The IMaR (Intelligent Mechatronics and RFID) Technology Gateway is part of the Institute of Technology Tralee in Ireland.
IT Tralee is a university level institution with a range of programmes from craft to postgraduate level. ITT has over 3,500 full-time and part-time students and 250 staff. Established in 1977, the Institute has operated autonomously since 1993. It provides education and training, facilitates industrial research and consultancy, and is a major engine of economic and social development in the region. The majority of the research conducted at the Institute is collaborative and focused on delivering innovative solutions to our industrial partners to promote cost competitiveness. The prioritised research themes at the IT, Tralee build upon the Institute's established and emerging strengths and directly align with national research priority areas and the Innovation Union flagship initiatives of the Europe 2020 strategy for a smart, sustainable and inclusive economy. Embedded at the heart of the Institutes is the IMaR research Center, which will perform the key requirements of the ProPAT initiative while also receiving administration support centrally from the institute.

Main tasks

IMAR will lead WP4 whereby their main tasks will include the further development of sensors from commercial sources being strategically conditioned and tailored to provide affordable monitoring capability and enhanced product monitoring additional to the novel sensors. Sensors functionality of strategic importance include,

- MicroElectroMechanical systems (MEMS)-based components;
- Smart sensors that have intelligent readout electronics and the microsensor in a single chip;
- Zero power 'flow tracker' sensors.

These sensors will provide an essential source of real time knowledge underpinning the attributes of the product. IMaR will also undertake the integration of each single sensor on a common, multifunctional thus meeting the requirement of a standardised output consistent for the use in the ProPAT platform.

How the legal entity profile matches the tasks in the project

IMaR provides an open access point to industry for the provision of applied research and technical expertise and aims to be industries partner of choice for innovative applied research using technologies including Radio Frequency Identification (RFID), auto ID, Sensors, Mechatronics and Industrial automation. IMaR tailors custom solutions to increase productivity and cost efficiency across a broad range of manufacturing industries including automotive, telecommunications, pharmaceutical, production/manufacturing, aeronautical, environmental and agri-business. IMaR exploits the use of smart Mechatronics via sensor integration, precision instrumentation and automation conditioning to enhance process automation, reduce lead times and ultimately deliver significant manufacturing cost reductions across industry partners.

The IMaR Technology Gateway is currently involved in a range of collaborative research projects with a variety of industrial partners in the fields of mechatronics, sensor development and RFID, including the co-ordination of a research grant of €1.6 million to work with partners across Europe in the development of next generation agricultural equipment. This project is supported by the European Union through the 7th Framework Programme (FP7) "Research for the benefit of SME's".

CV of the persons who will be primarily responsible for carrying out the work

DR. JOSEPH WALSH is the Head of School of STEM in the IT Tralee and a Director of the IMaR Technology Gateway at the IT, Tralee. Dr. Walsh has a Ph.D in Electronic Engineering from the University of Limerick and an M.Sc (Eng) in Bio-engineering jointly from Trinity Collage Dublin (TCD), University College Dublin (UCD) and the University of Limerick (UL). He has worked on national and international projects in both industry and research institutes. His research experience encompasses the field of mechatronics including dairy technology, next generation auto-ID systems, intelligent senor systems, robotics, automated milk quality analysis, bio-instrumentation and microelectronic circuit test technology. Dr. Walsh is a corporate member of the Institute of Physics (CEng CPhys MinstP), the Institute of Physics and Engineering in Medicine (CSci IPEM) the Institution of Engineering and Technology (MIET), the Institute of Electrical and Electronic Engineering (MIEEE) and the Institution of Engineers of Ireland (MIEI). He was been awarded the titles of Chartered Engineer and Chartered Physicist from the Institute of Physics and Chartered Scientist from the Institute of Physics and Engineering in Medicine

DR. PAT DOODY is an experienced Researcher and Lecturer in the School of Science and Computing in the Institute of Technology Tralee (ITT). Pat is the Co-founder and Executive Director of the Centre for Innovation in Distributed Systems (CIDS) at ITT. Pat was also Co-Founder and Co-Leader of the Wireless and Mobility Research Group at the ITT.His research interests include AutoID technologies, wireless technologies, intelligent knowledge discovery, data mining, and application of artificial intelligence based neural network algorithms, quality of voice systems and energy initiatives. In partnership with ALTOBRIDGE Ltd. (www.altobridge .com), a network communications start-up located in the Kerry Technology Park, Pat was the principal investigator for this project to develop an integrated end-to-end RFID tag to back office architecture. Pat obtained his bachelor degree in mathematics from the University of Limerick, Ireland in 1991 and PhD in mathematics focusing on artificial intelligence and neural networks from the University Of Limerick,

Ireland.

DR. ULTAN MC CARTHY received his Ph.D. degree in Biosystems Engineering from University College Dublin, Ireland in 2010. He worked at the RFID Center for Applied Research at the University of South Florida (USF). His areas of interest include Smart Supply Chain Management dynamics and security enhancement via RFID technology. He is also involved in the development of improvement strategies to enhance competitiveness across industries via the integration of Intelligent Mechatronics and data management techniques. He also works on the development of smart decision support systems adding value to the Supply Network and in the development of technological Return-on-Investment (ROI) models. Experience to date has spanned across a number of industries including food, pharmaceutical, automotive, defense and aviation.

MR. ANDREW SHIELDS is Strand Leader - RFID, Internet of Things & Sensors at the IMaR Technology Gateway in the Institute of Technology Tralee, Co Kerry, Ireland. Andrew has worked on numerous applied project with Industry over the past seven years. Including projects which conducted research into the area of RFID, public mobile platforms such as GSM /GPRS, Distributed Sensor Based Systems and Data Mining. He completed his Masters of Science degree at the Institute of Technology Tralee in 2007. He obtained his First Class Honours Bachelor degree in software development in 2002, also from the Institute of Technology Tralee. Andrew is currently undertaking a PhD. programme in ITT in the area of Artificial Intelligence with a particular focus on Reality Mining.

DR. DANIEL RIORDAN is Strand Leader - Mechatronic at the IMaR Technology Gateway at IT Tralee. Dr. Riordan has a Bachelor of Electronic Engineering from Dublin City University (2005), a Masters of Computer Science from IT Tralee (2008) and a Ph.D. in the area of Auto-ID Systems within CIMSS (2012) under an Irish Research Council (IRCSET) scholarship. Dr. Riordan has worked within the CIMSS / IMaR group since 2008. His research experience includes, artificial intelligence, computer modelling systems, sensor systems, bio-instrumentation, electro-magnetic interference analysis, digital signal processing and hardware and embedded electronics. He is a member of the Institute of Electrical and Electronic Engineering (MIEEE).

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Abdulkarim A., Elsibaei A., Jackson B., Riordan D., Rice J. (2012) “The use of sound analysis to guide femoral reaming in uncemented Total Hip Arthroplasty: A New Concept” Journal of Atlantic Orthopaedic, November 2012. *Winner of 1st medal prize in the Sylvester O'Halloran National surgical meeting
2. "Investigation of Bone Resonance during Femoral Reaming in Hip Replacement Surgery", O"Donoghue, P., Jackson, B., Riordan, D.,Walsh, J., Abdulkarim, A., Asibaei, A., Rice, J. 7th International Conference on Sensing Technology (ICST 2013) December 5th, 2013 *Runner-Up Best Paper in HealthCare Sector*
3. Doody P, Shields A. 2012. Mining network relationships in the internet of things. Proceedings of the 2012 international workshop on Self-aware internet of things (Self-IoT '12). ACM, New York, NY, USA, 7-12.
4. Shields A, Datta S, 2008. Can We Apply Principles From Social Networking To Healthcare Informatics For Intelligent Data Analytics? MIT Libraries DSpace
5. WSEM2, Doody P, Shields A, 2008. RFID, Sensors, GPS, GSM: Abundance of Data yet Starved for Knowledge?, CIDS Inaugural Symposium, Tralee, Kerry, Ireland.

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. Development of an On-Line Milk Quality Analysis System in a Dairy Milking Plant **(PI)**
2. Development of Environmental sensors for the Agricultural Industry **(Sub Contract)**
3. CyberBar© Smartfone delivered anti-tamper food traceability system based on direct on-food Data Matrix (DM) printing and chain integrity sensors **(Sub Contract)**
4. BOSCA© Smart Networked Sensing Systems in Agriculture (Poultry Industry Focus) **(Sub Contract)**
5. "Chirpmetrics - Monitoring the health and welfare of broiler chickens using smart vocalisation analysis techniques". **(Co-Applicant)**

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
IMaR is currently co-ordinator of a 7th Framework Programme (FP7) “Research for the benefit of SME’s” (€1.6 million) to work with partners across Europe in the development of next generation agricultural equipment. Currently the team consist 21 research staff (12 full time, 9 post graduate), 2 research directors and 8 technical academic staff across 2 strands. Equipment of relevance to ProPAT include but are not limited to the following; • 1 fully serviced Electronics lab (including development and demonstration areas) • 1 fully serviced Auto ID and Software lab (including development and demonstration areas) • Irelands only open device laboratory • Intelligent sensors & remote monitoring System Development • Advanced System test and evaluation • Horiba Jobin Spectrometer “linear Array Spectrometer VS 140”
▯ HaMAMATSU SWNIR Spectrometer ▯ Autolab Potentostat Galvanostat ▯ Melles Griot “Laser Light Neon Laser”

Participant no.	10	Name (short name)/Country	Spectral Engines Oy (SPECTRAL) / Finland
Description of the legal entity			
Spectral Engines Oy is a new start-up limited company in the field of novel optical sensors. Spectral Engines Oy is a spin-off from VTT Technical Research Centre of Finland (www.vtt.fi) which is the largest multidisciplinary contract research organization in Northern Europe. The motto of Spectral Engines is to take “spectrometer performance to sensor price point” by creating miniaturized, industry-grade spectral sensors utilizing VTT’s novel MEMS technology. We aim to change the way industries perform their material parameter control and automation by introducing new multiparameter, high-performance and reliable miniaturized sensors to replace old non-specific sensors and expensive spectrometers. Spectral Engines employs 4 people but aims to double the amount in a year. The founders have vast experience in working in large, international research projects. Working internationally is everyday business to Spectral Engines as our customers and partners are largely international. Spectral Engines was a semi-finalist in the SPIE Startup Challenge in 2014 and has been selected in the Top 25 companies in the Nordic Cleantech Open 2014.			
Main tasks			
SPECTRAL’s main role in this project is the development of the MEMS-based spectral sensors.			
How the legal entity profile matches the tasks in the project			
SPECTRAL’s core technical competence is in the design and realization of optical sensors based on MEMS Fabry-Perot Interferometers. SPECTRAL’s expertise is to match the end-user requirements all the way to MEMS fabrication specifications in order to meet cost and performance targets. We design the sensor concept, identify critical parameters by modelling, environmental tests and vast experience and use our knowledge in electronics design, embedded software, thermal design and spectroscopy to realize high-performance sensors. SPECTRAL’s customers are mostly in the field of process industry with a goal to save material, save energy and improve quality – all of which can be achieved through novel spectral sensor technology. SPECTRAL’s rank within the Top 25 in the Nordic Cleantech Open 2014 (out of 107 applicants) supports our role in bringing sustainability into process industries.			
CV of the persons who will be primarily responsible for carrying out the work			

Jarkko Antila is the Chief Executive Officer at Spectral Engines. He received his M.Sc. in electrical engineering from Helsinki University of Technology in 2003. He has 15 years of experience on optical measurement technologies, sensors and MEMS. Previously he has worked as a Team Leader and Senior Scientist at VTT. He has gained industrial expertise in sensor development, heading an R&D team at Inficon during 2006-2009. Mr. Antila's technical core competence is in interferometric sensors and miniaturised spectrometers based on Fabry-Perot interferometers with 5 patents and 6 patent applications in the field. Mr. Antila has a proven track record in leading and having major roles in several co-operational research projects.

Janne Suhonen is the Chief Commercial Officer at Spectral Engines. He received his MSc in Electrical engineering from the University of Oulu, Finland in 1995 and eMBA 2006. He started at VTT 1995. In his research career, Janne Suhonen has focused on the studies and development of optical analysers and measurements based on Near Infrared (NIR) and Raman spectroscopy. He has wide-ranging experience of optical measurement technologies in the different industrial applications covering the industrial fields of Pulp&paper, steel and pharmaceuticals industries. He has worked as Technology Manager, deputy Research manager and Key Account Manager roles at VTT.

Uula Kantojärvi is the Chief Technology Officer at Spectral Engines. He received his M.Sc.(tech.) from Helsinki University of Technology in 2006. Mr. Kantojärvi has worked at VTT Technical Research Centre of Finland for 10 years and he has specialized in system design, simulation and characterization of optical spectroscopic instruments. He has worked as researcher, senior scientist and team leader of 15 experts. He has prepared and lead projects related to optical instruments, MEMS and absolute calibration systems.

Jussi Mäkynen is a Senior R&D Engineer at Spectral Engines. He received the M.Sc. (Tech.) in electrical engineering from the Helsinki University of Technology in 2009. He has joined the research staff at optical sensors team at VTT Technical Research Centre of Finland in 2007. His core competence is in electronics design and embedded system development. He has 6 years of experience designing optical measurement devices and control electronics for FabryPerot interferometers, working as part of an international research**.**

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Product: Ultra-Compact IR Spectral Sensor with a USB2.0 interface (www.spectralengines.com)
2. Antila, J., Tuohiniemi, M., Rissanen, A., Kantojärvi, U., Lahti, M., Viherkanto, K., Kaarre, M. and Malinen, J. 2014.

   MEMS- and MOEMS-Based Near-Infrared Spectrometers. Encyclopedia of Analytical Chemistry. 1–36.)
3. Mannila, Rami; Tuohiniemi, Mikko; Mäkynen, Jussi; Näkki, Ismo; Antila, Jarkko; Hydrocarbon gas detection with microelectromechanical Fabry-Perot interferometer; Next-Generation Spectroscopic Technologies VI, 29 - 30 April 2013, Baltimore, MD, United States; Proceedings of SPIE. SPIE. Vol. 8726 (2013), Article number 872608
4. Antila, Jarkko; Rissanen, Anna; Mannila, Rami; Mäkynen, Jussi; Näkki, Ismo; Tuohiniemi, Mikko; Miniaturized MOEMS spectrometer technology for gas sensing; International Conference on Optical MEMS and Nanophotonics 2013, 18 - 22 August 2013, Kanazawa, Japan; Conference proceedings. IEEE (2013), 59 - 60
5. Jussi H. Mäkynen ; Mikko Tuohiniemi ; Antti Näsilä ; Rami Mannila and Jarkko E. Antila; " MEMS Fabry-Perot interferometer-based spectrometer demonstrator for 7.5 µm to 9.5 µm wavelength range ", Proc. SPIE 8977, MOEMS and Miniaturized Systems XIII, 89770U (March 7, 2014);

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. IQFUEL - Integrated sensor for determining the Quality of FUEL (Euripides EUR 07-408)
2. Imospe - Integrated adjustable MOems-SPEctrometer Platforms (Tekes 40423/09)
3. FABRY - Spectroscopic sensor devices based on novel FABRY-Perot interferometers (Tekes 40244/11)
4. VIAMOS - Vertically Integrated Array-type Mirau-based OCT System for early diagnostics of skin cancer (FP7-ICT2007-2, Grant No. 318542)
5. Spectral Engines - Commercialization of Fabry-Perot Interferometer technology as spectral engines and solutions (Tekes 40031/13)

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

SPECTRAL will invest in testing facilities, including wafer-level characterization tool, climate chamber and multichannel wavelength calibrator. The equipment is placed in conventional surroundings, i.e. no clean rooms are required or used.

Participant no.	11	**Name (short name)/Country**	VTT Memsfab Ltd (Memsfab) / Finland

Description of the legal entity

Contract manufacturing of micro- and nanoelectronic materials and devices.

Main tasks

Manufacturing of two lots of tuneable MEMS Fabry-Pérot filters

How the legal entity profile matches the tasks in the project

Perfect match, VTT Memsfab Ltd is one of the three tuneable MEMS Fabry-Pérot filter manufacturers in the world

CV of the persons who will be primarily responsible for carrying out the work

Martti Blomberg (male) obtained his licentiate's degree in Electrical Engineering in 1991 at Helsinki University of Technology. He has worked for 30 years in silicon technology, and more than 20 years in MEMS device design and manufacturing. He is currently the CTO of VTT Memsfab Ltd. His recent activities include process and device development and commercial manufacturing of various MEMS and MOEMS devices such as electrically tuneable surface micromechanical Fabry–Perot filters and thermopile detectors. He has written about 50 publications in the field and is an inventor in a number of patents, including several key patents in MEMS Fabry-Perot technology.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Micro-machined Fabry-Pérot interferometer for thermal infrared, IEEE SENSORS 2013 Conference, 4 - 6 November 2013, Baltimore, Maryland, United States
2. Monolithically integrated microspectrometer-on-chip based on tunable visible light MEMS FPI, Sensors and Actuators, A: Physical. Elsevier. Vol. 182 (2012) No: August, 130 - 135
3. MOEMS miniature spectrometers using tuneable Fabry-Perot interferometers, Journal of Micro/Nanolithography, MEMS, and MOEMS. SPIE. Vol. 11 (2012) No: 2, 023003-1-6
4. Electrically tunable surface micromachined Fabry-Perot interferometer for visible light, Sensors and Actuators A: Physical. Elsevier. Vol. 162 (2010), 184-188
5. Electrically tuneable micromachined Fabry-Perot interferometer in gas analysis, Physica Scripta. Vol. T69 (1997), 119 - 121

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. Spectroscopic sensor devices based on novel Fabry-Perot interferometers, FABRY, 2011 - 2014
2. Micro-Optical MEMS, micro-mirrors and pico-projectors, Lab for MEMS II (ENIAC project), pilot line development for MOEMS manufacturing, accepted, to be started in 2014

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
Micronova cleanroom with all necessary manufacturing tools are fully available

Participant no.	12	Name (short name)/Country	VTT Memsfab Ltd (Memsfab) / Finland

Description of the legal entity

DECHEMA - Society for Chemical Engineering and Biotechnology e.V., is an interdisciplinary scientific society with more than 5,500 personal and institutional members (including about 500 SMEs), whose activities cover wide areas relevant to the process industries. DECHEMA is dedicated to the support of R&D progress and implementation in various fields of chemical engineering, including reaction engineering, catalysis, materials, process efficiency and environmental protection. DECHEMA is board member of the European Technology Platform Sustainable Chemistry (SusChem) and hosts several secretariats of organisations in this field: European Federation of Chemical Engineering (EFCE) and its international equivalent (WCEC), the European Federation on Corrosion (EFC) and the German Catalysis Society (GECATS). DECHEMA is organiser of the International Exhibition-Congress on Chemical Engineering, Environmental Protection and Biotechnology (ACHEMA) and of around 50 conferences with over 10.000 participants per year, and offers more than 20 training courses on different topics. Co-ordination and support of DECHEMA's activities is carried out by about 190 employees.
DECHEMA hosts more than 50 websites including project websites and secure collaborative workspaces for information exchange and knowledge management of project consortia.

Main tasks

DECHEMA will lead the tasks related to standards: 1.4 Survey of standard quality control approach, as well as task 8.1 Comparison vs. existing quality control and recommendations for standardisation. They will also lead the training activities organising several workshops for their members (WP9) and act as dissemination manager (WP10).

How the legal entity profile matches the tasks in the project

Research coordination and management; strong network in chemical engineering and process industries; high expertise in linking industry and R&D; strong skills in international communication, dissemination and exploitation

CV of the persons who will be primarily responsible for carrying out the work

DR. RENATA KÖRFER is head of standardization department at DECHEMA and has 15 years of professional experience working in science and research management. Her background includes extensive knowledge of assessment and use of innovative process and methods in the field of process engineering especially in Food and Chemical industry. She has more than 8 years' experience in coordinating and participating in various EU projects (FP6/FP7) as well as EC Project Technical Advisor (PTA), proposal evaluator and project reviewer. She is also an executive secretary of technical committees DIN 055, CEN 332 and ISO 48 and a Member of the Managing Board, Working Committee DECHEMA/GDCh 'Process analytical technologies (PAT)'.

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Member of PPP SPIRE
2. Secretary of SusChem
3. Member of the Managing Board, task force DECHEMA/GDCh 'Process analytical technologies (PAT)'
4. Executive secretary of technical committees DIN 055, CEN 332 and ISO 48

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. EU project coordination and management of NanoBioPharmaceutics, Next-GTL, Synflow, CO2NET, E4Water, ChemWater, M4CO2, R3Water

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
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Participant no.	13	Name (short name)/Country	Grecian Magnesite (GM)/Greece

Description of the legal entity

Grecian Magnesite is a privately owned modern company established in 1959 possessing a substantial know how in mining, processing and marketing of industrial minerals. One of its mission is to exploit and process mineral raw materials supplying quality products to a broad spectrum of industries. It ranks among the top magnesia producers and exporters in the world being no. 1 exporter of CCM in the EU, with a staff around 320 people and turnover 45 million €. The company´s major deposits are located in Chalkidiki. The total production capacity for calcined products (caustic magnesia, dead-burned magnesia) is close to 200 000 mt a year whereas 95% output is exported. The annual production capacity of the monolithic refractories exceed 60 000 mt of basic monolithics. The company´s sales value to all EU countries exceeds 80% of its turnover.

Main tasks

GM will participate in WPs 1, 2,6 and 7. GM will evaluate/validate the performance of the sensor and its contribution to the milling capability/quality etc. GM will be responsible for making the necessary modifications to its milling facility in

order to accomodate the new technology (sensor).

How the legal entity profile matches the tasks in the project

Grecian Magnesite, is a major player in magnesia industry: the company produces and commercializes caustic magnesia (MgO), dead-burned magnesia, raw magnesite and basic monolithic refractories covering a broad spectrum of applications (ceramics, fertilizers, welding fluxes, metallurgical uses, rubber, construction, abrasives, glass industry, chemicals, fillers, pulp and paper industry, pigments, anticorrosion, printing inks, steel industry).
Grecian Magnesite is a modern company possessing a substantial know-how in mining, processing and marketing of industrial minerals. Grecian Magnesite is ISO 9001 certified and they have state of the art flowsheet ensuring exceptional quality and unique consistency of all their products. Their decades of experience in processing and milling of industrial minerals will be an essential contribution to the project's targets.

CV of the persons who will be primarily responsible for carrying out the work

THEOFILOS ZAMPETAKIS: (male) R&D Center Manager working with Grecian Magnesite for 20 years. He is a Chemical Engineer, holding a Ph.D. in Materials Science. He has a 25 year experience in industrial research and has participated in more than 10 EC funded projects.
KOSTAS DAGGILAS: (male) Yerakini Mine Quality Control Manager working with Grecian Magnesite for more than 20 years. He is a chemist with a long experience in the evaluation of magnesite ores.
POLYKARPOS PAPAGEORGIOU: (male) Production Manager of Grecian Magnesite's Yerakini mine. He is a Mining Engineer with long experience (over 20 years) in extraction, beneficiation, burning and final processing of magnesite ores.
PAVLOS VAGIONAS: (male) Mechanical Engineer and New Projects Manager working with Grecian Magnesite for over 20 years. He has a long experience in mechanical equipment related to beneficiation and final processing of industrial minerals.
SPYROS GALAZOULAS: (male) Metallurgist working with Grecian Magnesite for 8 years and is responsible for the final treatment of the products (CCM & magnesite). Experience in mineral processing (magnetic separation, screening processing, milling and beneficiation circuit).

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

1. Caustic calcined magnesite (CCM) sized (crushing, sieving) in various granulometries (relevant materials ActivMag, NOR-R, 88C, 83C etc)
2. Magnesium carbonate in granular and powder form (relevant materials are KERMA, R3 etc.)
3. Dead burned magnesite (DBM) sized in specific grain sizes (relevant materials are TANMA, MAGCAST etc.)

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

1. Caustic calcined magnesium oxide processing (e.g. magnetic separation, sieving, milling)
2. Magnesite ore beneficiation using photo-optic and laser sorters
3. Calcination of magnesium carbonate at high temperatures
4. Magnesium carbonate sizing (grinding, sieving, milling)
5. Dead burned magnesite processing: grinding, sieving, milling

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Optical and laser sorters, crushers, magnetic separators, sieves.
The milling installation includes an Esser-Neumann pendular mill for the milling of CCM and magnesium carbonate as well as dynamic classifiers and air bag filters.

Participant no.	14	**Name (short name)/Country**	GlaxoSmithKline (GSK) /UK

Description of the legal entity

GlaxoSmithKline (GSK) is a British multinational pharmaceutical, biologics, vaccines and consumer healthcare company headquartered in Brentford, London. It is the world's fourth-largest pharmaceutical company as measured by 2009 drug sales. The company was established in 2000 by the merger of Glaxo Wellcome plc (formed from the acquisition of Wellcome plc by Glaxo plc) and SmithKline Beecham plc (formed from the merger of Beecham plc and SmithKline Beckman Corporation, which in turn was formed by combining the Smith Kline French and Beckman companies).

GSK has a portfolio of products for major disease areas such as asthma, cancer, virus control, infections, mental health, diabetes and digestive conditions. It also has a large consumer healthcare division that produces oral healthcare and nutritional products, drinks and over-the-counter medicines. Stevenage, i.e. the company’s main API site, will be involved in the project.

Main tasks

Definition (selection) of typical real industrial production processes which could be clearly improved by and be compatible with the described sensors platform.

Provide practical hints and advices to the RTDs in order to develop an application-oriented PAT platform, as well as relevant samples along the process for offline calibration.
Provision of the facilities for installing and validating the developed PAT platform, contribution to its validation.

How the legal entity profile matches the tasks in the project

GSK is an end-user in the project and is expected to benefit from the installation of the PAT platform in its current production process leading to significant cost savings due to minimisation of drying times by detecting on each batch the drying end point, increase in productivity due to better control of the industrial processes as well as reduce the off-spec product quantities and ensuring high standards of final product quality. The main assets of ProPAT vs. existing systems available on the market or already present at GSK are the use of lower cost sensors with good accuracy combining chemical and physical information. The detection of particle size is important for product quality but also for operative questions, e.g. to avoid clogging filters in case of big agglomerates in the spray drying system. Finally a system like ProPAT would also allow reducing the time to market by assisting in new product development and allow the production facilities to be more flexible.

CV of the persons who will be primarily responsible for carrying out the work

NIGEL RICHARDSON (male), Director of future analytical technologies
LUISA FREITAS DOS SANTOS (**female**), Global Head Analytical Sciences, Product Development - PTS R&D
DR. THEODORA KOURTI (**female**), Senior Technical Director, Pharma Launch & Global Supply

List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content

GSK produces pharma products of very different types for which ProPAT may be relevant to improve process/quality.
In addition GSK reports the following publications related with PAT:

- JA Calderon De Anda (PhD student), XZ Wang*, X Lai, KJ Roberts, KH Jennings (GlaxoSmithKline-GSK), MJ Wilkinson (GSK), D Watson (Malvern Instruments Ltd - MIL), D Roberts (MIL), (2005) "Real-time product morphology monitoring in crystallisation using on-line imaging", AIChE Journal, 51(5), 1406-1065.
- JA Calderon De Anda, XZ Wang, X Lai, KJ Roberts, KH Jennings (GlaxoSmithKline-GSK), MJ Wilkinson (GSK), D Watson (Malvern Instruments Ltd - MIL), D Roberts (MIL), (2004), Real-time product morphology monitoring in crystallisation using on-line imaging. AIChE Journal
- JA Calderon De Anda, XZ Wang, KJ Roberts, (2004), Multi-scale segmentation image analysis for the inprocess monitoring of particle shape with batch crystallisers. Chemical Engineering Science.

A list of up to 5 relevant previous projects or activities, connected to the subject of this project

Among number of them, GSK reports the following relevant projects in terms of work with PAT mainly in terms of particle size monitoring and drying process control, in collaboration with UNIVLEEDS in several instances:

1. Project on Particle Breakage in Dryers (2006-2010): case award with UNIV LEEDS sponsored by the EPSRC and GSK. The project utilised DEM simulations to estimate stresses occurring in agitated dryers. Attrition occurring in a shear cell was analysed experimentally and coupled with the DEM simulations to give a prediction of attrition in agitated dryers.
2. On-line Imaging and Image Analysis for Shape Measurement and Control of Pharmaceutical Crystals Grown from Solution (2001-2005), sponsors: CONACYT, EPSRC, Malvern Instruments Ltd and GlaxoSmithKline
3. Control and Scale-up of Batch Crystallisation Processes for Organic Speciality Chemical Products (2000-2005), EPSRC, AEA Technology, AstraZeneca, Bede Scientific Instruments, BNFL, Clairet, GlaxoSmithKline, HEL Ltd, Malvern Instruments, Pfizer, Syngenta

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

The GSK New Product introduction site uses all typical formulation processes like granulation, tabletting, blending, etc. Facilities of different scales are available for installing ProPAT system as well as range of analytical techniques for performing reference quality controls.

Participant no.	15	**Name (short name)/Country**	Megara Resins S.A (MEGARA) / GREECE

Description of the legal entity

Megara Resins S.A. is a diversified manufacturer and supplier of raw materials for industrial and architectural coatings as well as rosin based and other synthetic resins for the paint, adhesive, paper and construction industry. For over 40 years, Fanis Megara Resins S.A. has been a pioneer in creating innovative technologies to help coatings formulators meet their customers' most demanding applications. Today, the company remains firmly committed to the pillars of innovation and new technologies and is widely regarded as being the most innovative Greek supplier to the coatings

industry through its continued investment in R&D, technical support and new product development. Megara Resins
offers its customers advanced and diverse products and technologies for surfaces with an emphasis on environmentally friendly products such as powder coating resins, rosin dispersions, alkyd resins, water-based acrylic dispersions, and unsaturated polyester resins. The company is committed to working with its customers to continuously developing environmentally advanced solutions and is dedicated to open communication concerning the safe handling, distribution, use and disposal of the products it makes.
Main tasks
Definition (selection) of two typical real industrial production processes (Polyester resin for powder coatings and Alkyl Succinic Anhydride) which could be clearly improved by PAT (Process Advanced Technologies) and be compatible with the described sensors platform. Provide practical hints and advices to the RTDs in order to develop an application-oriented PAT platform, as well as relevant samples along the process for offline calibration. Provision of the facilities for installing and validating the developed PAT platform, contribution to its validation
How the legal entity profile matches the tasks in the project
Megara Resins S.A. is an end-user in the project and is expected to benefit from the installation of the PAT platform in its current production process leading to significant cost savings due to minimisation of reaction times, increase in productivity due to better control of the industrial processes as well as reduction of the off-spec product quantities and ensuring high standards of final product quality.
CV of the persons who will be primarily responsible for carrying out the work
Nikos Agiasofitis (male) holds a degree in Chemistry from the Aristotle University of Thessaloniki and is Production Manager of the polyester resins for powder coatings Business Unit of Megara Resins S.A. since 2013. He has long industrial experience of more than 14 years during which he has provided leadership, coaching, mentoring and training while supervising and coordinating activities of +50 employees. Currently he is responsible for overseeing the entire manufacturing process of the polyester resins facility and making sure that production lines are running smoothly and efficiently. He works closely with supervisors and maintenance staff to plan work, set targets and make sure the finished products meet quality standards. Nikos is expert in the unit operations of the polyester resins facility. **Poppy Krassa** (**female**) has a BSc degree in Chemistry awarded from Ioannina University in 1999 and a MSc degree in Catalysis and Environmental Protection awarded by the Hellenic Open University in 2007. Poppy joined Megara Resins in 2013 and was appointed as Research Project Manager. She strives to develop research capacity and capability and is committed to embedding research into core Megara business. She has more than 14 years of research and management experience in EC-funded collaborative research projects with extensive knowledge of polymers for paints, coatings, structural materials and adhesives, in particular alkyd resins, unsaturated and saturated polyester resins and aqueous dispersions. During her past experience she has managed numerous funded and in-house research projects related to polymer synthesis and characterization which led to the development of new innovative products.
List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content
MEGARA produces resins of very different types for which ProPAT may be relevant to improve process/quality
A list of up to 5 relevant previous projects or activities, connected to the subject of this project

Megara has been only lately involved in 4 European and 1 national projects, all closely related to improving their products and concern either the optimisation of the production process, the development of alternative environmentally friendlier products, or substitution with bio-based alternatives to increase their sustainability.
The most relevant one concerns the "New weather-stable low gloss powder coatings based on bifunctional acrylic solid resins and nanoadditives" which is an IAPP project (Grant agreement no.: 324410), 4-year project which started in November 2013 and is coordinated by Megara. The project deals with acrylic resin and cover one application in common (powder coatings).

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work

Megara Resins S.A. is equipped with manufacturing facilities comprising industrial scale reactors with capacities ranging from 2.000 kg to 10.000kg capable of production of a wide range of solvent-based and water-based emulsions, polyester resins and other synthetic resins such as rosin dispersions, alkyd resins, acrylic resins etc. used in the manufacturing of paints and coatings. A pilot - scale reactor is also available with 200 kg production capacity.

Participant no.	16	**Name (short name)/Country**	MBN Nanomaterialia S.p.A.(MBN)/ITALY

Description of the legal entity

MBN Nanomaterialia S.p.A. (Italy) (http://www.mbn.it/) was established in 1994 with core business being concentrated in the development of mechanochemical reactors for the production of nanophased materials produced by high energy ball milling. With an industrial production capacity of 150 ton/range, MBN's mission is to bring nanotechnology from research environment to everyday life products offering to the market innovative nanomaterials adaptable to conventional and

new powder metallurgy and deposition processes. The industrial production is focused on environmental friendly (Co free) alloys for diamond grinding tools and high performance thermal spraying powders (Ti, WC-Co, MMC's). All materials produced by MBN represent the direct outcome of scientific and technological research initiatives performed at National and European level in collaboration with Universities, Research Institutes and Industries. MBN is also working in the development of innovative material solutions in many application fields such as nanostructured metal and polymeric powders for high resolution additive manufacturing, polymer based alloys for textile and consumer goods and advanced Mg and Ni based alloys for hydrogen storage.

Main tasks

- Development of powders with proper size distribution based on metals or ceramic through High Energy Ball Milling
- Suppling of nanostructured micro particles of metals or ceramic produced by mechano-chemical technology
- Morphological characterization of metal powders, particles size distribution, by SEM analisys, DLS, BET and XRD analysis
- Validation of the particle size analyzer in industrial environment for in line/at line monitoring of dry particles or particles in solution

How the legal entity profile matches the tasks in the project

They are a qualified and experienced end-user partner for the project.

CV of the persons who will be primarily responsible for carrying out the work

PROF. ING. PAOLO MATTEAZZI. (Male) Professor of Metallurgy at mechanical engineering of University of Udine, he founded the company MBN in 1994. With a consolidated and well proved experience in promoting, coordinating and managing research projects at National and European level, Prof. Matteazzi is also Chairman of European Technology Platform on Nano-Micromanufacturing (MINAM) and acting Chair of NANOfutures ETIP (European Technology Innovation Platfform) He is author of more than 200 scientific papers in materials science, more than 140 in the field of powder technology and nanophased materials. **DR. IVAN LORENZON**. (Male) Graduated in chemistry at Venice University, he worked as product quality responsible and as reference solving person in a water treatment laboratory. In 2008 he joined MBN as responsible of materials research and product developer especially as regards the development of new hybrid organic-inorganic composites in the frame of National and European project initiatives.
List of up to 5 relevant publications, and/or products, services (including widely-used datasets or software), or other achievements relevant to the call content
1. MBN currently produce 50tons per year of nanostructured materials
A list of up to 5 relevant previous projects or activities, connected to the subject of this project
1. NANODEFINE project- 604347- "Development of an integrated approach based on validated and standardized methods to support the implementation of the EC recommendation for a definition of nanomaterial" 2. NANODETECTOR project - 280478 - " Ultrasensitive plasmonic detection of single nanoparticles" 3. NANOFORART project - 282816 - "Nano-materials for the conservation and preservation of movable and immovable artworks" 4. COOLCOVERING project - EeB.NMP.2010-1- Development of novel and cost-effective range of nanotech improved coatings to substantially improve NIR (Near Infrared Reflective) properties of the building envelope
Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work
MBN has developed its own High Energy Ball Milling equipment at different scale for the industrial production of NanoMaterials, Nanostructured Materials, Composite, Solid and Liquid dispersions. MBN has all the equipment needed for secure handling nano-structured powder, facilities for re-agglomeration by spray drying, facilities for powder classification in liquid and in controlled atmosphere, as well as equipment for suspension homogenization by ultracavitation. MBN utilize standard equipment for physical and morphological characterization of the material produced: specific surface area measurements (BET), average crystal size (XRD), average particle size distribution (Laser Diffraction), elemental analysis (EDS, LECO), imaging (SEM), optical properties (UV-Vis-NIR spectrometer), thermal properties (TGDTA).

4.2. Third parties involved in the project

Participant no. 8- Universitat de Barcelona (UB)/ Spain

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	N
If yes, please describe and justify the tasks to be subcontracted	
Does the participant envisage that part of its work is performed by linked third parties	N
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	

Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	Y
Fundació Bosch i Gimpera (FBG) is a third party of Universitat de Barcelona (UB). FBG was created and is controlled by UB, and has an authorisation to administer International funds on behalf of UB. According to the "Annotated General Grant Agreement", FBG falls under the specific case "Foundations, spin-off companies, etc., created in order to handle the administrative/financial tasks of the beneficiary", and it is in charge of the administrative and financial management of European projects at the UB. FBG does not perform scientific/technical work in the project.	

Participant no. 10- Spectral Engines Oy (SPECTRAL) / Finland

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted)	Y
If yes, please describe and justify the tasks to be subcontracted Spectral Engines will subcontract the following tasks for the realisation of test rigs: • MEMS-detector packaging (this is outsourced). • Electronic fabrication • Optomechanics design and fabrication • Thermal design and simulations These tasks have been valued at €42,500. The justification for this subcontracting is that Spectral Engines does not have all the necessary in-house expertise, design software or installations (clean rooms) in order to perform this work.	
Does the participant envisage that part of its work is performed by linked third parties	N
If yes, please describe the third party, the link of the participant to the third party, and describe and justify the foreseen tasks to be performed by the third party	
Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement)	N
If yes, please describe the third party and their contributions	

5. Ethics and Security

5.1. Ethics

In keeping with the required Ethics Appraisal Procedure, an Ethics Self-assessment was performed with the completion of the Ethical Issues Table. It can be concluded that this project does not involve any ethics issues.

However, in relation to potential safety issues derived from sampling along the process, which can at times be hazardous, e.g. when handling hot resins, unreacted monomers, powdery particles, etc.), any environmental harm that could be caused by the research will be minimised at all times by the implementation of necessary measures that will be taken to mitigate the risks. The Coordinator will work closely with the consortium in order to ensure that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in the project.

It is fitting to first of all mention that every partner related to chemical activities in the consortium already follows strict rules according to the local legislations as well as more global regulations such as REACH both for the synthesis of the compounds and their handling, and that the project will of course respect the pre-established precautions. When needed the Environmental, Hygiene and Safety responsible of each location could also be involved in assessing that the practice followed is adequate. The different RTDs partners involved in the WP2 and 3 will also respect strictly the Material Safety Data Sheets in terms of how to use the respective products in a safe manner after receiving, if necessary, additional training from the producer. Although sampling handling is expected to be required for the calibration of the new techniques, it is in fact similar to what is currently done for the off line controls of the same products already manufactured in each site. On the long term, when implemented as a substitute of the current off line analysis, a clear improvement of the safety of the environment and lowering of worker's exposure will be derived from ProPAT.

In relation to the validation and demonstration work that will be performed during the project, a statement is provided by each of the process industries.

Megara Resins S.A (MEGARA) will demonstrate the ProPAT system for controlling the polymerisation process in at least one of the following: ASA resins production and UPR production, whereby in relation to the precautions for safe handling of polyester resins and ASA (alkenyl succinic anhydride) produced at Megara Resins:

- Saturated polyesters for powder coatings are condensation polymers formed by the reaction of polyhydric alcohols and saturated dibasic acids. The final product is an odourless, solid (flakes) material which contains zero VOCs (100% solvent-free), is chemically stable and does not require a hazard warning label in accordance with GHS criteria.
- ASA (alkenyl succinic anhydride) is an amber-brown viscous liquid at room temperature which is produced from the reaction of isomerized olefins and maleic anhydride. It is used as a sizing agent in the alkaline production of paper. The final product is classified as Xi (Irritant), R43 (May cause sensitisation by skin contact) due to its maximum concentration of 0.3% maleic anhydride.

All employees involved in the handling of powder coating resins and ASA and specifically during the sampling process wear special protective equipment such as heat insulating gloves, safety glasses, appropriate respiratory protection and personal protective clothing to prevent contact during mechanical processing and/or hot melt conditions.

Megara Resins S.A. has been certified according to the international environmental standard EN ISO 14001:2004 for its Environmental Management System & Services. Moreover, the company, acknowledging the importance of maintaining and improving environmental quality, has implemented a range of measures and actions in its Environmental Management System (EMS) that assure the continual improvement of its environmental performance. The necessary infrastructure for the monitoring of environmental aspects has been installed, and the relevant environmental audit procedures are followed.

Concerning the **Grecian Magnesite** (GMs) contribution to the project, which is related to the milling of magnesium oxide (Magnesia), the following is stated:

1) Magnesium oxide is not classified as hazardous material.
2) GM's production plants operate under Environmental Permits which regulates according to the Greek legislation the dust emission and the safety and health conditions in the working areas.

Concerning **MBN Nanomaterialia** S.p.A. (MBN), they will perform demonstration and validation work of the ProPAT system in a ceramic/metal milling process for ceramic-metallic blends production, whereby they state that:

1) The ceramic and the metallic part of a cermet contribute differently to the toxicity of the material, in the typical case of Tungsten Carbide plus Cobalt as a binder (WCCo) the toxic effects are caused by the small percentage of cobalt within the material. WCCo is classified as probably carcinogenic and can cause a lung disease known as "Hard metal disease" in exposed workers. Humans and the environment mainly get in contact with tungsten carbide-cobalt particles only when it is in its powdered form during handling or during the grinding of workpieces since it is very stable as a bulk piece or as a coating. In general, the human body as well as the environment are only exposed to very small amounts of tungsten carbide-cobalt particles, therefore only a very low risk has been associated with this material.

2) Users are safe when handling finished workpieces containing tungsten carbide-cobalt nanoparticles since there is no particle release from these products. Modern factories producing this material, as MBN, are obliged by the national and European law to have filter facilities installed to prevent an environmental release of these materials. This is accomplished also by MBN nanomaterialia, by using the required personal protection and all waste products are collected and disposed of as hazardous waste.

GlaxoSmithKline (GSK) will demonstrate the ProPAT system in a pharma mixing and granulation process. The exact formulation manufactured in the process where ProPAT will be validated is not selected yet, and will be defined in WP1 whereby safety aspects can be included as one of the criteria. While most excipients are safe, some concentrated active principles (APIs) must be handled with due precautions. Nevertheless the risks associated with such sampling of mixed powders and granulates, as well as the characterization of the upcoming substances, are already carried out with required precautions in the factory by trained analysts. As ProPAT will be mainly introduced in a step where the different compounds are already mixed and APIs are present at a relatively low concentrations, the associated risk is even further mitigated. Such samples are not typically dusty but in any case, the required Personal Protective Equipment will be used.

5.2 Security[72]

Please indicate if your project will involve:

- activities or results raising security issues: (NO)
- 'EU-classified information' as background or results: (/NO)

[72] Article 37.1 of Model Grant Agreement. *Before disclosing results of activities raising security issues to a third party (including affiliated entities), a beneficiary must inform the coordinator — which must request written approval from the Commission/Agency;* Article 37. *Activities related to 'classified deliverables' must comply with the 'security requirements' until they are declassified; Action tasks related to classified deliverables may not be subcontracted without prior explicit written approval from the Commission/Agency.; The beneficiaries must inform the coordinator — which must immediately inform the Commission/Agency — of any changes in the security context and — if necessary —request for Annex 1 to be amended (see Article 55)*